art science

# pliance-x system

# manual

v.20







novel.de

# **Welcome to** novel pliance-x **system**

This manual provides instructions for the installation and operation of the pliance-x system. Also included are tips/guidelines for the proper use of the system to obtain the most accurate data results.

The pliance-x is an in-shoe dynamic pressure distribution measuring system which utilizes capacitive sensors. The system is designed for a variety of applications including medical, ergonomic and biomechanical testing scenarios.

The names of software and hardware components as well as proprietary names of companies are in general protected by laws of trademarks and brands or patent law.

All information in this manual was carefully compiled and written; however, errors may be present. novelgmbh cannot be held responsible for consequences of incorrect information in the manual.

If you find any errors when using the manual, please notify us at one of the following locations:

novelgmbh
Ismaninger Straße 51
81675 München
Germany

Tel.: 089 / 41 77 67 – 0
Fax: 089 / 41 77 67 99
email: novel@novel.de
http://www.novel.de

novelelectronics inc.
964 Grand Avenue
St. Paul, MN 55105
USA

651-221-0505
651-221-0404
novelinc@novelusa.com
http://www.novelusa.com

# Table of Contents

**1 PRECAUTIONS ..... 4**

**2 SETUP OF THE PLIANCE-X SYSTEM ..... 5**

2.1 INSTALLATION OF THE SYSTEM DRIVERS ..... 5
2.2 LOADING THE NOVEL SOFTWARE ..... 5
*2.2.1 Loading Software that did not previously exist on the Computer ..... 6*
*2.2.2 Upgrading pre-existing Software ..... 6*
2.3 CALIBRATION AND CONFIGURATION FILES ..... 7
2.4 CONNECTING THE PLIANCE-XSYSTEM ..... 8
*2.4.1 The pliance-x system components ..... 8*
*2.4.2 The pliance-x Box Components and Connections ..... 9*
*2.4.3 The pliance-x Data Collection Modes ..... 10*
*2.4.4 Connect the pliance-x system components ..... 11*
*2.4.5 Fiber optic cable (FOC) Setup (valid for all pliance-x systems) ..... 12*
*2.4.6 Bluetooth Setup (valid for pliance-x systems with Bluetooth) ..... 13*
*2.4.7 Preparing the Subject ..... 23*
2.5 BATTERY CARE ..... 24
*2.5.1 General info ..... 24*
*2.5.2 Charging the battery ..... 24*

**3 DATA COLLECTION WITH NOVEL DATABASE AND PLIANCE ..... 26**

3.1 DATABASE SOFTWARE ..... 26
3.2 PLIANCE ONLINE DATA COLLECTION SOFTWARE ..... 26
3.3 EXPORTING DATA FROM THE NOVEL DATABASE MEDICAL ..... 28

**4 THE PLIANCE-X ONLINE PROGRAM ..... 30**

4.1 DATA ACQUISITION ..... 30
*4.1.1 Collecting data ..... 30*
*4.1.2 Viewing data ..... 32*
*4.1.3 Window descriptions ..... 33*
4.2 MENU AND TOOLBAR OPTIONS ..... 38
*4.2.1 File Menu ..... 38*
*4.2.2 View Menu ..... 39*
*4.2.3 Data acquisition Menu ..... 40*
*4.2.4 Options Menu ..... 46*
*4.2.5 Expert Settings ..... 56*
*4.2.6 Help menu ..... 57*
4.3 THE MAT CONFIGURATION FILE ..... 58
*4.3.1 Configuration of Measurement Tab ..... 59*
*4.3.2 Sync Setup ..... 59*
*4.3.3 Sensor type ..... 60*
*4.3.4 Description of Configuration ..... 60*
4.4 SETTINGS AND CALIBRATION SOFTWARE AND CALIBRATION HARDWARE ..... 61
*4.4.1 The Calibration Device ..... 61*

*4.4.2 Settings + calibration online ... 63*

**5 PLIANCE-X SYNC BOXES AND SOFTWARE ... 77**

5.1 pliance-X Sync Box Components and Connections ... 78
*5.1.1 Wired Sync Box Components ... 78*
*5.1.2 Wireless Sync Unit Components ... 79*
*5.1.3 Connections of the Sync Boxes and pliance Box ... 80*
*5.1.4 Schematic diagram of the electrical ports of the Sync Box ... 81*
*5.1.5 Connecting the wired sync box to the pliance device ... 82*
*5.1.6 Connecting the wireless sync box to the pliance device ... 83*
5.2 The Sync Setup Tab in the pliance online Software ... 85
*5.2.1 Master and Slave Frequency ... 85*
*5.2.2 Setting the Trigger Output ... 85*
*5.2.3 Setting the Trigger input ... 86*

**6 TROUBLESHOOTING BLUETOOTH CONNECTION ... 88**

6.1 General Bluetooth Information ... 88
6.2 Possible Solutions for Dropped Frames ... 89
6.3 If Bluetooth COM Ports are Not Created Automatically ... 91

**7 TESTING SENSOR MAT AND CABLE FOR ERRORS ... 93**

7.1 Testing the mat for errors ... 94
7.2 Testing the mat cable for errors ... 94

**8 DEFINITION OF VARIABLES ... 96**

**9 DEFINITION OF FILE TYPES ... 98**

9.1 pliance Data Files ... 98
9.2 pliance Calibration and Configuration Files ... 98
9.3 pliance Exported files ... 98
9.4 Other pliance Files ... 100
9.5 Measurement conversions ... 100

**10 ISSUES WITH THE FIBER OPTIC CABLE DRIVER INSTALLATION ... 101**

# 1 PRECAUTIONS

- The pliance-x system should not be used in extremely high temperatures or high humidity. The recommended operating conditions are 0-40°C (32-100°F) and 20-80% humidity.

- The flexible mat sensors should be handled with care. Pressure ranges outside the designated loading should be avoided as well as use of sharp objects on the sensing device.

- Folding or wrinkling of the mat sensors is not advised and could damage the sensors.

- Please review the section on battery care. The batteries supplied with the pliance-x system must be charged/discharged in the manner instructed to maintain the longest possible lifetime.

- Any attempt to service the pliance-x system or components by unauthorized personnel will void the system warranty.

#####

# 2 SETUP OF THE pliance-x SYSTEM

## 2.1 Installation of the System Drivers

**IMPORTANT**

Administrator privileges are necessary to install drivers and software.

If novel has loaded the software on your computer, skip this section.

1. Insert the CD containing the novel software and drivers into the CD/DVD drive of your computer.
2. If the Auto-run-menu opens automatically:
   a. Install the following drivers and components:
      i. the hardlock driver,
      ii. the Fiber Optic Cable Drivers,
      iii. Firebird (required for novel database software)
      iv. Video filter (only if novel player with video is being used)
3. If the menu does not open automatically:
   a. Select the following setup files from the CD using Windows Explorer:
      i. HASPUserSetup.exe
      ii. nvl_ftdi_setup.exe
      iii. Firebird-1.5.x.exe
      iv. Avisynth_258.exe (only if novel player with video is being used)

## 2.2 Loading the novel Software

**IMPORTANT**

The pliance-x system comes with a Hardlock dongle that connects to the USB port of the computer. The Hardlock must be connected to use any of the novel software programs.

If novel has loaded the software on your computer, skip this section.

### 2.2.1 Loading Software that did not previously exist on the Computer

1. Insert the CD containing the novel software and drivers into the CD/DVD drive of your computer.
2. If the Auto-run-menu opens automatically:
   a. select the pliance-x software option and follow the setup wizard. The installation wizard will recommend to install the novel software on the computer's default system drive (typically "C:") in the subfolder "\novel\". It is highly recommended to follow this recommendation because it helps the novel staff to streamline your support. . The following sections refer to the C:\novel\ path.

      If the software is installed in a different location please replace the above mentioned location by the path which you have chosen.

3. If the Auto-run-menu does not open automatically:
   a. Select setup.exe (will be located in a folder called pliance-x with the version number) file from the CD using Windows Explorer.
   b. Double click on setup file to run the setup wizard.
4. The factory default calibration and configuration files will install automatically into the appropriate folders.

After installation of the software, a program group called pliance-x will be located in the **Start** menu under **Programs** and on the desktop in the novel shortcut folder.

### 2.2.2 Upgrading pre-existing Software

**IMPORTANT**

Always make sure the calibration and data files are backed up prior to uninstalling software.

Please note the following procedures are described for a PC with Windows XP (SP3). On Windows Vista or Windows 7 the locations for starting specific tasks or the displays on the screen may look slightly different.

1. First uninstall the old software.
   a. Go to **Start menu | Control Panel | Add/Remove Programs.**
   b. Select the pliance-x program list.
   c. Click Uninstall.
2. Install the new software following the directions in **Section 2.2.1**.
3. All files should still be located in the same directories as previously.
   a. Please note that after version 14 of pliance-x software the calibration and configuration file locations have changed:

i. The old directory for the configuration files was
C:\novel\NOVSYST\pliancex,
while the new directory is
C:\novel\NOVSYST\pliancex\mfg
ii. The old directory for the calibration files was
C:\novel\NOVSYST\pliancex\cab
while the new directory is
C:\novel\NOVSYST\pliancex\cab\pliance

b. It may be necessary to manually move these files to the appropriate directory.

4. Connect the novel pliance hardlock to the PC and test all programs to make sure they open and run and ensure that all previously collected data and configuration/calibration files are accessible.

If any problems occur, please contact your novel representative.

## 2.3 Calibration and Configuration Files

The factory default calibration and configuration files will automatically be copied during the installation of the software on the computer. It is also possible that a separate CD containing calibration and configuration files will be provided. If so, these files should be copied into the following directories on the computer:

1. Configuration Files – extension .mfg. All files should be copied to C:\novel\NOVSYST\pliancex\mfg.
2. Calibration Files – there are seven associated files with each calibration with the following extensions: *.cab, *.set, *.tbl, *.def, .*.geo, *.chl and *.zpt. All files should be copied to the directory C:\novel\NOVSYST\pliancex\cab\pliance.

For further information on calibration and configuration files please see **Sections 4.3 and 4.4**.

## 2.4 Connecting the pliance-x system

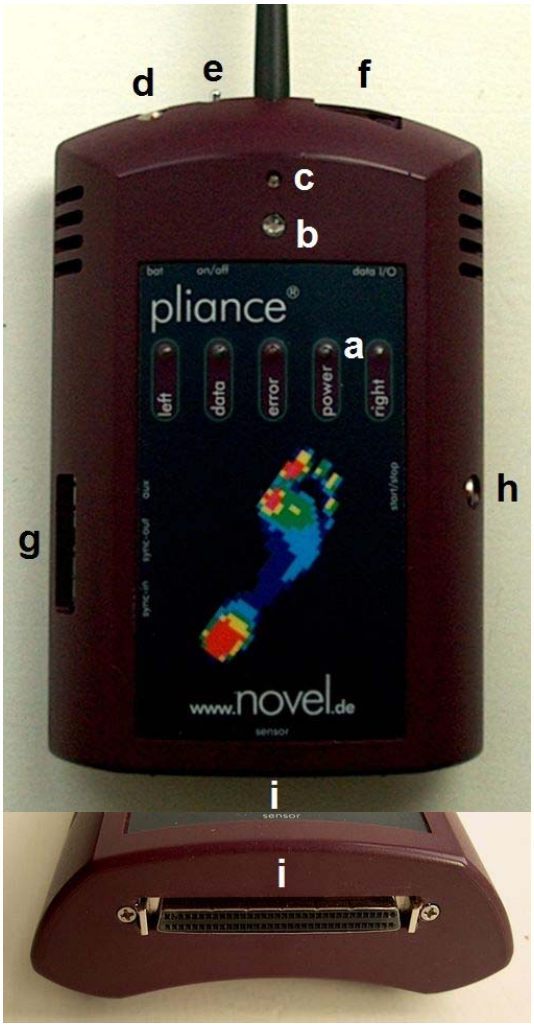
### 2.4.1 The pliance-x system components




The pliance-x system components include:

- pliance-x sensor mat (A)
- pliance-x sensor cable (B)
- pliance-x box ( C)
- pliance-x fiber optic cable with fiber optic/USB adapter (D)
- USB cable (E)
- novel belt (F)
- pliance-x battery (G)
- pliance-x battery cable (H)
- pliance-x battery charger (I)
- start/stop trigger (J)
- Bluetooth dongle (K)

## 2.4.2 The pliance-x Box Components and Connections







- Front Panel:
  five LEDs (a):

| | |
|---|---|
| left | is used during the zero measurement of the left mat |
| data | indicates transmission of data between computer and pliance-x box |
| error | indicates an error during the measurement |
| power | indicates if the system is powered on |
| right | is used during the zero measurement of the right mat |
| top green “flash LED” (b) | During booting the pliance-x system: This LED flashes when the pliance-x box is turned on and when the internal boot process of the pliance x box is finished<br><br>If the pliance x is in the measuring mode: It flashes when a measurement is started. |

| | |
|---|---|
| top blue BT LED (c) | if it is on, it indicates an active Bluetooth connection between the pliance-x and the PC |

- Top Panel:

| | |
|---|---|
| 4pin lemo connector (d) | connects the pliance-x box with the battery |
| on/off-switch (e) | switch to turn the pliance-x box on or off |
| 2 pin fiber optic cable Connector (f) | connects the pliance-x box via fiber optic cable with the PC |

- Left side Panel (g):

| | |
|---|---|
| 3 pin fiber optic connector | connects the pliance-x box with optional synchronization unit |

- Right side Panel (h):

| | |
|---|---|
| 3pin lemo connector | connects the pliance-x box with the start/ stop trigger |

- Bottom Panel (i):

| | |
|---|---|
| mat cable connector | connects the pliance-x box to the sensor cable |

- Back Panel (j)

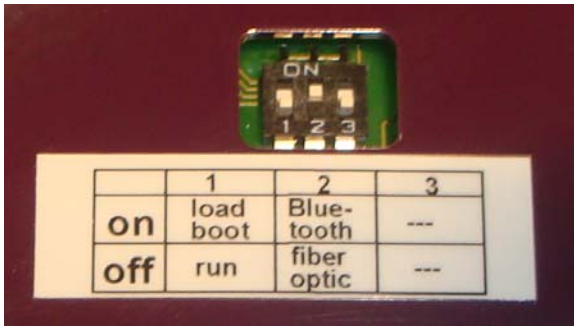
| | |
|---|---|
| dip switches | Open the panel to locate the pliance-x serial number, and DIP switches used to select between the Bluetooth telemetry mode and fiber optic cable data transmission mode. |

### 2.4.3 The pliance-x Data Collection Modes

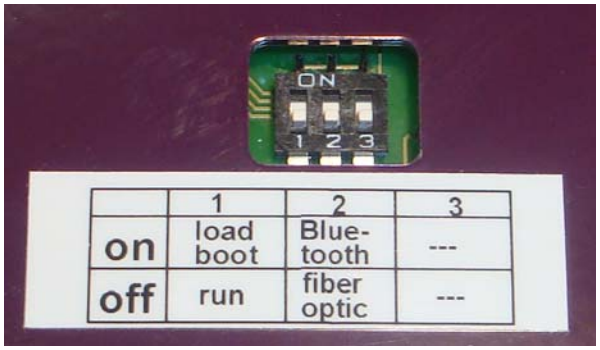
There are three modes of data collection with the pliance-x system:

1) **Online –** in this mode data is transmitted real-time and displayed to the computer. The computer controls the data collection of the device. There are two possible methods to connect the pliance-x box to the computer:

   **a) Bluetooth telemetry** - This setup utilizes Bluetooth to transfer data via wireless communication to the computer. When using this mode the three dip switches on the back side of the pliance-x box have to be set to 1=“OFF”, 2=”ON”, 3=“OFF”.



b) **Fiber Optic Cable** – With this method a fiber optic cable tethers the pliance –x box to the computer. When using this mode all the three dip switches on the back side of the pliance-x box must be set to the "OFF" position.




**2) Online + Flash** – in this mode data is transmitted real-time and displayed to the computer and is also stored simultaneously to the flash memory of the device. The computer still controls the pliance-x system within this mode. This mode of data collection can be helpful when the subject reaches the limits of the Bluetooth data transmission and some data might be not received from the PC. Please see **Section 6** for additional information on using this mode with an unreliable BT transmission.

3) **Offline measurement (Monitor Mode) pliance-** in this mode data transmission is controlled solely by the start/stop trigger switch connected to the pliance-x box and the data is stored to the internal flash memory of the pliance-x box.

If the computer receives data from the pliance-x box, the pliance-x online program will display the data using the Monitor mode. The computer does not control the data acquisition.

### 2.4.4 Connect the pliance-x system components

Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

1. Connect the pliance-x box to the novel belt as shown below. Please note that a version of the belt in which the box sits horizontally is also available. Please contact your novel representative to obtain such as belt:



Thread the two lower horizontal straps through the plastic loops of the device, each from inside to outside.

Close the two upper horizontal straps. The device should now be secure and all LEDs should be visible.

Close the two lower horizontal straps, then move the device to the upmost position and secure the two lower vertical straps.

2. Connect the sensor cable (B) cable to the pliance box (C) via the mat connector (i). Ensure that the clips on the cable are fully attached to the box. Make sure to go straight in and straight out with this connection and do not torque the connector when attaching it or removing it from the box. It can be helpful to coil the pliance cable and attach it to the Velcro strap provided to the left of the box.
3. Insert the battery (G) into one of the sleeves on the pliance belt. Connect the battery via the battery cable (H) to the pliance box (C) via connector (d). Ensure that the red dots align on the cable and connector before pushing the cable in. Always ensure to pull on the release neck to release the cable without damage.
4. Connect the sensor mat (A) to the sensor cable (B). Connect the Right or Left mat to the Right or Left cable connector as shown on the connector. The sensor mat can only be connected one way to properly transmit data - i.e. Match the "R" (Right) on the mat plug to the "R" on the cable. Please see **Section 2.4.7**.
5. Switch the pliance-x box on.

## 2.4.5 Fiber optic cable (FOC) Setup (valid for all pliance-x systems)

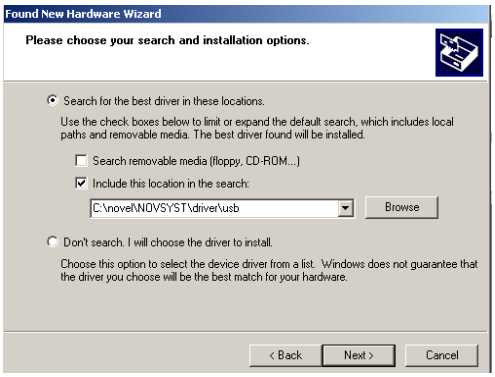
### 2.4.5.1 Installation of drivers (occurs with initial connection to computer)

Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

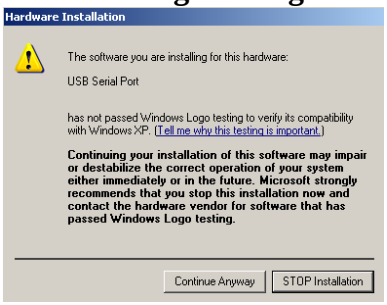
1. Connect the USB cable (E) to the fiber optic/USB adapter (D) and connect the USB cable to the computer.
2. The **Found New Hardware Wizard** will appear
   - NOTE: If the **Found New Hardware Wizard** does not appear and Windows automatically installs drivers, or the connection is not made within the pedar online software, please check that the proper drivers have installed following the instructions in **Section 13**.
3. Choose **Install the software specific location (advanced)**




4. Choose the checkbox **Include file location in the search:** and refer to the directory c:/novel/novsyst/driver/usb




5. Click **Next**
6. Confirm the message for the **Windows Logo testing** with **Continue anyway**.




7. The process will repeat itself for the second required driver.

### 2.4.5.2 Making the FOC connections

1. Connect the fiber optic cable (D) to the pliance-x box (C) via connector (f).
2. Connect the other side of the fiber optic cable to the fiber optic/USB adapter (D).
3. Ensure that the dip switches are set appropriately (see **Section 2.4.3 – 1a**)
4. Switch on the pliance-x device

The pliance-x system is now ready for online measurements and the pliance-x box should be switched on.

## 2.4.6 Bluetooth Setup (valid for pliance-x systems with Bluetooth)

#### 2.4.6.1 Installation of Bluetooth-USB-Adapter under Windows XP Service Pack 2 or Service Pack 3 or Windows Vista

Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

> **IMPORTANT**
>
> Administrator rights may be necessary to setup the Bluetooth connection.

With Windows XP Service Pack 2 and later versions of Windows Microsoft has included its own Bluetooth stack. novel recommends using these drivers in place of those drivers which come with the Bluetooth dongles. Different Bluetooth dongles have been provided by novel, but each should work with the Windows BT drivers. It may be possible that your computer has an internal Bluetooth connection. This software may interfere with the Windows software. It may be necessary to disable the internal Bluetooth and uninstall the drivers provided with the system if the Windows drivers are not available. For ease of use and connection novel recommends at least Service Pack 2 or 3 be installed on all computers running the pliance system.

To determine if Windows XP Service Pack 2 is installed on your computer either go to the control panel and click on **System**, or right click on the **My Computer** icon on the desktop. The following will show the system information:




1) After connecting the Bluetooth dongle (K) to the computer Windows will display a blue and white “B”, as circled in red in the Windows task bar:

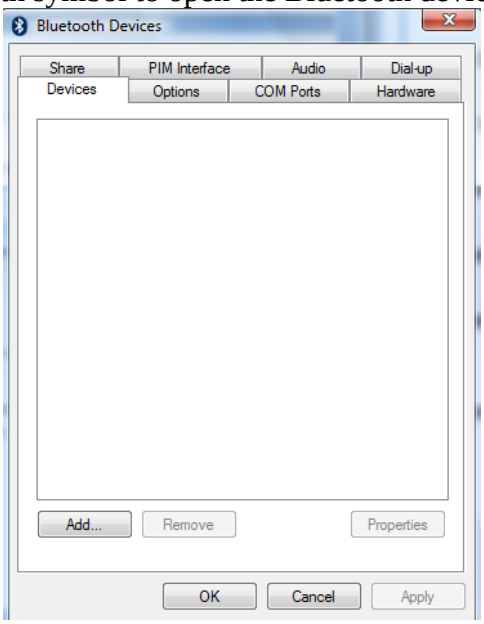



If an additional Bluetooth symbol is present in the task bar, it may be necessary to disable the internal Bluetooth of the computer and/or uninstall the additional Bluetooth software which is on the PC:




Please note that by default Windows will eventually hide the Bluetooth icon. If the icon is not found within the task bar, the Bluetooth device menu can be accessed from the same symbol in the Control Panel. novel recommends creating a shortcut from this item on the desktop for easy access to the Bluetooth devices.

2) Double click on the Bluetooth symbol to open the Bluetooth devices window

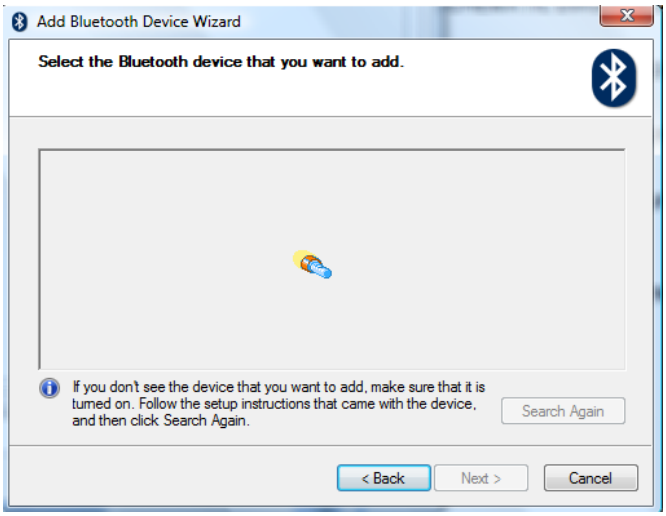



Click **Add** to add the pliance device.

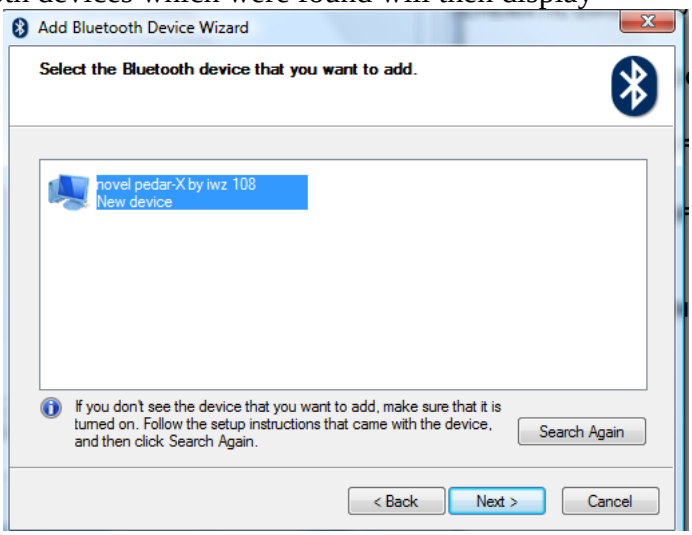
3) The **Add Bluetooth Device Wizard** will begin. Choose **My device is set up and ready to be found** and then click **Next**




4) The computer will begin to search for Bluetooth devices




A list of the Bluetooth devices which were found will then display

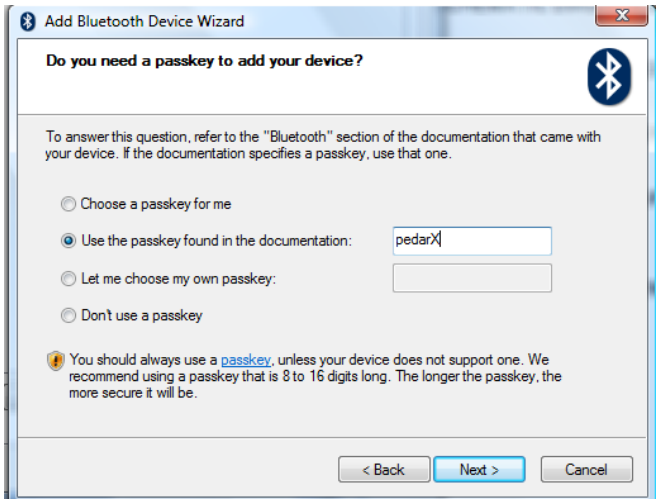



Choose the pliance-x device and click **Next.**

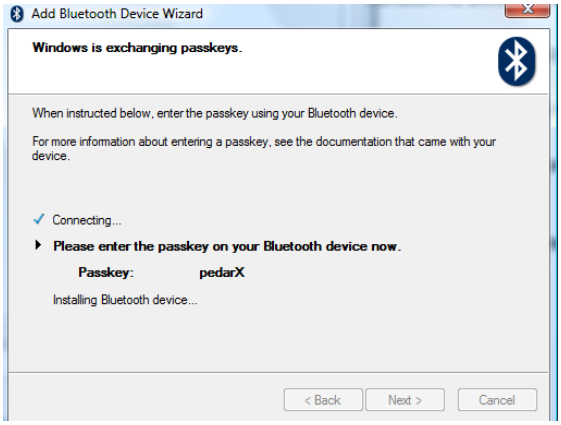
If the pliance –x box is not listed please choose **Search Again** to initiate the search once again. If the box is not found please ensure that the dip switches are set properly (See **Section 2.4.3 – 1b**). Please refer to **Section 6** for additional Bluetooth troubleshooting information.

Please note that if a large number of devices are found within the area, this can interfere with data transmission. If devices such as cell phones, computers, etc are present, and it is determined later that the Bluetooth is not transmitting all frames, it is recommended to turn off the Bluetooth transmission on these devices, if possible.

5) Choose the radial button **Use the passkey found in the documentation** and enter the phrase **pedarX** as the passkey. Click **Next**.



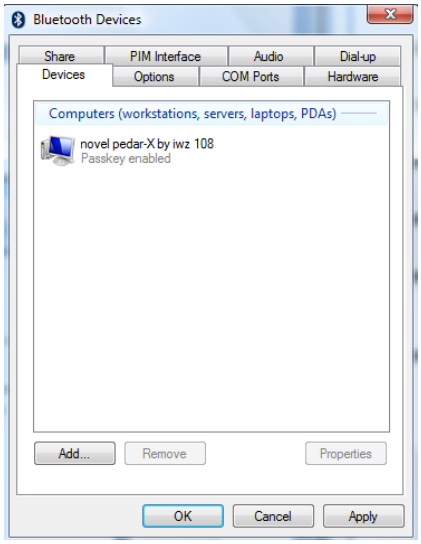

6) The Bluetooth connection will be established and the virtual COM ports will be created. Make note of the **Outgoing COM port**, as this will be needed within the pliance-x online software. Click **Finish**.



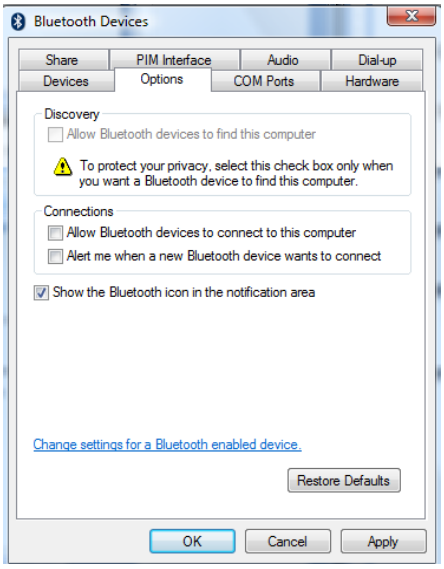




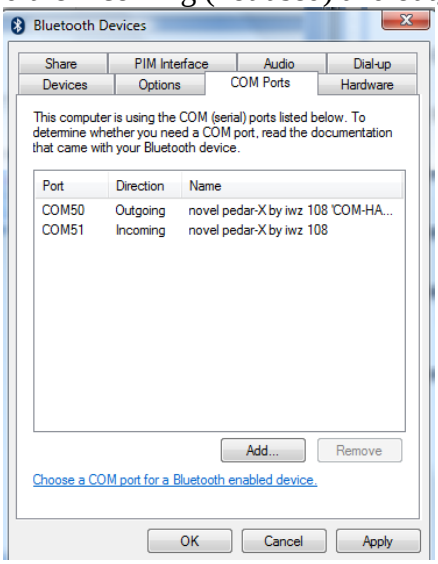
7) The pliance device will now be listed within the Bluetooth Devices window:



To ensure proper data transmission between the pliance and the computer the ability of other Bluetooth devices should be turned off. Click on the **Options** tab and uncheck the options **Allow Bluetooth devices to connect to the computer** and **Alert me when a new Bluetooth device wants to connect**.




8) If later it becomes necessary to determine which Bluetooth port is the Outgoing port you can choose the **COM Ports** tab and the incoming (not used) and outgoing ports will be listed:




9) Please refer to **Section 6** for additional information on the Bluetooth connection including troubleshooting a connection, manually creating COM ports and what to do if Bluetooth transmission is not reliable.

### 2.4.6.2 Installation of Bluetooth-USB-Adapter under Windows 7

Please refer to the previous section for additional information. Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

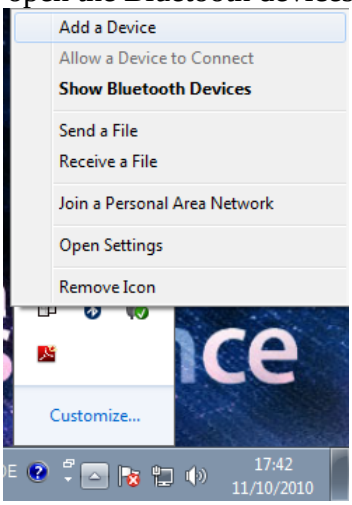
1) After connecting the Bluetooth dongle (K) to the computer Windows will display a blue and white “B”, as circled in red in the Windows task bar:




If an additional Bluetooth symbol is present in the task bar, it may be necessary to disable the internal Bluetooth of the computer and/or uninstall the additional Bluetooth software which is on the PC:

Please note that by default Windows will eventually hide the Bluetooth icon. If the icon is not found within the task bar, the Bluetooth device menu can be accessed from the same symbol in the Control Panel. novel recommends creating a shortcut from this item for easy access to the Bluetooth devices.

2) Click on the Bluetooth symbol to open the Bluetooth devices window




Click **Add a device** to add the pliance device.

3) The **Add a Device Wizard** will begin. The computer will begin to search for Bluetooth devices. A list of the Bluetooth devices which were found will then display

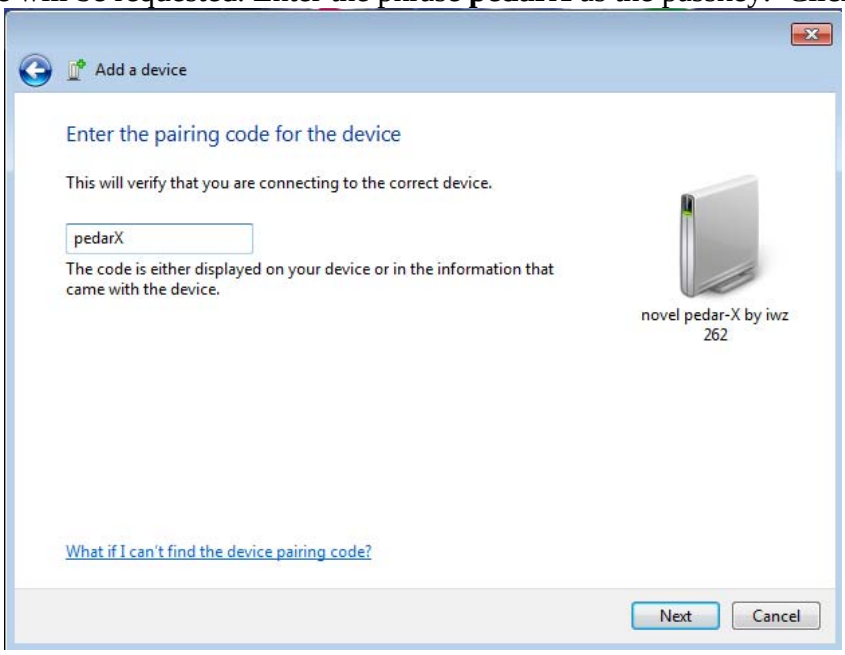



Choose the novel pedar-X by iwz device and click **Next.**

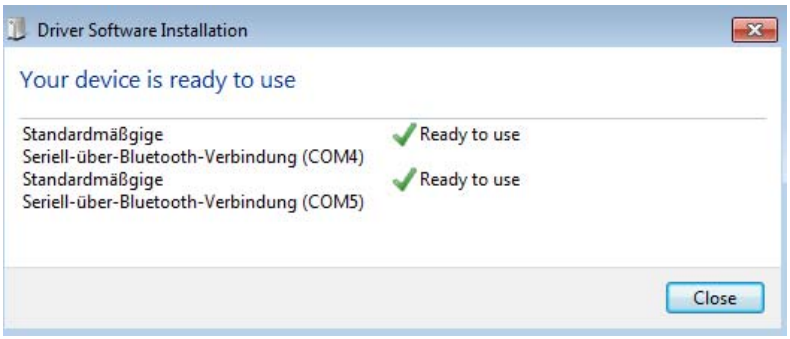
If the box is not listed please choose **Search Again** to initiate the search once again. If the box is not found please ensure that the dip switches are set properly (See **Section 2.4.3 – 1b**). Please refer to **Section 6** for additional Bluetooth troubleshooting information.

Please note that if a large number of devices are found within the area, this can interfere with data transmission. If devices such as cell phones, computers, etc are present, and it is determined later that the Bluetooth is not transmitting all frames, it is recommended to turn off the Bluetooth transmission on these devices, if possible.

4) A pairing code will be requested. Enter the phrase **pedarX** as the passkey. Click **Next**.

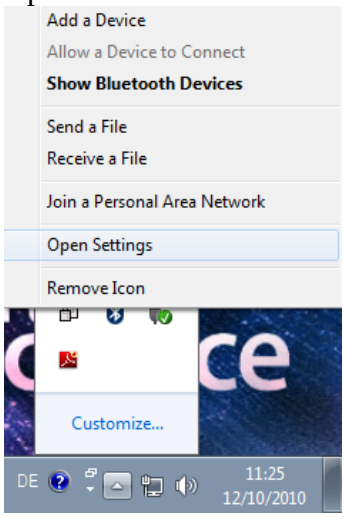


5) The Bluetooth connection will be established and the virtual COM ports will be created. Make note of the **Outgoing COM port**, as this will be needed within the pliance-x online software. Click **Close**.

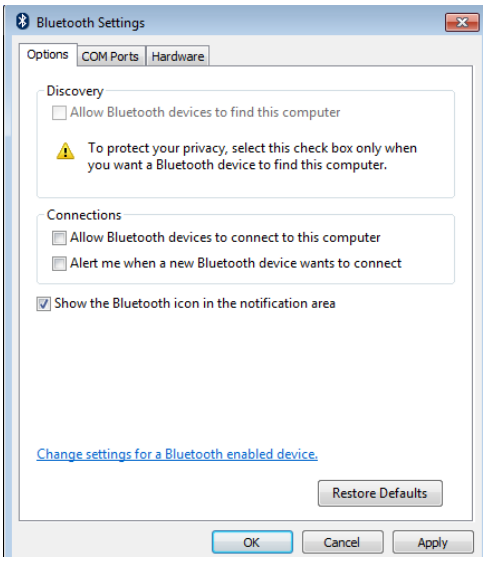



6) The pliance device will now be listed as a Bluetooth Device.

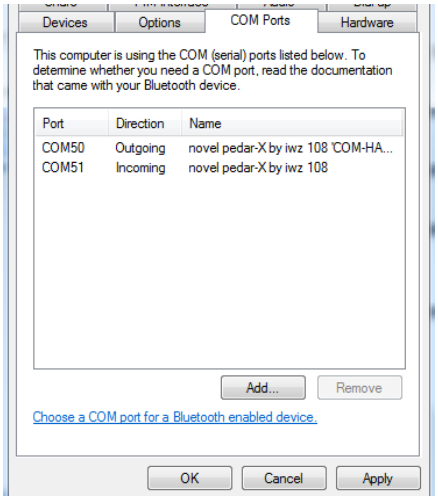
7) To ensure proper data transmission between the pliance and the computer the ability of other Bluetooth devices should be turned off. Choose Bluetooth menu again from the task bar and choose the **Open settings** option.




8) An options menu will appear. Please uncheck the option **Allow Bluetooth devices to connect to the computer** and **Alert me when a new Bluetooth device wants to connect.**




9) If later it becomes necessary to determine which Bluetooth port is the Outgoing port you can choose the **COM Ports** from the **Open Settings** menu item and the incoming (not used) and outgoing ports will be listed:



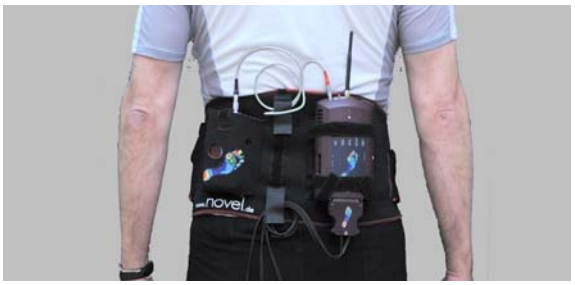

10) Please refer to **Section 6** for additional information on the Bluetooth connection including troubleshooting a connection, manually creating COM ports and what to do if Bluetooth transmission is not reliable.

### 2.4.6.3 Making the Bluetooth Connection

Once the Bluetooth connection has been setup, each time the pliance-x device is turned on the connection will automatically be made between the pliance device and the computer. Please note that that the Bluetooth connection is specific to the USB port on the computer, so ensure to connect the Bluetooth adapter to the same port with each data collection.

### 2.4.7 Preparing the Subject

1. Connect the pliance-x system as described in **Section 2.4.4**.
2. If the testing protocol requires that the subject wear the pliance device, attach the pliance-x belt to the subject's back and ask them to sit down on a chair. If the chair has a back, ensure that they do not lean back. A backless chair is preferred.
3. Detach the mat cable from the Velcro on the belt and place the cable in the appropriate position for attached to the subject.
4. Place the sensor mat on the subject's body. Ensure that the mat fits properly, i.e. that the mat is not too big or too small for the application. A mat which is too big and is confined within a space can be damaged and an mat which is too small may underestimate pressures/forces.
5. Attach the mat to the sensor cable. Ensure that the "L" and the "R" on the sensor mat and connectors match.
6. An instrumented subject is shown below:



7. These instructions are general and not applicable to all pliance applications. Your pliance system may come with an Appendix which describes specific connections for your purchased sensors/cable. Please refer to this Appendix when making connections.

# 2.5 Battery care

## 2.5.1 General info

The batteries provided with the pliance-x system are rechargeable NiMH batteries.




Usage times:

Standard NiMH battery – approximately 4 hours
Lightweight NiMH battery – approximately 10 minutes

These batteries have effect called “lazy battery effect”. Partial charging and discharging them often will cause the capacity to decrease. The battery should be completely charged before use. It is recommended once every 3 months (or 50-60 charge cycles) to completely discharge the battery before charging it using the Ansmann charger provided. If this guideline is not followed, batteries may not charge to their full capacity on subsequent use. Thus, battery use time and the lifetime of the battery may be diminished.

If the batteries will not be used for a long length of time it is important to maintain the charge of the batteries. Deep discharge of the batteries can damage them. Batteries with model numbers 2300 – 2700 should be recharged at least once/month. Batteries with model numbers 2000 should be recharged at least twice/year. Please follow the instructions below.

If this process is used for each charge/discharge cycle the batteries will last for 3-5 years (1,000-1,500 charge cycles). Subsequently, the cells inside of the battery may need replaced.

## 2.5.2 Charging the battery

The following is the process of charging the batteries with the Ansmann charger:
If battery charge is low (as indicated by the LEDs on the battery), perform the following:

1. Connect the battery to the charger and the charge process will begin automatically.
2. Completely charge the battery. The charge LED will turn off and the green “ready” LED

on the Ansmann charger will blink. It may be possible that 5 LEDs are not received on the battery, however the battery will be fully charged. If this process occurs regularly, and it is hard to know the actual status of the battery's charge. Please contact your novel representative for further information.

As described above, once every three months (or after 50-60 charge cycles) the battery should be completely discharged.

1. Completely discharge the battery using the Ansmann charger (even if the battery was completely discharged while running the pliance device). Press and hold down the yellow "discharge" button on the Ansmann charger until the orange discharge LED illuminates.
2. The charger will discharge the battery and then immediately begin to recharge the battery. NOTE: The green "ready" LED may blink during the discharge process. Ignore this.

If the battery does not last long enough to perform measurements after charging your battery to full (as indicated on the Ansmann charger), please contact your novel representative for further technical assistance.

**Note: If the battery is low during usage with the pliance-x the first LED on the battery will blink.**

#####

# 3 DATA COLLECTION WITH novel database AND pliance

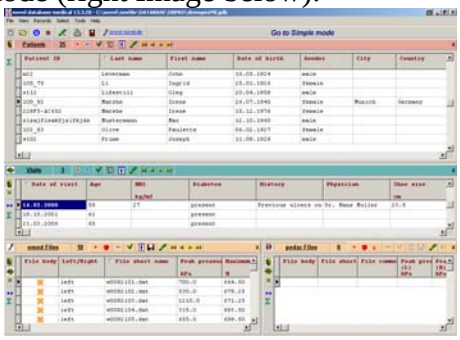
## 3.1 Database Software

The database novel database organizes patient information with pliance data files. It is possible to calculate biophysical parameters and to create user specified reports.

1) You can begin the novel database software by clicking on the 

icon located in the novel folder on the desktop. The software will begin in one of two viewing modes, the simple (left image below) or advanced mode (right Image below).




   a. To begin entering patient information in the simple mode, simply click on the 

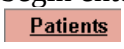
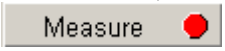
button.
   b. To begin entering patient information in the advanced mode, click the  icon in the 

table.
2) Enter the patient information (a patient ID is required) and then click the 

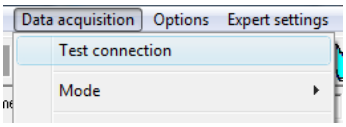
button.
3) Within the visit window the current date will be added. You may also add in any clinical data within this window. For a complete description of the information within this window please see the database medical manual.
4) Once the patient information has been entered, choose the  (pliance) tab.
5) To begin data collection click the 

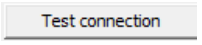
button and the pliance collection software will begin.

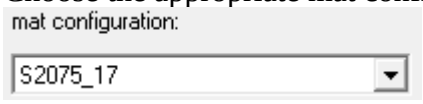
**For a complete explanation of all database options, please refer to the novel database manual.**

## 3.2 pliance online Data Collection Software

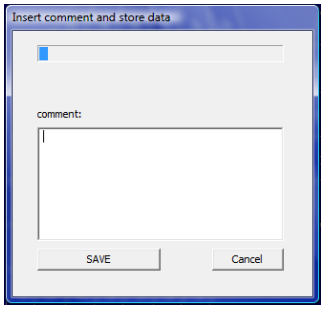
1. The pliance data collection software will open and it is first recommended to test the connection between the pliance box and the software.

2. Go to 

and choose the appropriate port from the drop down menu and click 

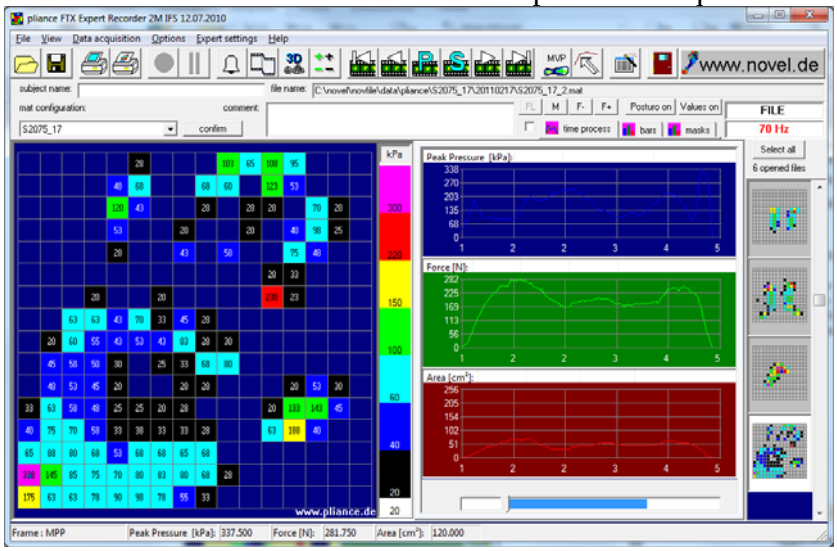
(NOTE: older version of software - enter the appropriate Bluetooth or Fiber Optic Cable port manually. See Section 2.4.6.2 to locate the Port number). You should receive a response in the "flash configuration" Window (a "connection OK" message will appear in older software versions).
3. Instrument the subject according to the defined protocol for your study while sitting in a chair. Please see Section 2.4.7 for details on instrumenting the subject.
4. Choose the appropriate mat configuration from the drop down menu 

and click the 

button.
5. Unload the sensor mat by ensuring that there is no load on the sensor surface. The sensor mat should also be in a flat position, as performed during calibration.
6. Begin recording by clicking on the  button. Ensure to record the appropriate number of steps/repetitions for your activity and click the  button.
7. Enter a comment within the following window and click Save: 

8. Record a second trial by clicking the  button again.
**For more specific collection options, please refer to Section 4.**

Once data collection has stopped all collected trials will be located in the right hand section of the collection screen as thumbnail pictures for quick access/review.

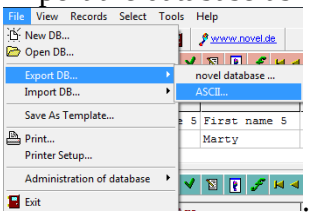


9. To play back an activity, choose the thumbnail on the right hand side of the screen and click the  button.
    a. To move forward or backward one frame, use the  and  buttons, or move to the first or last frame with the  and  buttons
10. You can choose how to view the pressure picture by using the  ,  , or  (isobar) buttons.
    a. You can zoom in and look at the individual pressure values of each sensor by clicking on the pressure picture in the 2D mode. Right click to zoom back out.
11. The values of pressure, force, and area can be viewed in the graphs on the right hand side of the screen
    a. You can zoom in on these graphs by clicking on the  button.
    b. You can track (move) the cursor along the graph by clicking the  button.
    c. Alternatively you can view the values of pressure, force, and area in bar graphs by clicking the  tab.
12. Once finished collecting trials, exit the pliance software by clicking the  or X button.

## 3.3 Exporting data from the novel database medical

The pliance software most often includes the novel database light. The following chapter describes features of the more advanced database, novel database medical. This database is available as a standalone program or is also included in the multiprojects –ip software package.

1. Once data analysis is complete all parameters can be exported from the database to an ASCII format, which can be opened in Microsoft Excel (or the like).
    a. It is important to view what parameters you would like to export within the novel database medical. Select the parameters of interest within each table (Patients, Visits, pliance Files, etc) by using the  button in each table.
    b. You may save this exported list by using the  button and choosing Save list to DB.
2. Export the database using the menu item:




. Choose a location to save the database to and then click Export.
3. Files will be saved for each table: .avs – contains information from the Patients and Visits Table, .adt – form the emed table, .apd – from the pliance table, .apl – from the pliance

table, .aur – from the user table. Open these in Excel (or the like) to view them in table delimited format.

**For a complete explanation of all database options, please refer to the database medical manual.**

######

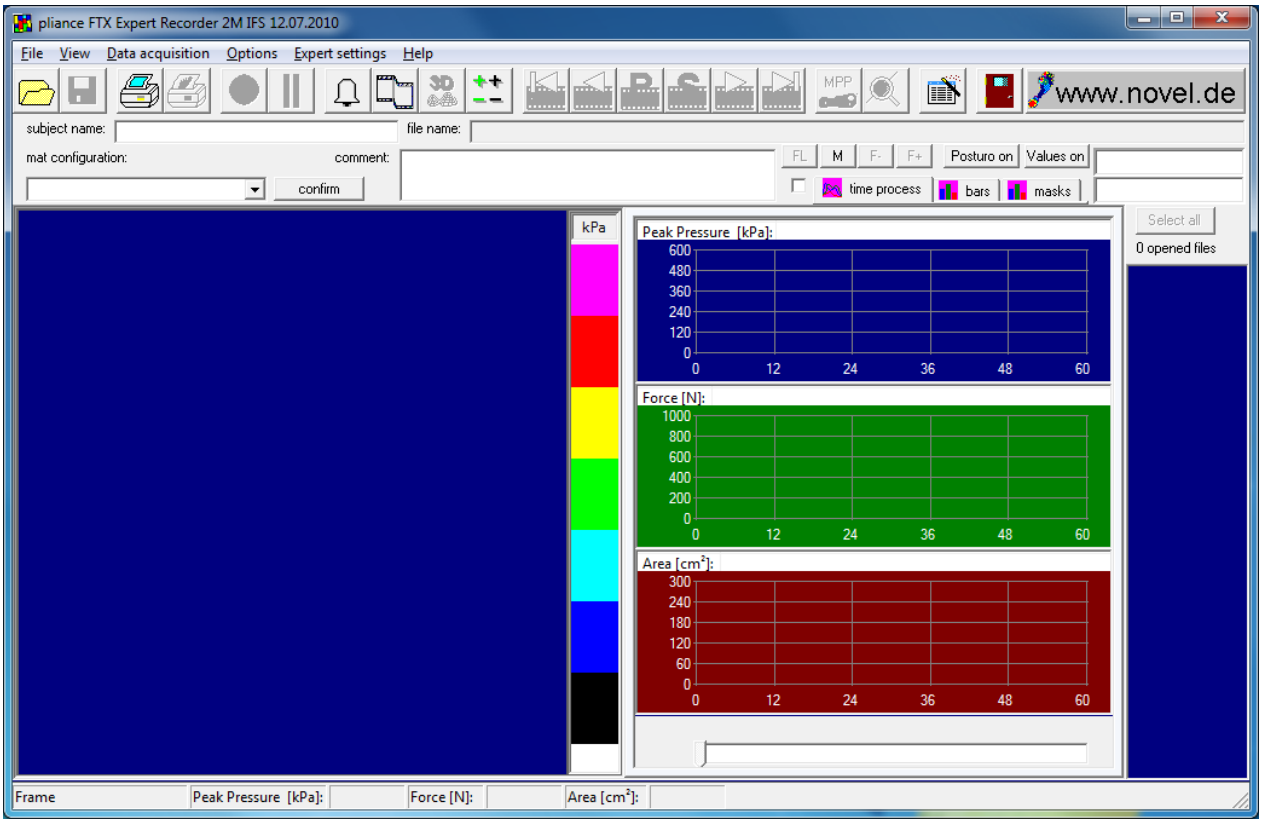
# 4 THE PLIANCE-X ONLINE PROGRAM

The pliance online program collects and displays the data captured with the pliance-x sensor mat. The program also provides the ability to calibrate the mat sensors (purchase of the trublu calibration device is required for this option). Many different collection and display options are available within the program and will be reviewed in the following sections.

## 4.1 Data Acquisition

### 4.1.1 Collecting data

1. Opening the pliance –x online program either from the desktop shortcut or from the novel database (as described in **Section 3a**) and the following window will appear.



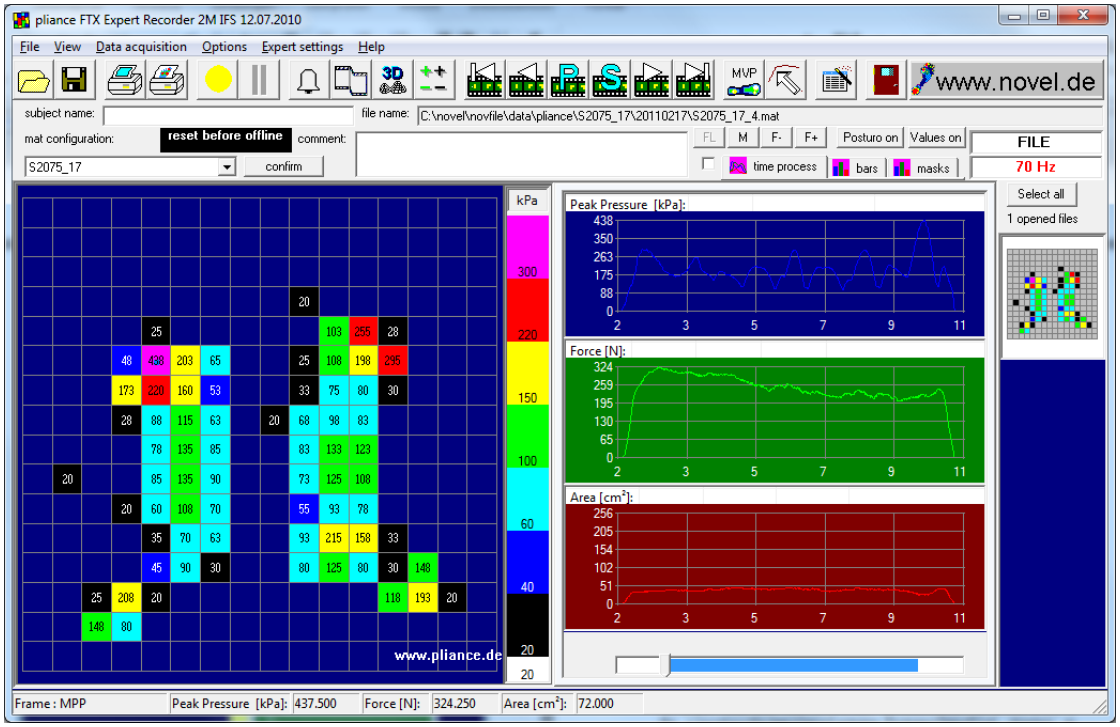

2. Test the connection with the pliance-x system by going to **Data Acquisition | Test Connection**. Please see **Section 4.2.3 –a** for additional information.
3. Select mat configuration file from the **mat configuration** pull down menu.
4. Instrument the subject, as described in **Section 2.4.7**.
5. Click the **confirm** button. This process tells the pliance box which mat configuration you have chosen.
6. A message will appear to unload the mat. Unload the mat, ensuring that the mat is in a relatively flat position, as prompted and then click **OK**.

**NOTE:** The zeroing process is done with no load on the sensor mat. This includes the preload that may occur from placing the mat in an extreme curved position (i.e. around on object). If a pre-load does exist which you would like to ignore, a trial should be captured with this preload and used in analysis later.

The sensor mat should be re-zeroed each time the mat's position is changed. This can be done by going to Data Acquisition | Zero Settings | with zero. To use the last zero measurement that was performed (i.e. the box was turned off), go to the menu Data acquisition | Zero settings | previous zero. Please see **Section 4.2.3-d** for additional zeroing information.

7. To begin recording, click the record icon. 
8. To stop the measurement, click the stop and save icon. 
9. By default the automatic save option is enabled within the pliance online software. The measurement is therefore saved as soon as the collection is stopped. By default the data is saved to the c:\novel\novfile\data\pliance\ directory. If this option is not selected, click the save icon  and follow normal Microsoft Windows saving procedures. Once saved, a MPP picture of the data file will be shown on the right side of the screen as seen in the following picture.

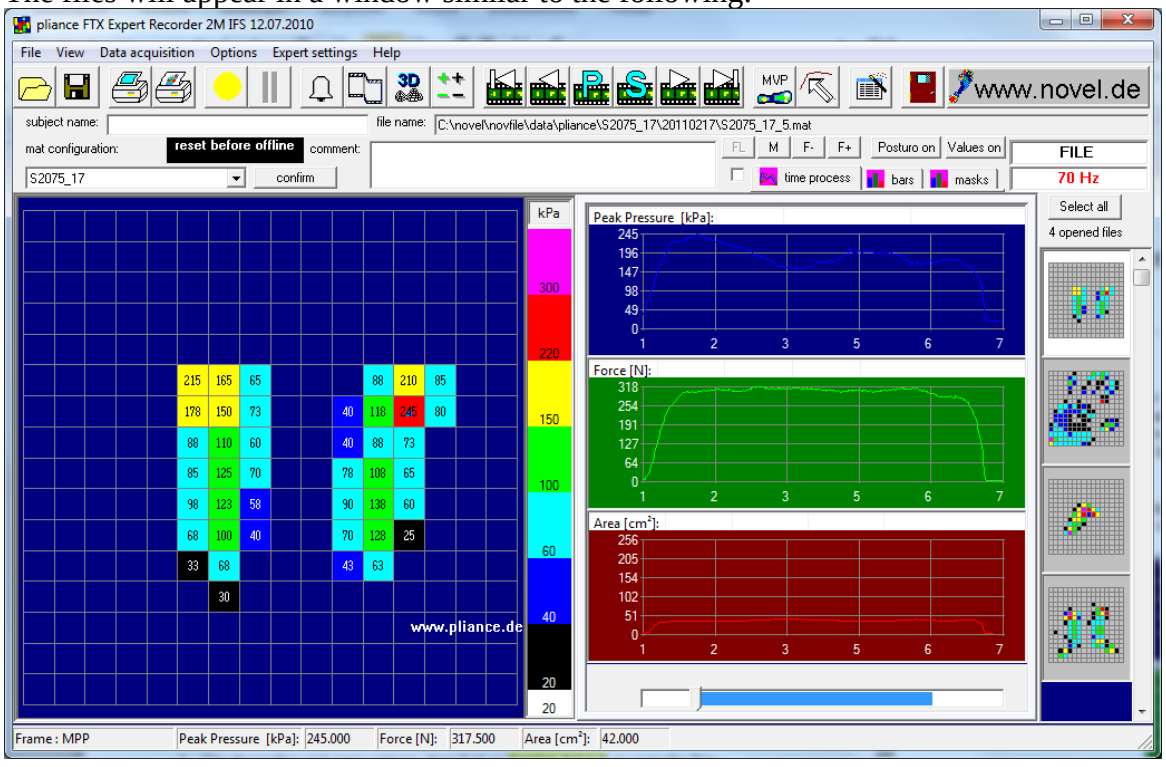
   **NOTE:** If the data collection was begun through one of the novel databases, the subject's name will automatically be assigned to the file name and will display under the **Subject Name** field within the software. The measurement will save by default to the c:\novel\novfile\data\pliance\Lastname_Firstname\DateofVisit\ directory as Lastname_Firstname_#.mat, where # is the incremental measurement number. If the database is not being used a subject name may be manually entered into the **Subject Name** field.



10. Also by default, the option to enter a Comment for the trial will appear. This step is very useful to make note of what occurred within the trial. This can be helpful in post-processing later.
11. To continue collecting data, follow steps 7-10.
12. It is possible to view the data files that were collected during this session simply by clicking on the MPP thumbnail located on the right of the screen. The selected file will appear in the pressure picture window.
13. To close the program, click the exit icon. 

### 4.1.2 Viewing data

1. Open the pliance –x online program and select the open file icon.  A standard Microsoft Windows open file box will appear. Select the file(s) to open and click **open**.
   **Alternatively**
   Select the data you would like to view from the novel database and open the pliance-x program from the database (refer to the novel database manual for specific instructions)
2. The files will appear in a window similar to the following:

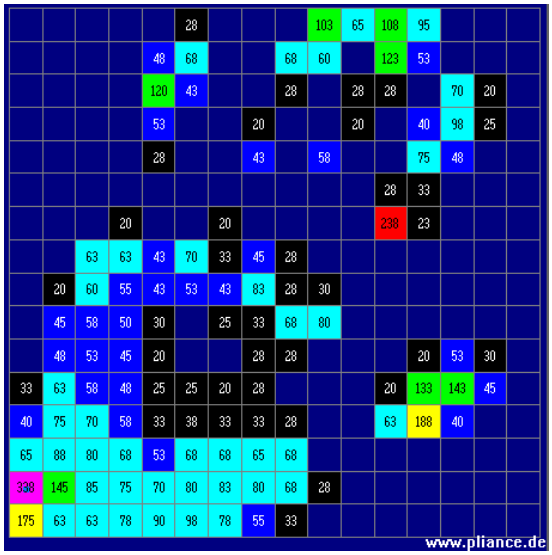



3. Use the toolbar and menu options, described in **Section 3.2.1.9**, to view the files.
4. Close the program. 

### 4.1.3 Window descriptions

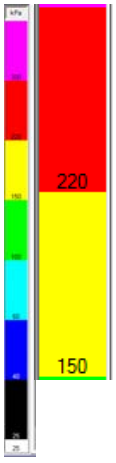
All windows are synchronized to describe the same point in time.

1. **Pressure Picture** – displays the pressure distribution. Located on the left portion of the screen.

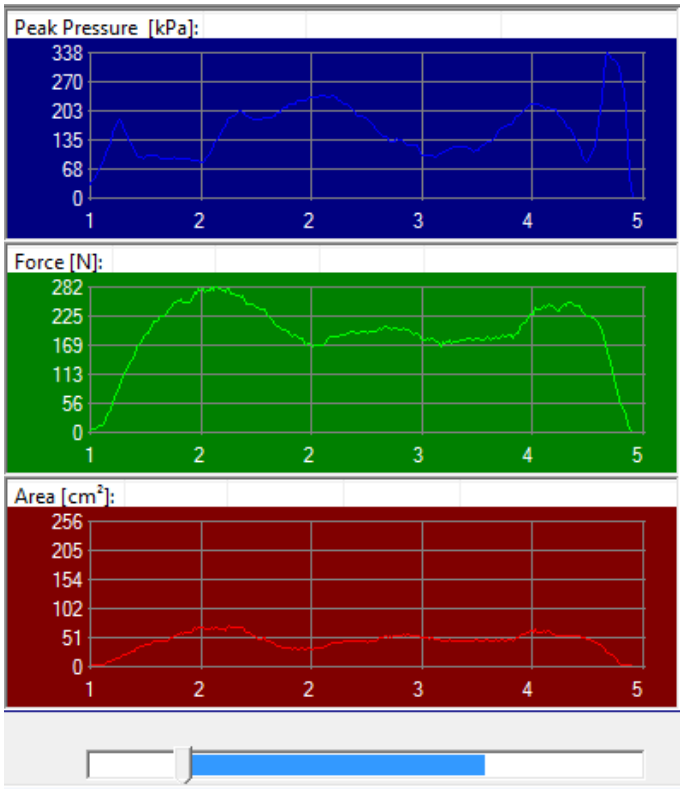



   a. Pressure values/colors for each sensor of the mat are represented here in the 2D mode.
   b. The left foot is displayed on the left of the screen, the right foot on the right.

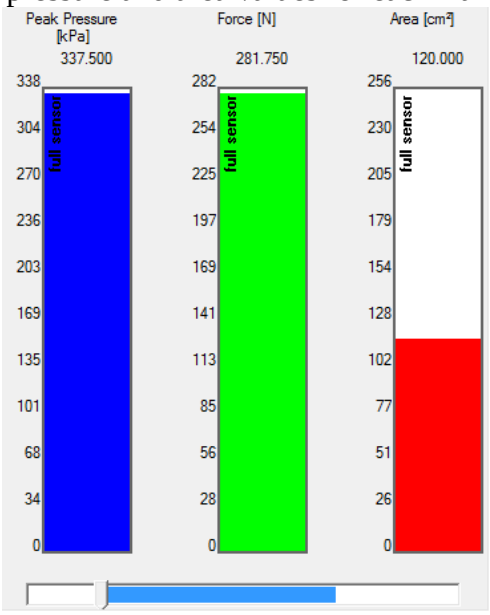
2. **Color Code** – displays the color code relating the pressure values to a color. The units of pressure and the cutoff are also displayed here. The value given for each color is the lowest value for the color level. For example, all pressure from 150 up until 220 kPa will display yellow.



3. **Time process tab** - displays force, pressure and area versus time curves for each foot. The values at the time frame are displayed at the top of each graph.

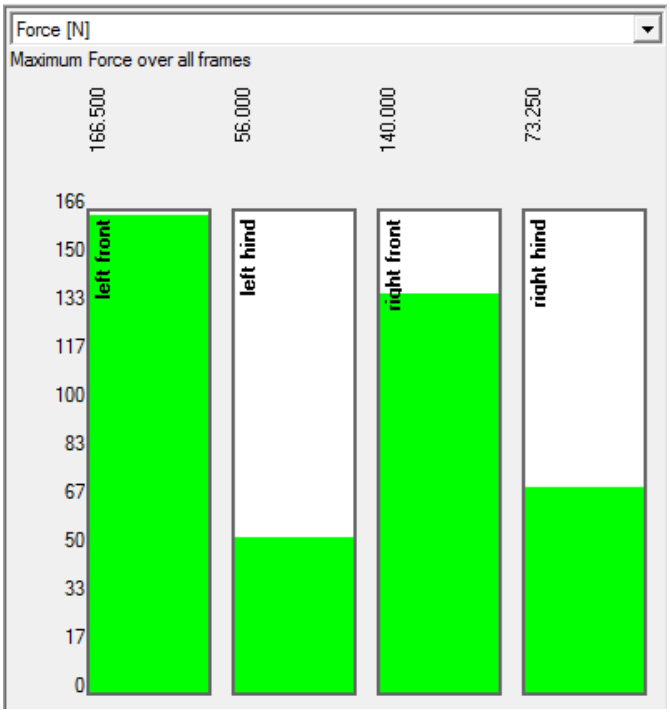


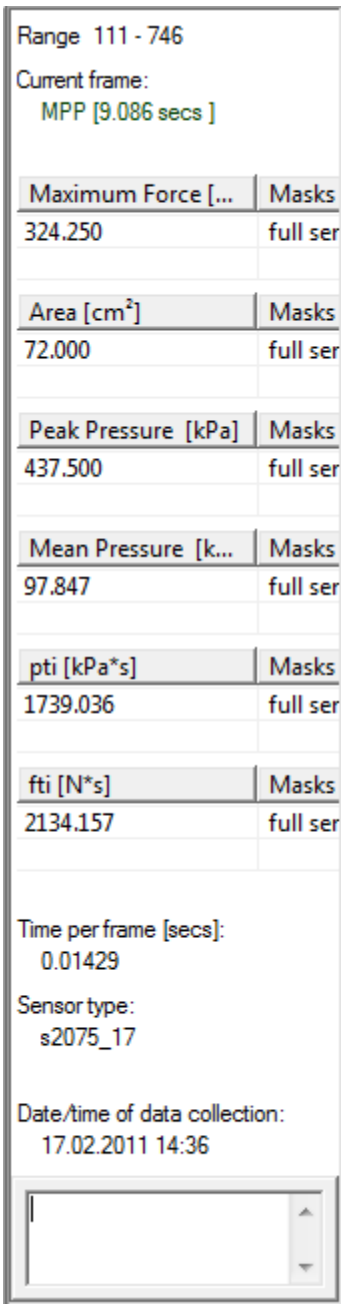

4. **Bars tab** – displays force, pressure and area values for each frame in a bar graph format.




5. **Masks** - When a mask is applied the values for each mask are displayed in this window. Parameters can be calculated and displayed for each mask. For further information on creating these masks, please see **Section 4.2.3-f**. The parameters are Force (N), Peak pressure

(kPa), Mean pressure (kPa), Contact Area ($cm^2$), Pressure-time integral (kPa*s), Force-time integral (N*s) and can be selected from the pull down menu at the top of the window.

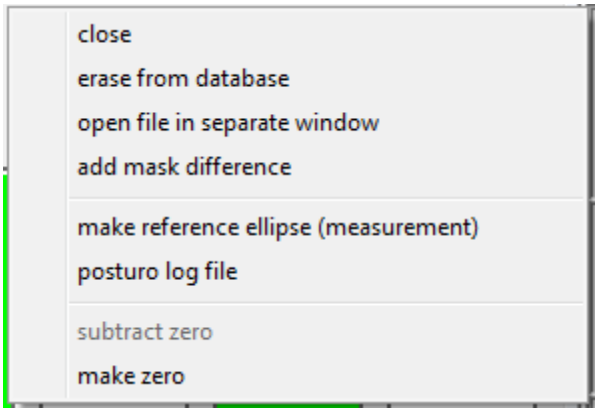





6. **Values On** – displays information relative to each frame of data.
    a. **Range:** Shows the number of data frames which are being viewed in the time process window.
    b. **Current frame**: Displays the current frame number and the corresponding time.
    c. **Force (N):** Displays total force for the left and right mat.
    d. **Area ($cm^2$):** Displays the total contact area for the left and right mat.
    e. **Peak Pressure (kPa):** Shows the maximum pressure that occurred in one sensor element for the displayed frame for the left and right mat.
    f. **Mean Pressure (kPa):** Shows the average pressure for the displayed frame for the left and right mat.
    g. **PTI (kPa*s):** Shows the pressure-time integral value for the selected time period for the left and right mat.
    h. **FTI (N*s):** Shows the force-time integral value for the selected time period for the left and right mat.
    i. **Time per frame (sec):** Shows the time duration of each frame. This value is dependent on the sampling rate.
    j. **Mat type:** Shows the calibration file used to collect the data file.
    k. **Date/time of data collection:** Shows the date (in DD.MM.YYYY format) and time of data collection.
    l. **Comment Box:** Will display any comments which were saved with the measurement.

7. **Selected Trial Menu**
    **a. Close:** Select to close the selected trial
    **b. Erase from database:** Select to exclude the selected trial from the novel database. This may be necessary if a trial should be repeated and should not be included in analysis later. Please note that the trial will still exist on the computer hard drive. Only the copy within the novel database will be removed.
    **c. Open file in separate window:** Opens the selected file in another window of the pliance online software.
    **d. Add mask difference:** Applies a mask to the selected trial and subtracts the difference between the masks.
    **e. Make reference ellipse (measurement)**
    **f. Posturo log file**

g. **Make Zero:** Select a file to make as a zero, or offset for the data. All additional trials can be subtracted from this trial. This can be helpful when an offset is present and would like to be ignored.
h. **Subtract Zero:** Select to subtract from a file which was made zero. Each sensor element's reading in the zero trial will be subtracted from the selected trial.



## 4.2 Menu and Toolbar Options

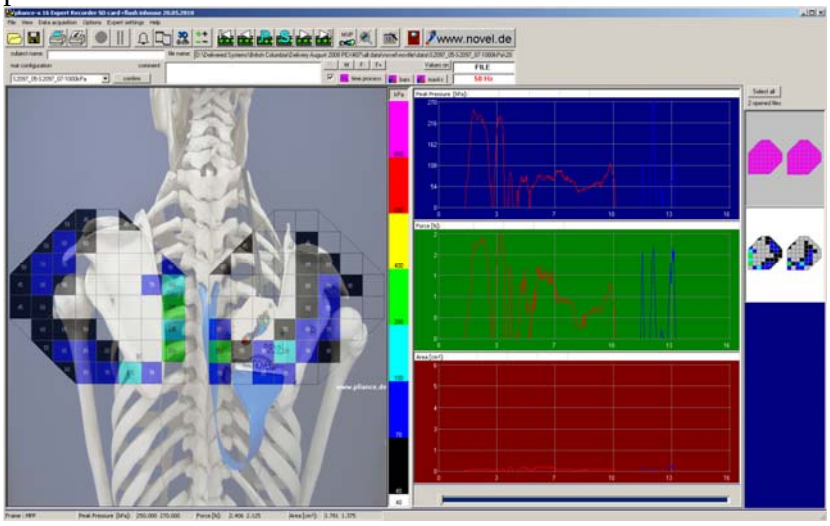
### 4.2.1 File Menu

a.  Open — Use to open data files. This will display a typical Windows open screen. If using the novel database, data files may be opened directly into the pliance online software.

b. Open Desktop Files — Opens desktop (.dsk) files. Desktop files are files which are created when digital video is captured within the pliance online software. This file contains information on the synchronization of the video (.avi) and pliance (.mat) file.

c. Open FAR Files — Opens an archived flash file. The FAR file may contain a set of .mat files which were downloaded from the pliance's flash memory.

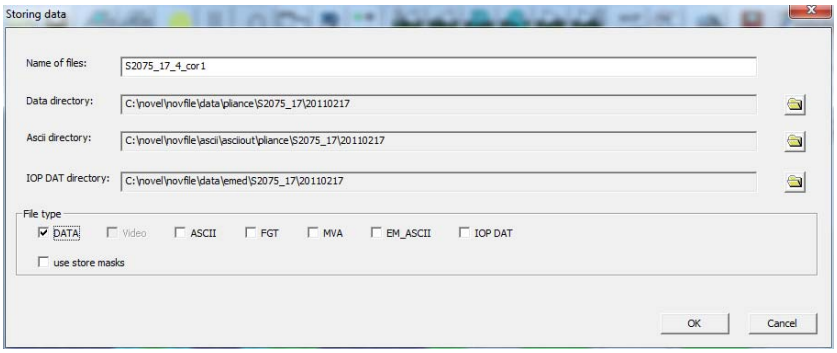
d. Open Image File — Opens an image file (supported file types are bmp, gif, jpg, ico, emf, wfm). The image is used as a background image for the pressure data window in the online software:




The transparency of the picture can be set under the Options menu. See **Section 4.2.4 - n**.

e. Open Audio instruction file — When starting a pressure measurement, an audio file can be played simultaneously. This can be helpful when instructing multiple patients/subjects to perform a specific repetitive task. The supported audio file is wav.

f.  Save — Use to save data files. Can save in the following formats: mat, asc, fgt, mva, em_ascii, dat. More than one file format can be selected and saved at one time. Also, the default folder may be changed. Please note that an original .sol file may not be overwritten. A new name must be assigned for this file.
See **Section 9** for descriptions of each exported file format.




| | | | |
|---|---|---|---|
| g. | | Close All | Use to close all data files. |
| h. |  | Print | Use to print data file in real size. Just the left or right can be selected at the same time and can be printed in black and white or color. |
| i. |  | Print screen | Use to print the entire computer screen. |
| j. | | Capture and send to clipboard | A selected area of the screen can be captured and sent to the clipboard. |
| k. |  | Exit | Use to close the program. |
| l. | | Exit with storing changes | Close the program and also save any changes to the display options. Please see **Section 4.2.4** for additional information. |

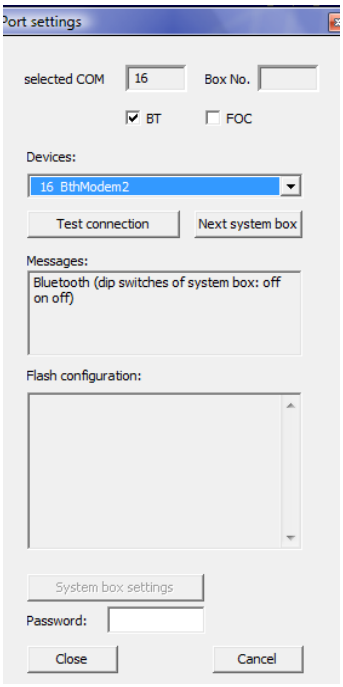
## 4.2.2 View Menu

| | | | |
|---|---|---|---|
| a. |  | First frame | Move to first frame of data file. |
| b. |  | Previous frame | Move one frame back in data file. |
| c. |  | Next frame | Move one frame forward in data file. |
| d. |  | Last frame | Move to last frame in data file. |
| e. |  | MPP/MVP | Toggle between the Maximum Pressure Picture (MPP) and Mean Value Picture (MVP) of the file. |
| f. |  | Play | Play back the data file. |
| g. |  | Stop | Stop playing the data file. |
| h. |  | Zoom | Toggle button to switch between:<br>1. being able to select a frame by clicking on a time curve and<br>2. being able to zoom on the time process graphs |

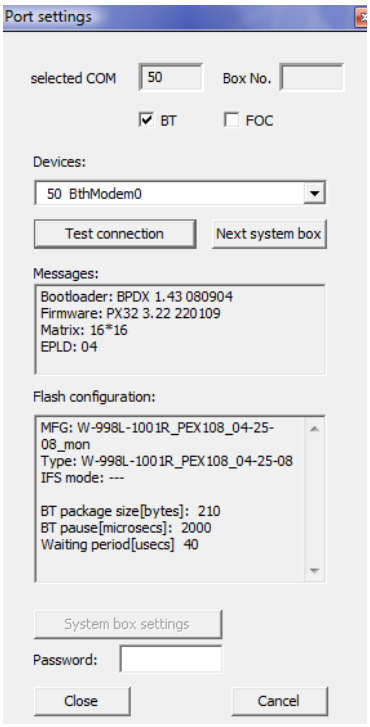
### 4.2.3 Data acquisition Menu

a. Test Connection

Select this option to test the connection between the computer and the pliance-x box. The connection should be tested prior to data collection. This window will automatically appear if the connection is lost, or the box was turned off. A window similar to the following will appear:



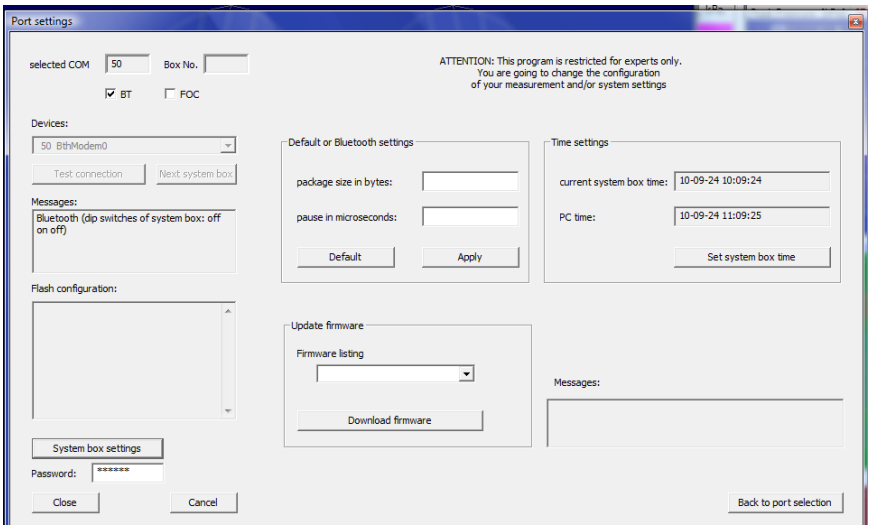

1. Select how the pliance-x is connected to the computer: select BT if using Bluetooth or FOC if using the Fiber Optic Cable. A message will appear at the bottom specifying how the three DIP switches on the pliance-x box should be set. Please refer to **Sections 2.5.4 and 2.5.5** for information on making the initial FOC and BT connection.
2. Choose the appropriate port from the **Devices** menu. With a BT Connection, this port will be the Outgoing COM port listed when the port was created and will contain the statement "BthModem". For a FOC Connection, the statement "VCP".
3. Select the test connection button. A message will appear indicating that the connection was found or if a problem exists.




A response in which the information shown above is listed indicates that the connection has been made. This information may be helpful when discussing or troubleshooting with your novel representative.

A message such as **serial port locked**, or **box may be in menu mode** indicates that the virtual COM port (either BT or FOC) is not set up properly. Please refer to **Sections 2.5.4 and 2.5.5** to troubleshoot this connection. If using the FOC, please also refer to **section 10** to ensure that the proper drivers are installed.

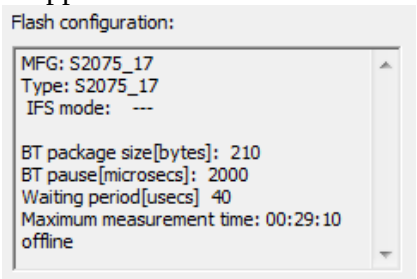
4. Once the connection has been established click **Close.**
5. Additional system box settings may be set with the help of your novel representative. After a connection has been made, enter the password **system** into the dialog box and click **System box settings**. A window similar to the following will appear:




The only option that can be altered without the help of your novel representative are the Time Settings. This allows the user to set the pliance box time to match the system box time. This can be useful when making offline measurements. All other options should not be changed unless instructed to do so.

b. Offline/ Online Measure-ment Settings

This option will display the measurement configuration settings within the flash memory of the pliance device. A window similar to the following will appear




c. Mode

Different modes of operation can be selected depending on the task to be performed. See **Section 2.4.3** for details on these data collection modes:
*Online*
*Online + flash*
*Monitor + Test Connection*
*Monitor*

d. Zero settings

This menu item allows the user to choose different options for zeroing the system. Please refer to **Section 4.1.1 #6** for information on zeroing philosophy:
**With**: select to make a zero measurement before collecting data. This option is useful if a footwear change has been made. The system should be re-zeroed

**For objects**: it is possible to take the zero measurement for each individual sensor matrix (i.e. "object") when using a combination of different sensor matrices. See **Section 4.2.4 – p** for additional information.

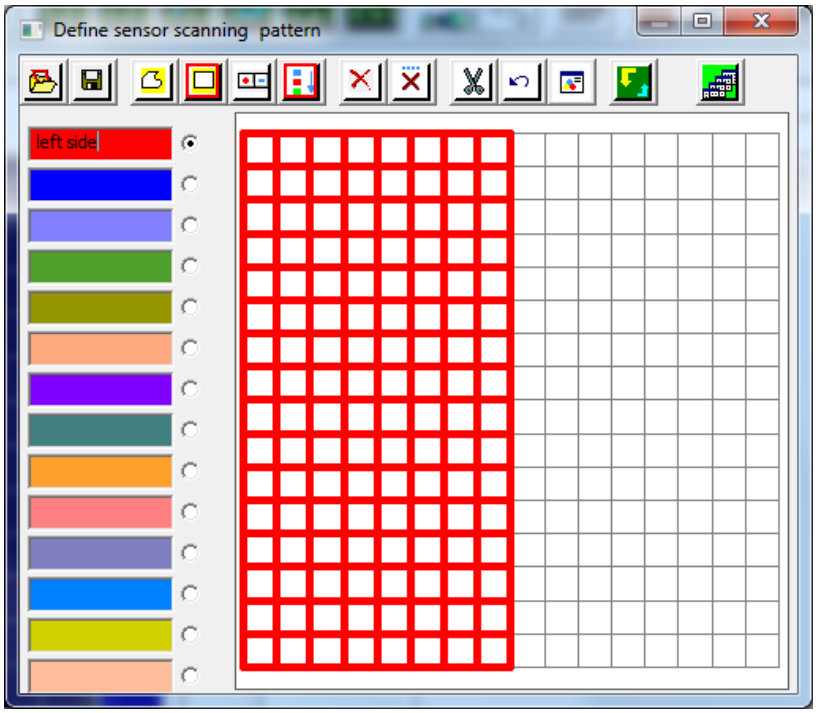
**Without**: select to collect data without using a zero measurement.
**Previous**: select to collect data using the last zero measurement made. This option is typically selected if the box was turned off and back on. The user must re-confirm the connection and the software will automatically prompt for the unload to occur. Click **Cancel** and then select this menu option to use the previous zero.

e. Stop System — Stops communication with the pliance box.

f. Define Sensors to be Scanned — This menu allows the user to both turn off sensors which will be scanned and also define masks for display during data collection.

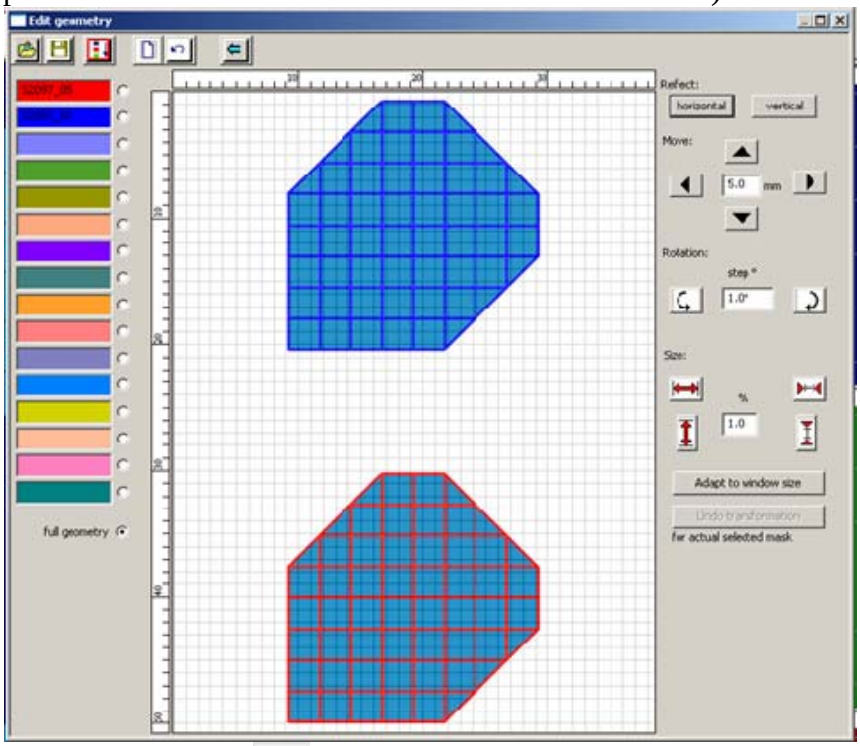
Sensors are commonly turned off to increase the sampling rate of the pliance system. For example, the maximum scanning rate of the full 16x 16 sensor mat is 78Hz. By turning off the right half of the sensor area, the frequency can be raised to double scanning rate, for instance 156 Hz.

Masks may be defined prior to data collection if they are to be displayed during data collection for online feedback, or automatically exported as part of the MVA file. Please see **section 9** for further details on the MVA file.



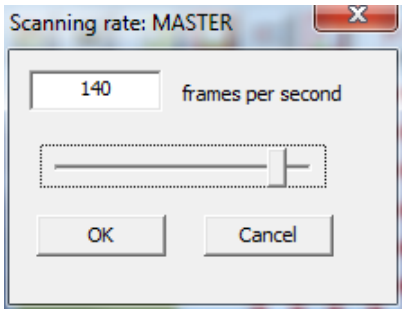

- The **Open Mask** icon opens a previously defined mask/sensor
- The **Save Mask** icon saves a mask/sensor definition
- The **Free Drawing** icon provides the ability to draw a mask in any shape.

- The **Rectangle** icon draws a square box around the mat and selects all sensors within the box.
- The **Sensor on/off** icon turns off/on an individual sensor within the mat
- The **Show all masks** icon shows all masks, or only the selected mask
- The **Delete single mask** icon deletes the currently selected mask
- The **Delete all masks** icon deletes all masks
- The **Remove unmasked sensors** icon deletes all sensors which are not included in any mask. Therefore, deactivated sensors will not appear in the pressure measurement screen screen.
- The **Undo removing sensors** icon restores the removed sensors.
- The **Edit geometry** icon allows the user to change the position, rotation and size of the sensor matrices (sensors which were masked in the previous screen) for display in the pressure measurement screen. NOTE: This will not influence any sensor parameters which are relevant for measurement results.)




- The **Change color** icon allows the user to change the color of a mask

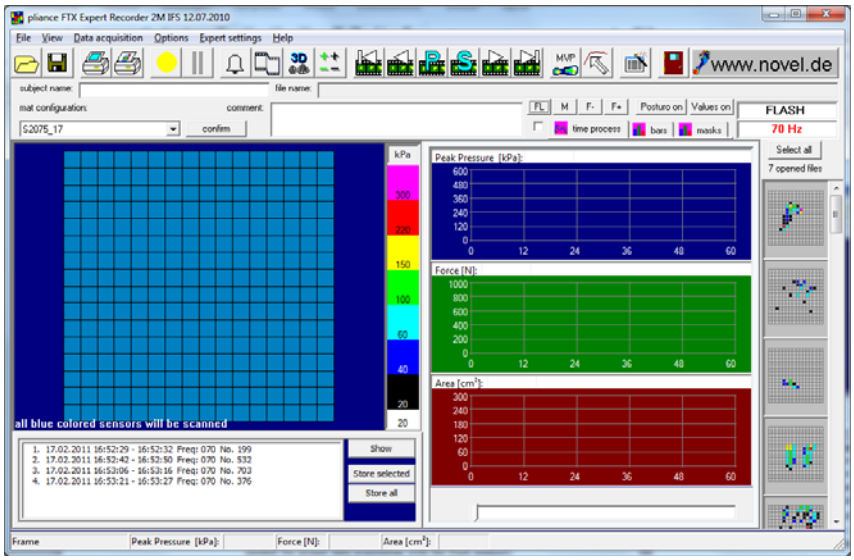
- The **Download all masks**  icon allows the user to create a new configuration file containing the settings that have been defined here. A typical Windows Save dialog will appear. The user may then select the measurement rate for the new configuration file:




**Create a new mask:**

To create a new mask select a radial button alongside an empty color  and insert a name for the mask into the text field. Select the appropriate draw tool and draw the mask

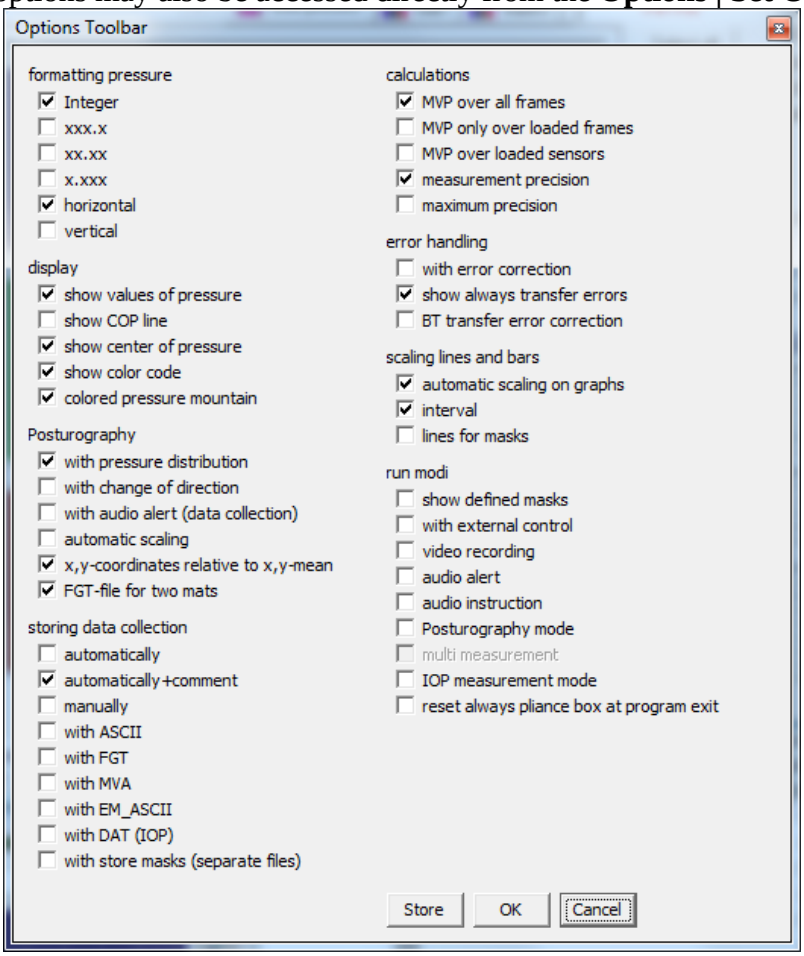
| | | |
|---|---|---|
| g. | Video Time Limit | This menu allows the user to set a maximum recording time for Video. |
| h. | Video Preview | This menu initiates the video camera and displays a preview window for the video |
| i. | Configure Flash | **Load current mat configuration to flash:** select this option to download configuration and calibration files to the flash card. This is needed for proper data acquisition with the flash memory.<br>**Erase measurements in flash:** select this option to delete all data from the flash card memory. |
| j. | Read Flash | **Read from device:** This menu item will display a Window containing a list of measurements contained on the device. The user may select to **store all** measurements, only **store selected** measurements or **show** the measurements within the pliance software. |



**Append new series:** This menu item is used after measurements have been read within the software. This will access the memory of the device again for download of additional trials.
**Save to FAR File:** This menu item saves the entire contents of the flash memory to a file with the extension .FAR. This file can be opened within the pliance software later and individual trials may be downloaded at that time via the same method as described under **Read from device**.

### 4.2.4 Options Menu

Many of options available within the **Options Menu** may be access by clicking the **Options** button  within the main window toolbar. The options will be described as listed within this dialog box. The options may also be accessed directly from the **Options | Set Options** Menu.




a. Formatting Pressure — Select how many decimal places will display when the pressure values are displayed on the 2D pressure picture. The user may also select whether the values display horizontal or vertical.

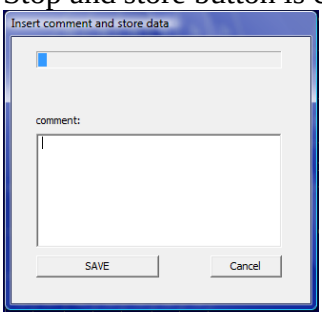
b. Display
- **Show values of pressure:** if selected the values of pressure will display on the 2D pressure picture
- **Show COP line:** if selected the COP line for the entire trial will

display on the 2D pressure picture

- **Show the center of pressure:** displays a white and black circle of the COP during dynamic playback
- **Show color code:** display the color code block, or turn this option off
- **Colored pressure mountain:** if selected the 3D pressure picture will display in color

c. Storing data collection

- **Automatically:** the data will be stored automatically when the Stop and store is clicked. The data will be stored to the path defined within the Configuration program – typically C:\novel\novfile\data\pliance. If the pliance online program was started through the database a copy of the data will also be returned to the database.
- **Automatically + Comment:** a window will appear when the Stop and store button is clicked:




A comment may be entered regarding what occurred within the trial (i.e. the condition). This comment can be useful later when reviewing the data, or when searching for data within the novel database. As with the setting above, the data will be stored automatically (see above for more details).

- **Manually:** data will remain in the computer's RAM memory until stored. The user must manually go to File |Save | SOL file to save the trial.
- **With ASCII, FGT, MVA, EM_ASCII, Export:** Select these options to automatically export the trial into the exported file formats. These will be saved to the path defined within the Configuration program – typically C:\novel\novfile\ascii\asciiout \pliance. Please see **section 9** for definitions of each file export format.

d. Calculations

The Mean Value Picture (MVP) can be calculated three ways.

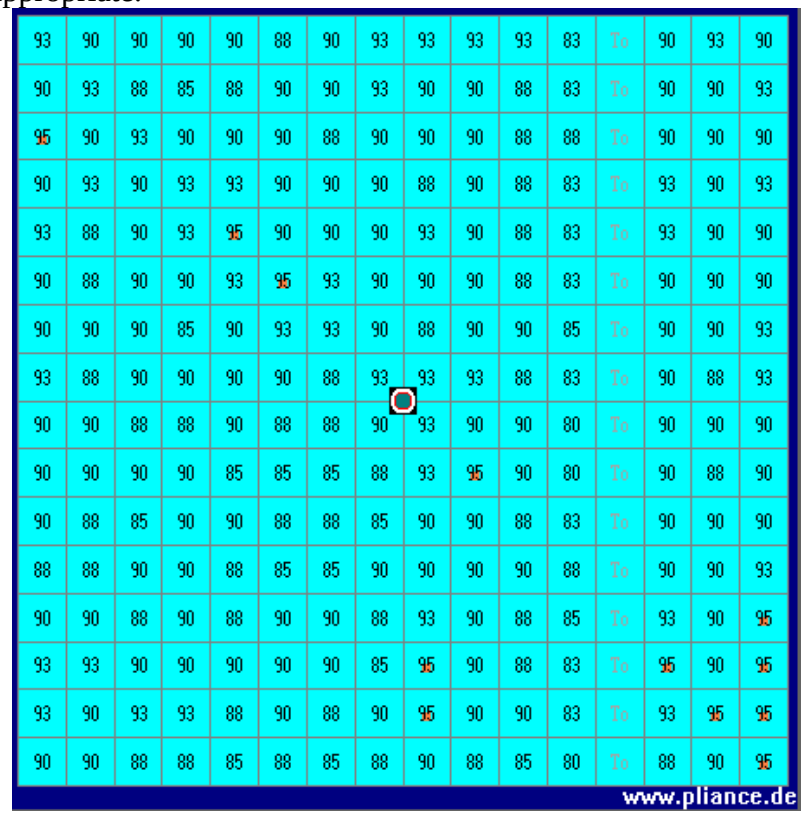
1. **over all frames:** average is calculated including *all* data, even zeros.
2. **only over loaded frames:** entire data *frames* that contain *all* zeros will not be included in the calculation.
3. **over loaded sensors:** *no* zero values will be included in the calculation.

**Measurement precision vs. Maximum precision:** When pressures are averaged the resolution which results will be higher

than the actual resolution of the measurement. The maximum precision will average floating numbers, where measurement precision will only display averaged numbers which are valid with the measurement resolution.

e. Error Handling

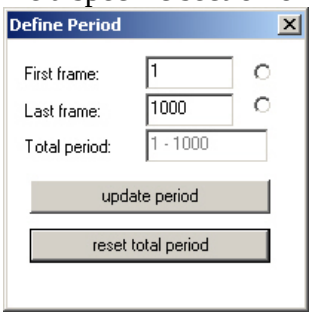
- **With error correction:** If an error occurs the software will use the sensor elements around the sensor with the error to approximate the value within the element with the error. If this option is turned on, there is no way to undo the correction. It is recommended that this error be left off unless errors are commonly seen and it is determined that the estimation is appropriate.




In the example above a Timeout error has occurred in a column on the right side of the mat. The sensors around this column are used to estimate the pressures. Please see Section **Point o** below and **Section 7** for additional information on error codes.

- **Show always transfer errors:** If an error occurs, or if using Bluetooth and dropped frames occur, this will display a message regarding the errors. It is recommended that this option ALWAYS be on. Error codes are not stored with the data file. Therefore, if an error occurs, the error code should be reviewed immediately after collection and a note be made. If the BT signal has dropped frames, this option will display the number and percentage of dropped frames which have occurred.
- **With BT transfer error correction:** When using the online + flash mode, this option will automatically fill the dropped frames

| | | |
|---|---|---|
| | | which occur to the software from the flash memory of the device. Please note that this option is not recommended for larger lapses in data or when the connection via BT is very unreliable. For additional information on BT dropped frames and troubleshooting, please see **Section 6**. |
| f. | Scaling of lines and bars | This menu determines how the time and bar graphs are scaled. The menu item **Options \| Scaling of lines/bars** allows you to set these settings, as well as the numbers used of the automatic scaling is turned off.<br>• **Automatic scaling on graphs:** will change the scale to the maximum pressure, force, area etc which was captured during the trial.<br>• **Interval:** will auto-scale the time graph to the entire trial.<br>• **Lines for masks:** This option will display a separate line within the time process window for each of the defined masks. Please see **Section 4.2.3–f** and **Point p** below, for additional information on creating and displaying masks. |
| g. | Run Mode | **Show defined masks:** This option will display the **Masks** window by default (instead of the time process graphs). Please also see **Section 4.1.3 #5** for further information on this window.<br>**Video Recording:** This option turns on the video recording. This option can also be turned on and off by using the Video on/off button in the main toolbar. A  button indicates that video is on; where a  button indicates that video is off. Please also see **Point q** for additional information on video recording.<br>**Audio Alert:** This option turns on the audio alert. This option can also be turned on and off by using the Audio on/off button in the main toolbar. A  button indicates that the audio alert is on; where a  button indicates that the audio alert is off. Please also see **Point r** for additional information on the audio alert feature.<br>**Reset pliance box before exit:** This option resets the pliance box upon exiting. Please note that the pliance box must still be on and connected to the computer for this option to function properly. |
| h. | Language | Switch between supported languages. The available languages are English, German and Italian. |
| i.  | Show 3D/2D/isobar pressure picture | With each mouse click the pressure picture will toggle between the 2D, 3D and isobar displays. |
| j. | Show number of | When selected, displays the assigned sensor identification number that corresponds to the ASCII output file. |

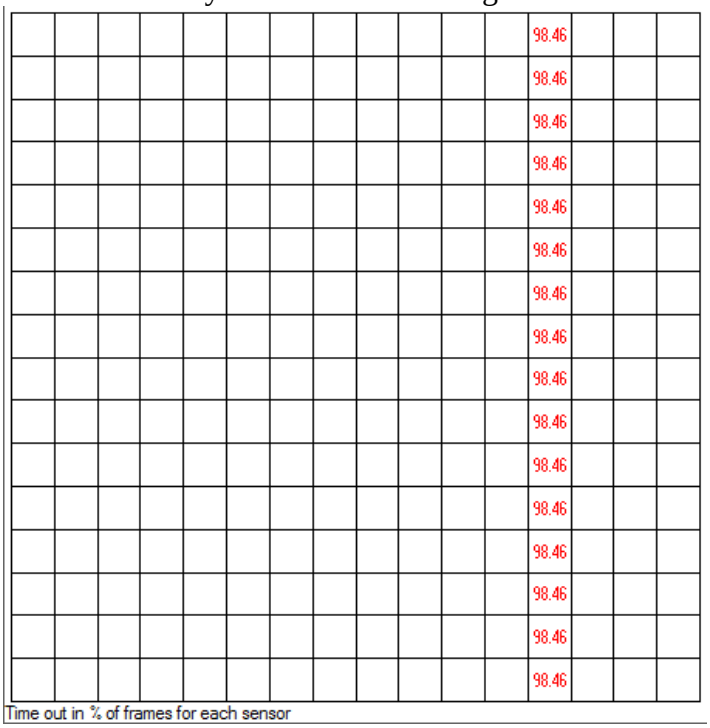
| | | |
|---|---|---|
| | sensors | |
| k. | Show area of sensors | When selected, displays the physical area of each sensor in $mm^2$. The information on this display may be useful when using the ASCII file. |
| l. | Define period | Allows the user to define a specific section of data to view or save.<br>  <br>Select the first and last frame by either typing in a frame number, or by selecting the radial button and then clicking on a point within the time process graph. Once selected, click **update period**. The new data and calculations will display. If the new trial should be saved, use the menu item **File \| Save As \| MAT file** (see **Section 4.2.1 - f** for additional information. Use the **reset total period** to view the entire data file. |
| m. | Speed of roll over process | Allows the user to change the rate at which the data file is played back.<br>  <br>The dialog will display one frame for the time which is defined and is dependent on the measurement rate that the file was collected. In the example above the software will play back one frame every 20 milliseconds. Therefore, if the data was collected at 100 Hz, or 1 frame every 10 milliseconds, this will display the play back at half speed. In this example a definition of 5 milliseconds would play back the trial at double speed. Click **OK** to apply the change. |
| n. | Simultaneous scrolling | With this option enabled and a second pliance file is opened in a separate window, when the play button is clicked both trials will play back simultaneously. Please note that this does not mean that the trials will be synchronized. This option can also be enabled/disabled within the main toolbar. The  button indicates that simultaneous scrolling is disabled; where a  button indicates that the feature is enabled. |
| o. | Image transparency | It is now possible to have an image in the background of the Pressure picture window. This image may be selected from the **File \| Open Image** menu. The transparency of this image can be |

modified with this menu item.

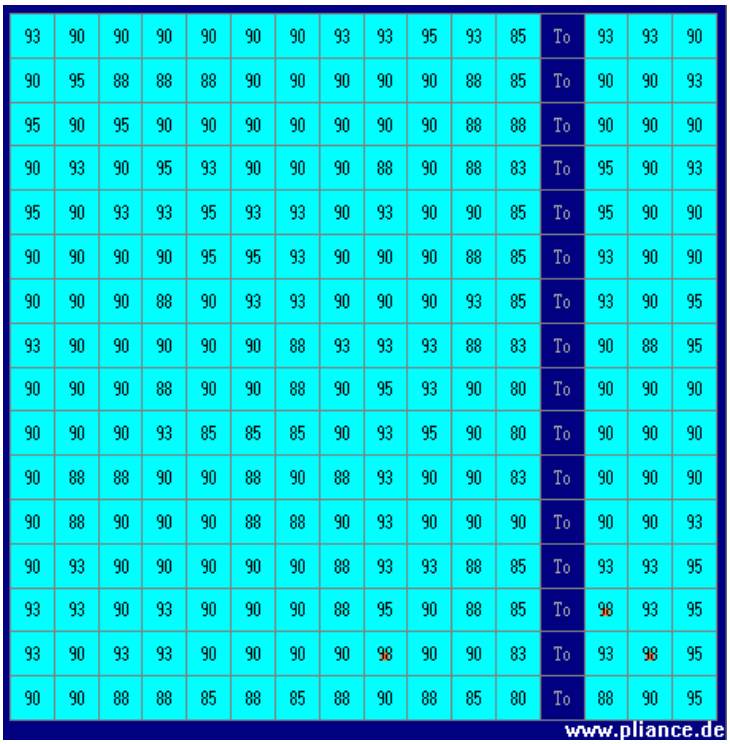
p. Error codes

If an error has occurred during data collection, one of the following menu items will become available to view the error. Please note that the error code is NOT saved with the trial, so as soon as the trial is closed, or another measurement is started the error code will be gone. It is important to ensure that the option **Show always last transfer errors** is enabled. This will prompt the user to view the error and make note of it before continuing on to the next measurement.

Once one of the following is chosen the pressure picture window will display in which sensor(s) the errors have occurred and for what percentage of the trial collected. If this is a very low number, the error can likely be ignored. If the time in which the error occurred is higher, or is occurring quite frequently, steps should be taken to determine why the error is occurring.




- **Converter overload:** This error will occur if the system received data which is above the amplification of the sensor. This error commonly occurs if high loads have been placed on the sensor. For additional information please see calibration **Section 4.4.2.2**.
- **Timeout:** This error indicates that the pliance box did not receive a reading from a sensor(s) when it expected to.



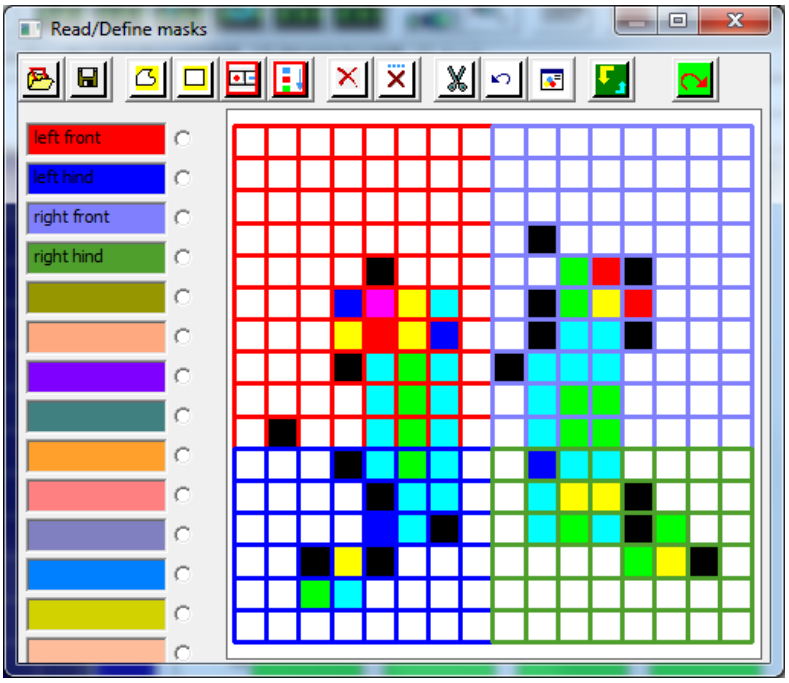

This error will occur if there is insufficient time for the computer to communicate with the pliance-x, or if a sensor element has become damaged. Please see **section 7** for information on testing the pliance sensor mat for errors.

- **Sensor overload:** This error occurs when one or more sensors have exceeded their maximum capacity.
- **Last transfer error:** This option will display the last measurements error codes. If no error codes were reported the message will state this.

q. Read/define masks

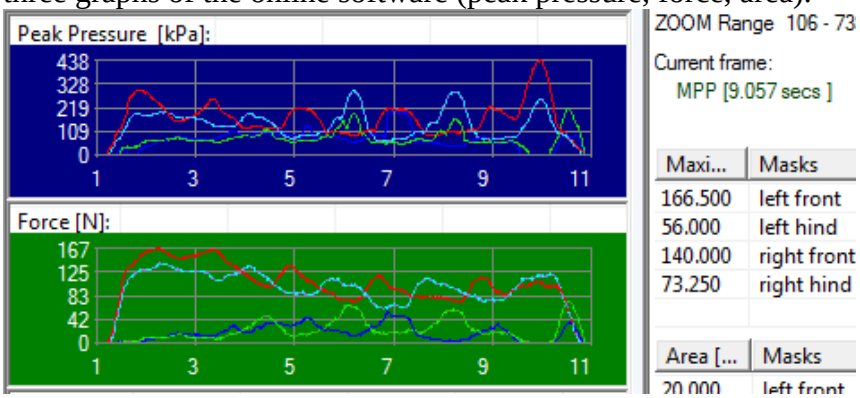
Select this option to define or display a mask file for use with the collected trial(s).

- **Define Masks:** A window similar to the following will appear. All options are the same as that described in **Section 4.2.3-f**. The masks may be applied by clicking the **overlay changed masks** button  .




- **Calculation and store object setup**: The user can define:
  - the different sensor matrices – called “objects”
  - for which sensor matrix/object a zero measurement is required
  - if a separate .mat file should be saved for each sensor matrix/object.

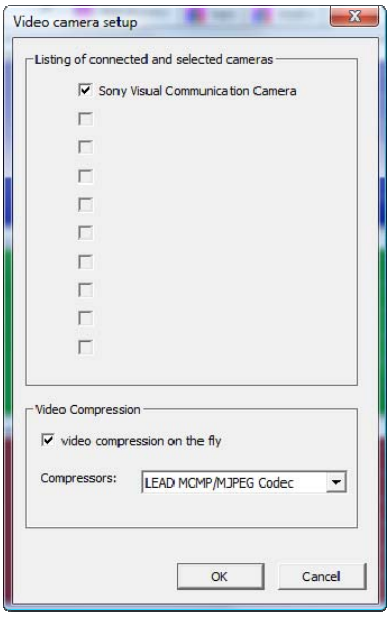
  Once the different sensor matrices/objects have been defined, each object will be displayed with a separate curve within the three graphs of the online software (peak pressure, force, area).




- **Read masks:** Allows the user to apply a mask to the currently selected file.
- **Read masks for all files of same type:** Allows the user to apply a mask to all currently open files with the same mat setup.

r. Video Camera Setup

The video camera setup dialog allows the user to select and deselect the cameras which are connected to the computer. Only those selected will be captured. The user may also select a video compressor. Please note that codecs may be required to view the collected videos on other computers:

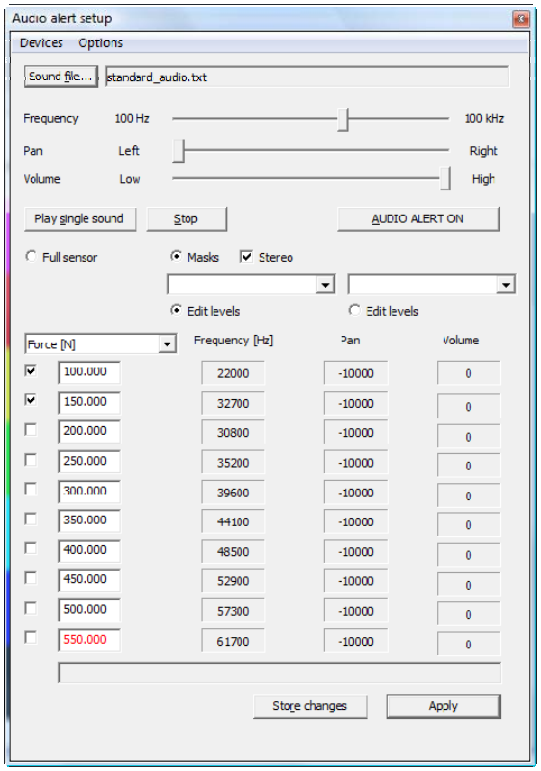



s. Audio Alert Setup

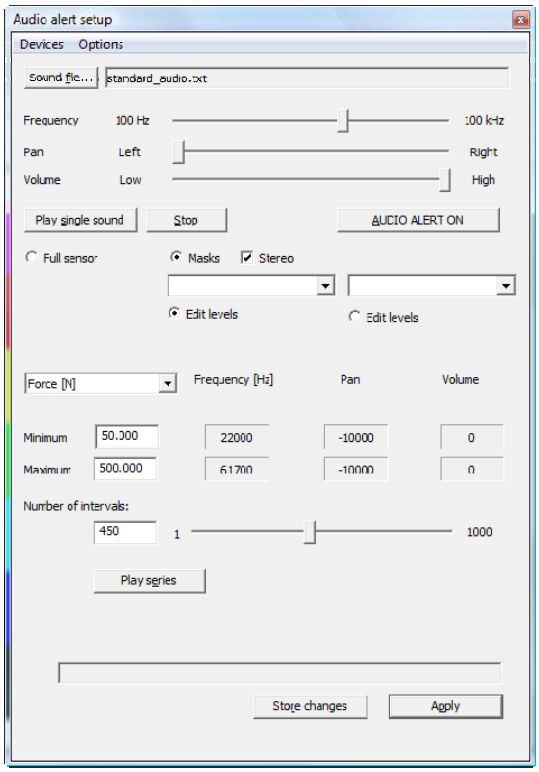
The audio alert function provides the user with the ability to provide audio feedback for data being collected or played back. This option can be useful during biofeedback training sessions. Force, peak pressure, mean pressure and contact area can all be input variables for the determination of the output sound.

- Each variable may be adjusted for the left and right foot independently. This can be done by selecting the **Full sensor** or **Masks** radial button and then the subsequent left or right mat. **Edit levels** should then be selected to edit the level for that particular mask. In some versions of the pliance software you will have the ability to apply the sound to other masks which have been defined along with the left and right mat
- The frequency, pan (left vs. right) and volume may be adjusted for each level selected. To play the sample sound click the **Play single sound** button and then the **Stop** button to stop the sound.
- In some versions of software you will also have the ability to apply a proportional sound, as opposed to discrete sound levels. This option is selected from the **Options** menu.

**Discrete Sound:**




**Proportional Sound:**




The proportional sound series may be sampled by clicking the **Play series** button.

- Once the options have been selected the settings may be saved as a new souds file by clicking the **Store changes** button. A previously saved audio alert file may be opened by clicking the **Sound File...** button.
- The audio alert can be turned on by clicking the **AUDIO ALERT ON** button and the **Apply** button will apply the settings to the software for the next data collection.

| | | |
|---|---|---|
| t. | EM_ASCII export setup | This option is used in conjunction with orthotic development. The export directory and default name for the export file can be defined. |
| u. | Pressure range setup | This option allows the user to change the color range settings of a previously collected trial. This change may be applied to only the currently selected file, or all open files. |

### 4.2.5 Expert Settings

Please note that not all options are available for all versions of software. The Basic and Standard versions of software will not be able to access all of these features.

The pliance software uses an mat configuration file for measurement. This configuration file contains the calibration file, as well as information on how the data will be recorded. This includes the measurement rate, the length of the trial which will be recorded, synchronization options, etc. Please see **Section 4.3** for a complete description of the configuration file.

The calibration file, which is contained within the configuration file, describes the relationship between an applied known pressure and the electrical signal from the sensor mat. This procedure is completed prior to system delivery, but may also be completed by the user with the aid of the trublu calibration device. Please see **Section 4.4** for a complete description of the configuration file.

| | | |
|---|---|---|
| a. | Edit actual mat configuration | The mat configuration which is selected under the **mat configuration** drop down list will be opened for editing. Please see **Section 4.3** for a complete description of the configuration file. |
| b. | Edit mat configuration | A mat configuration can be selected from a typical Windows Open screen for editing. Please see **Section 4.3** for a complete description of the configuration file. |
| c. | Create New mat configuration | Select to create a new configuration file for a calibration. Please see **Section 4.3** for a complete description of the configuration file. |
| d. | Settings + calibration Online | Select to complete the calibration process for sensor mat. The pliance box must be connected to the software prior to selecting this menu item. Please see **Section 4.5** for a complete description of the calibration procedure. |
| e. | Settings + calibration Offline | Select to review a previously completed calibration. Please see **Section 4.4** for a complete description of the calibration procedure. |
| f. | Directories | The default directory paths may be changed with this menu item. |

### 4.2.6 Help menu

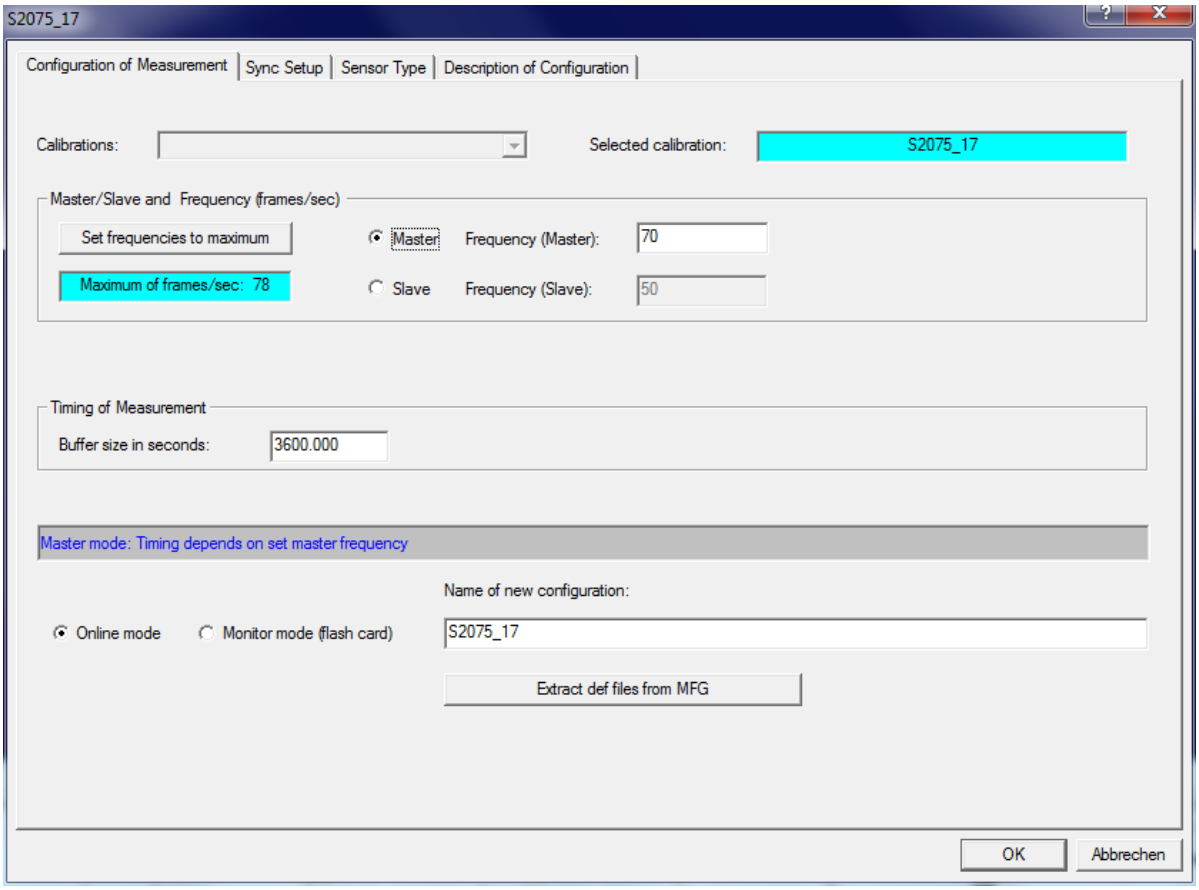
| | | |
|---|---|---|
| a. | novel (www.novel.de) | Opens the novel.de website in the user's default browser. |
| b. | About | View software version information. |

## 4.3 The mat configuration file

Please note that not all options are available for all versions of software. The Basic and Standard versions of software will not be able to access all of these features.

The pliance software uses an mat configuration file for measurement. This configuration file contains the calibration file, as well as information on how the data will be recorded. This includes the measurement rate, the length of the trial which will be recorded, synchronization options, etc.

The configuration file may be accessed via any of the methods described in **Section 4.3.5-a-c**. Once the configuration file is accessed, the following Window will appear:



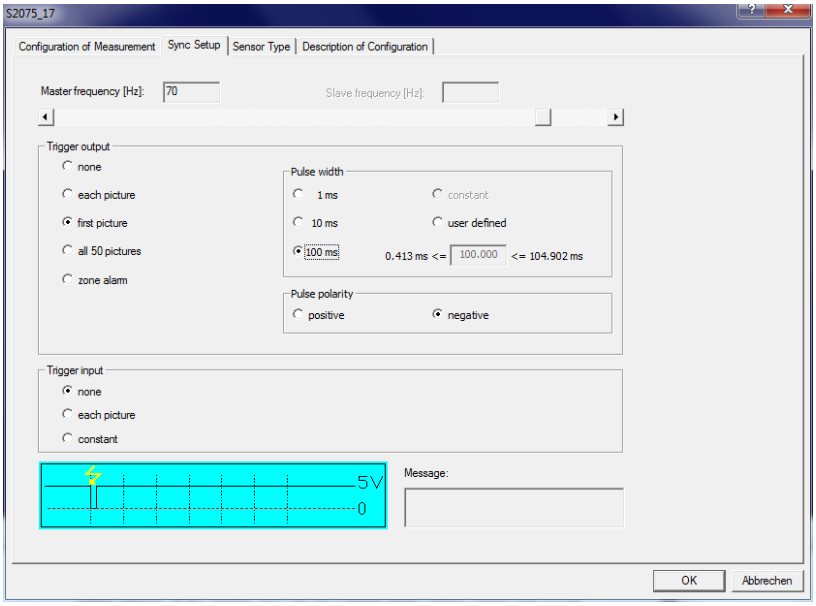

If a new configuration is created the same window will appear, but with the option to select the calibration from the **Calibrations** drop down menu:



### 4.3.1 Configuration of Measurement Tab

- **Selected calibration:** will display the calibration file which is associated with this configuration.
- **Master/Slave Frequency (frames/sec):** This section defines the measurement rate of the pliance system. The max sampling rate can be selected by clicking the **Set frequencies to maximum** button. (Please note that typically a frequency is choosen which is a bit lower than the maximum, perhaps 5%. Otherwise you will see the red error LED being on during data collection, and after stopping the measurement a message will appear saying "measuring frequency too high".) Alternatively the sampling rate may be entered under the **Master** field. The slave device (if used) can also be entered.
- **Timing of Measurement:** The total collection time may be entered within this field. By default the maximum measurement rate is set to 3600 seconds (one hour). Typically users will stop the data collection manually using the **Stop and store** button, or the system is triggered externally.
- **Mode:** The configuration must be set to either an online or flash card mode. If online + flash is used, this is considered an online configuration. Please see **Section 2.4.3** for additional information.
- **Name of new configuration:** Enter or change the name of the configuration file here. It is recommended that the configuration file contain the information on the sensor mat used, the date of calibration and any other important information, such as online vs. flash mode etc.
- **Extract def files from MFG:** This button will extract the calibration files contained within the MFG. This can be useful if the configuration files were transferred to another computer without the calibration files.

### 4.3.2 Sync Setup

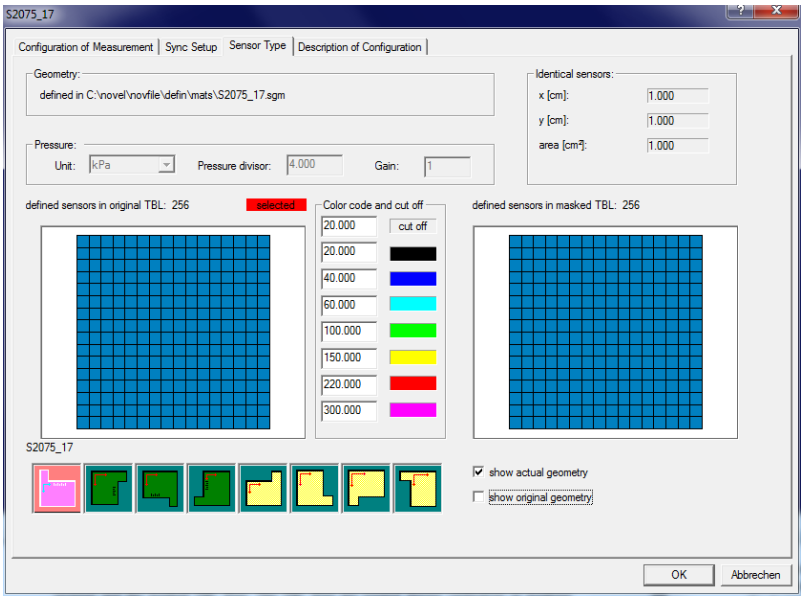



- This window displays the options for a trigger output and trigger input. These options may be used with either the wired or wireless synchronization device (Please note: only certain

versions of the wireless sync can use a trigger input – please contact your novel representative if you are unsure if your sync device supports a trigger input).

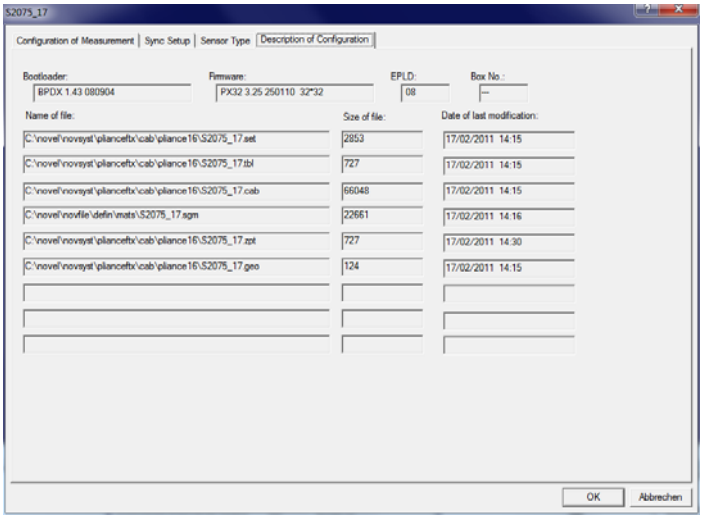
- For a complete description of the synchronization hardware and software see **Section 7**.

### 4.3.3 Sensor type




This window displays the pressure color code and cutoff value and the layout stored in the original and the masked TBL files. The TBL files are created during calibration or masking and are used to determine the maximum sampling rate of the mat pair.

### 4.3.4 Description of Configuration




This window displays the paths for the set of calibration files. These paths can't be modified and are for reference only.

## 4.4 Settings and Calibration Software and Calibration Hardware

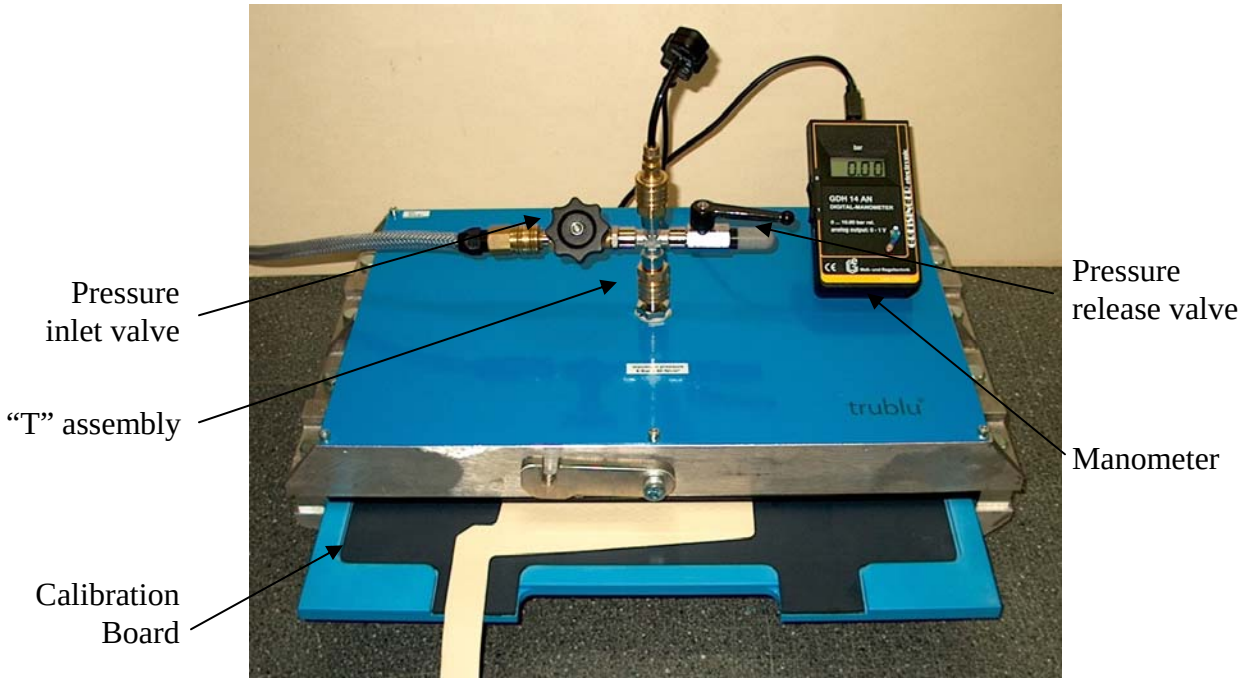
The calibration file describes the relationship between an applied known pressure and the electrical signal from the sensor mat. This procedure is completed prior to system delivery by novel and the calibration files are installed during installation. This procedure may also be completed by the user with the aid of the trublu calibration device. **Calibration must take place in a trublu calibration device supplied by** novel**.**

**IMPORTANT**

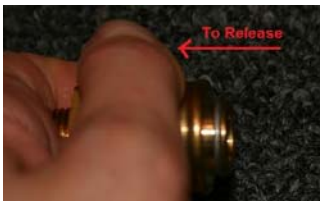
novel recommends checking calibration of each mat approximately every three months by applying known pressures to the sensor mat(s) in the calibration device and comparing with the pressure obtained from an online measurement. Please see **Section 4.4.2.1 #13** for instructions on checking the calibration.

### 4.4.1 The Calibration Device

The calibration device has been developed specifically for calibration of pliance sensor mat and consists of a rubber membrane, housed within a secure unit. Compressed air is fed into the device thus exerting a pressure on the inserted sensor mat. The rubber membrane ensures a homogeneous pressure distribution.



- **Place the calibration device on a flat, stable surface and secure to prevent it from slipping off the surface.**
- Connect the supplied brass air pressure "T" assembly component and the manometer to the top of the calibration device as shown in the picture above. Please note that all connections utilize a quick release connection. Pull back on the connector to release.



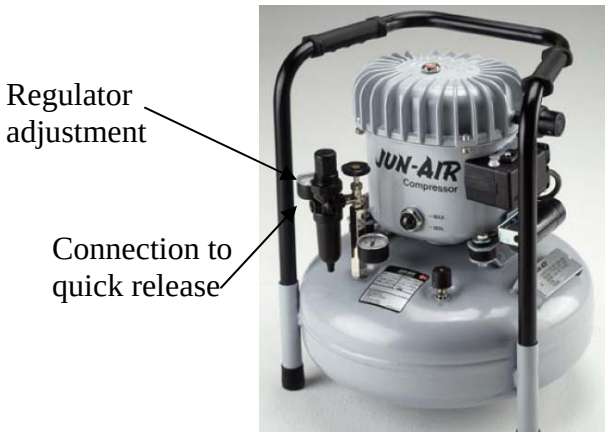

- Connect the air supply (compressor) to the calibration device. The air must flow from the compressor to the calibration device, thus making it possible to read the air pressure on the manometer pressure scale. For the pliance calibration a 90 PSI (6 bar) air supply is required. An extra female fitting (pictured on the left below) is typically provided for connections to the air compressor/supply. This is a standard ¼" NPT connection and can be made utilizing plumber's (Teflon) tape.







- The calibration device must only load pressure with the calibration board pushed completely into the device. Two stoppers on the bottom of the calibration board will indicate when the board is in the proper position
- To allow pressure into the calibration device close the release valve by turning it 90 degrees clockwise. SLOWLY open the inlet valve by turning the knob counterclockwise. The manometer should begin to read pressure. Ensure that you can obtain the required 6+ bar for calibration.
- High levels of pressure should not be left inside the calibration device for long periods of time, as this can cause slow leaks in the bladder system. Ensure that the steps performed with calibration are performed systematically and quickly.
- Commonly novel will provide a Jun-air air compressor with the calibration device. This is a low noise compressor and is designed for laboratory/office applications. The female quick release picture above should be connected to the regulator on the air compressor, as pictured below. The air compressor should be regulated to just above 6 bar. Please refer to the compressor manual for complete instructions.




**IMPORTANT**

It is extremely important to close the inlet valve from the compressor (clockwise) and open the outlet valve (parallel to the outlet pipe) when not using the calibration device. If this is not performed, it is possible that air can leak from the compressor into the calibration device over time and rupture the rubber bladder.

**IMPORTANT**

The manometer is calibrated in bar.

1 bar = 10 N/cm$^2$ = 100 kPa

## 4.4.2 Settings + calibration online

Select this option to complete the calibration process for sensor mat. The computer must be connected to the pliance device to select this option. The process includes setting the amplification of each sensor element within the matrix (the "Settings" procedure) and then performing a multi-point calibration. The final step of the calibration procedure is checking the calibration within the pliance online software.

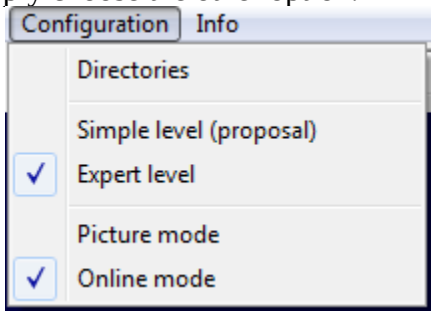
There is no need to use this program during daily routine use of the system. Calibration should be repeated or checked approximately every three months.

### 4.4.2.1 Calibrating Sensor mat (Simple Level)

1. Select the menu item **Expert Settings | Settings + Calibration Online** from the pliance online software.
2. Connect the pliance system and sensor mat as described in **Section 2.4.4**.
3. Place the sensor mat within the calibration device.

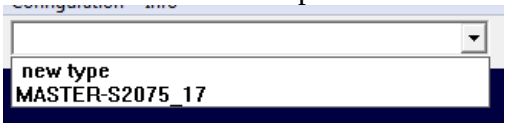
a. The sensor mat should be placed within the calibration device as shown in the first picture in **Section 4.4.1** above.
b. Place the sensor mat in a way that the pliance mat label (showing a serial number as for instance S2075_17) should be UP and visible.
c. If you can connect several sensor mats to your system, ensure that the sensor mats are properly spaced within the device. They should not be touching each other and should be about 1 cm from the edge of the board.
d. Push the calibration board into the device. Ensure that it is pushed in completely in and the two stoppers on the board are touching the front of the device.

4. The following instruction is related to the Expert level of the calibration program. You can select this mode from the Configuration menu. If you prefer the "Simple mode" (less buttons and numbers), simply choose the other option.



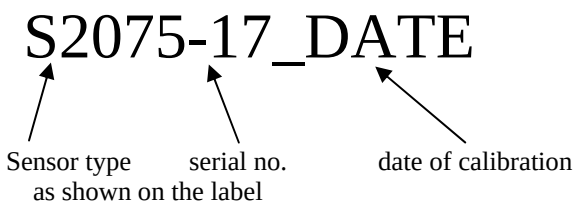

5. You may begin the calibration via two methods. Within the Calibration software:

a. Go to the menu **File | NEW calibration** or select the  button from the toolbar.
b. A MASTER Calibration file may have been provided with your sensor. If so, select this from the drop down menu:



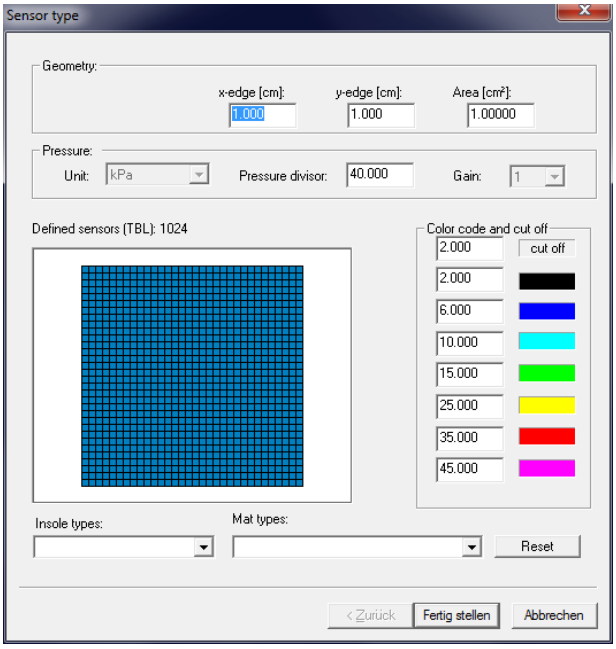

and go to the menu **File | Use EXISTING sensor type and calibrate** or select the  button from the toolbar.

6. A Windows Save As window will appear. Give a name to the calibration file. The calibration file should be named like the following:



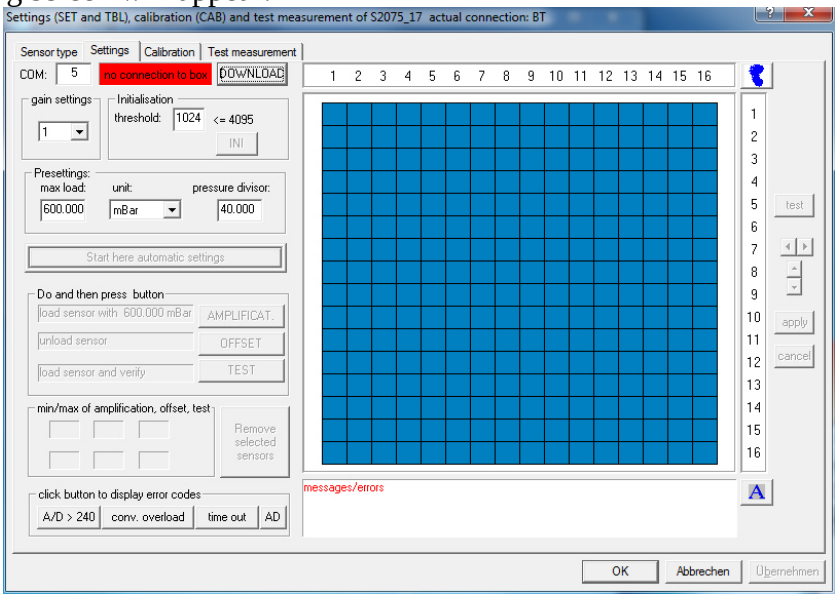

Do NOT save over the Master calibration file. Click **Save**.

7. The following screen will appear when a **new sensor** is created:

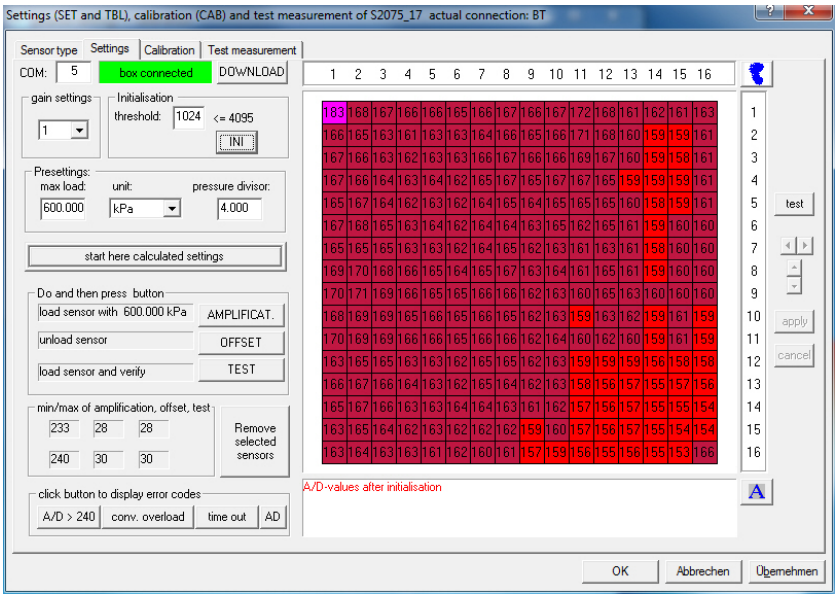



Check and change, if necessary, the Color code, the x-edge and y-edge (size of one sensor element in centimeters), and click **Finish**.

8. The following screen will appear:

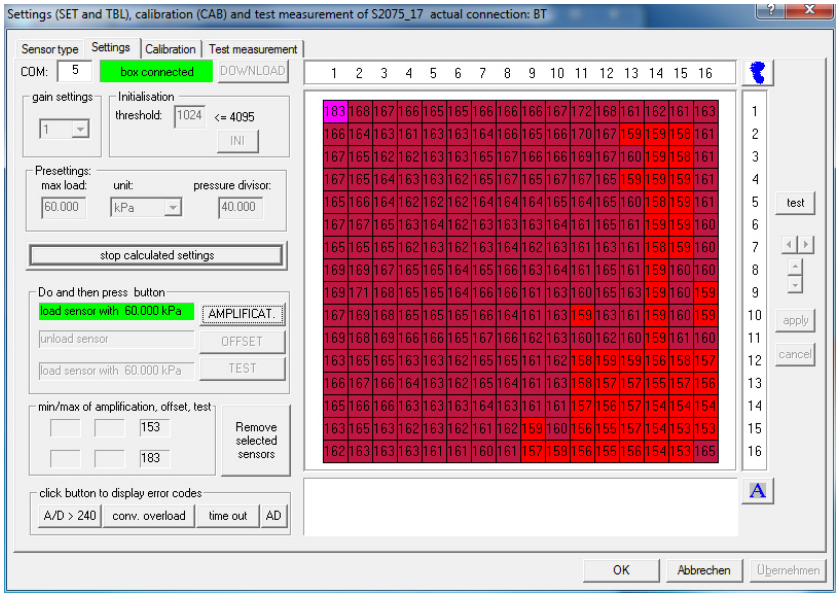



9. Verify the system is connected to the specified COM port and click **DOWNLOAD**.
10. You will then see that the message box connected and the following screen




The numbers and colors shown after the Download button is clicked are not important. Any messages (error codes, warnings, etc) regarding the progress of calibration can be found in the message field in the bottom right of the window.

11. Click **Start here automatic settings**. You will receive a message within the message window on the bottom right of the screen which says “Please wait: preparing automatic settings”. The software will then indicate with a green window what should occur. The first step will be to load the sensors with the maximum load. This value is different for many pliance mats, but a typical load is 600kPa, or 60N/cm², or 6Bar.




a. The software will then indicate with a green window what should occur. The first step will be to load the sensors with the maximum load, which is 60 N/cm$^2$, or 6 bar. To ensure that amplification of the sensors occur properly at high loads, it is

recommended that you load the sensor mat slightly higher than the amplification number mentioned above, i.e. 0.1 bar higher.

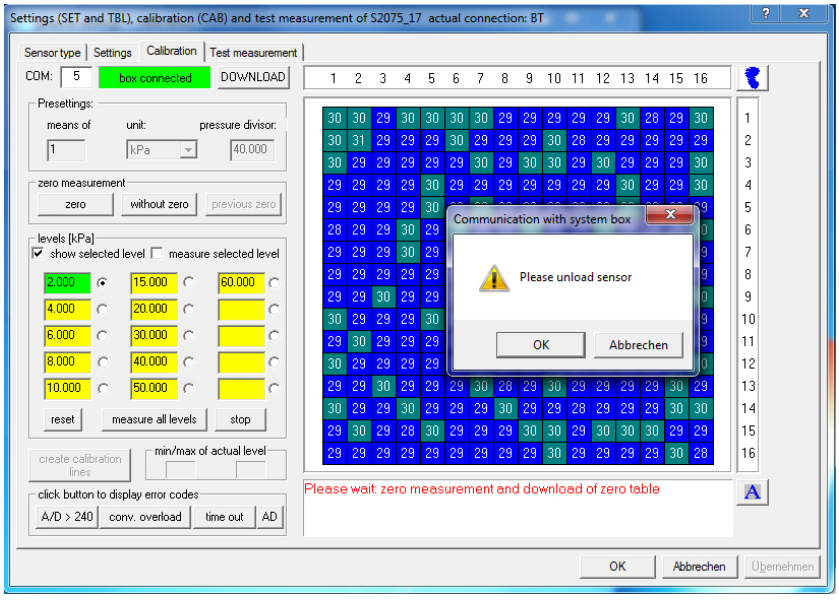
i. Load the sensors in the calibration device to the level indicated. Follow the instructions for loading pressure described in **Section 4.4.1**. Please note that the manometer will display in bar and the software is, by default, set to display in N/cm$^2$. Please make this conversion (a factor of 10, accordingly.
ii. Click the **AMPLIFICATION** button.
iii. The min and max values under Amplification should read between 235 and 245. If a value is greater than 240 it will be highlighted in yellow. This is not an issue.

b. The **unload sensor** field will turn green.
i. Release the pressure by opening the release valve and release all pressure.
ii. Click the **OFFSET** button. The software will wait 10 seconds and then capture two unloaded states. Do NOT load the system with any pressure during this time.
iii. The min and max values should read between 25 and 35.

c. The **load sensor with 60 N/cm$^2$** field will turn green.
i. Load the sensors in the calibration device to the level indicated (60 N/cm$^2$).
ii. Click the **TEST** button.
iii. The min and max values under Test should read between 235 and 245. If some sensors read a gray > sign, this is OK.
iv. Unload the calibration device.

d. Click **store settings** to save the settings. If any errors have occurred they will be listed under the messages/errors box and the next step will not be able to be performed. Please refer to the Expert Mode described in **Section 4.4.2.2** for further information.

12. Click on the CALIBRATION tab. A window similar to the following will appear:



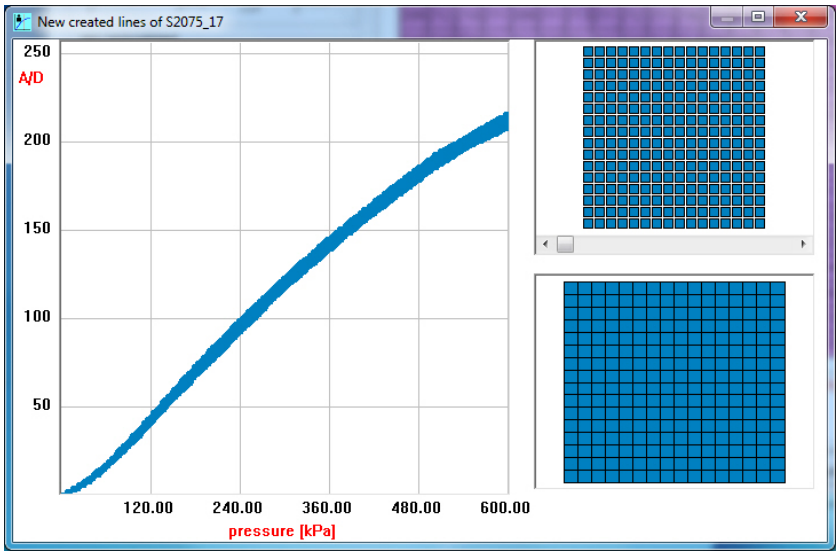
The first level, 2 N/cm2, will be green. Click **measure all levels**.

a. A message will appear to unload the sensor mat. Ensure that no pressure is in the calibration device and click **OK**. Please note that this step is similar to the unload requested when the pliance online software requests that the sensor mat be unloaded.
b. The software will begin to read the data in real-time from the pliance device. Apply the specified pressure (the value will be green). Once the sensors in the calibration device are loaded to the specified level, press the space bar or Enter key to acquire the A/D values corresponding to the pressure level.
c. As soon as a level is taken the next level will turn green. You should also hear a "ding" if the volume is turn on. Increase the pressure placed on the sensor mat to the next level and click the space bar. Continue the process until the last level has been acquired.

**NOTES:**

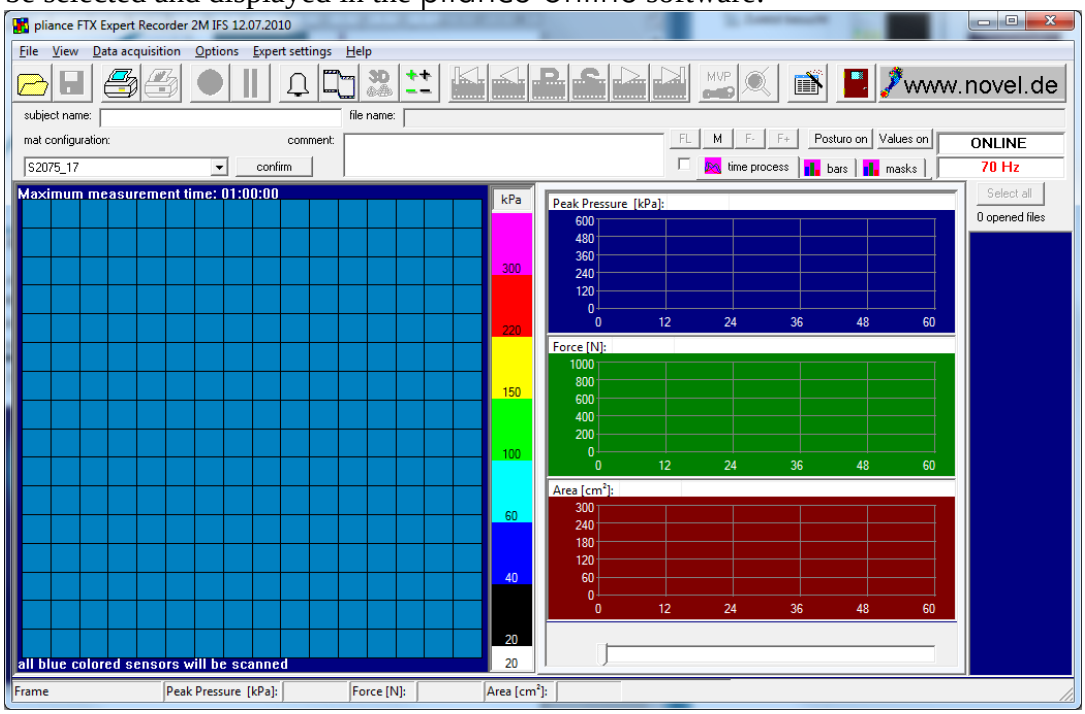
- The pressure should be increased to each level; you should not go over a level and come back down. The process should occur smoothly and not take a long period of time (less than a few minutes), so as not to introduce hysteresis on the sensors. If a level is over shot, release the pressure to below the level and come back up, but this can cause a slight change in curvature of the lines. It is recommended that the process be restarted:
  - **Stop** button: click to stop the calibration process.
  - **Reset** button: click to reset the calibration process.
  - **Zero** button: re-zero the sensors (no pressure should be applied).
  - **Measure all level** button: click to begin the calibration process.
- Minimum and maximum levels: this displays the min and max value that occurred at each pressure level. These values may be important if having problems with the calibration. Errors concerning these values will be displayed in the messages/errors box. Please contact your novel representative for assistance.
- Font icon  : use to specify the font of messages.
- The foot icon  in the upper right corner is used to switch the view of the sensor layout between the electronic layout and the physical layout.

13. When the calibration is over click **create and store calibration lines**. A window similar to the following will appear.




Close the calibration lines window.

14. Click the **OK** button within the calibration window. A message will appear asking if you would like to create a new configuration file. Click **YES**. A new configuration file will be created for use within the pliance online software. The configuration file will now be selected and displayed in the pliance online software:



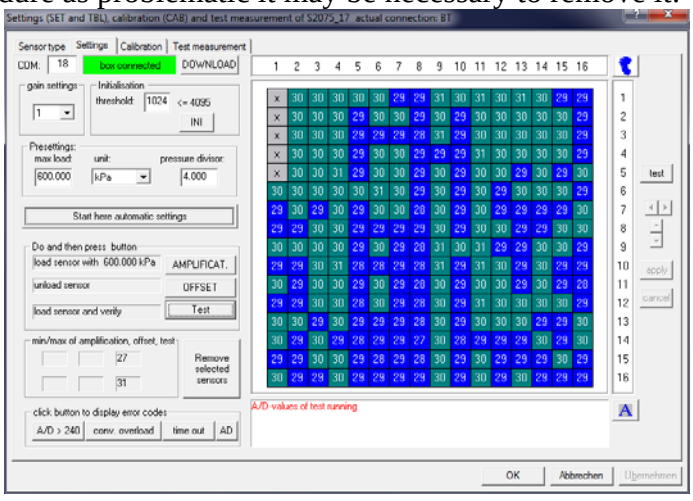

15. The sensor mat will now be tested within the calibration device to ensure that the calibration is accurate. This process is also the process that should occur when testing whether or not the sensor mat need recalibrated.
    a. Choose the appropriate **mat configuration** file from the drop down menu, if not displaying already.
    b. Click the **Confirm** button to confirm the use of this mat configuration.

c. Click **OK** to unload the left and right mat. Ensure that no pressure is within the calibration device.
d. Click the **Record** button and a trial will begin.
e. Click the **Values on** button
f. Load the sensors to any pressure level; choose something lower than 1 bar to begin with.
g. Read the **Mean Pressure** value for the left and right mat. This value should read similar to that of the manometer. Also ensure that the **Peak Pressure** value is not far off from the mean pressure value.
**NOTE:** Keep in mind that the values displayed in the online software are most likely kPa and the manometer displays in bar. These have a factor difference of 100, i.e. 100 kPa = 1 bar.
h. Check additional pressure levels up to the maximum calibrated value (600 kPa / 6 bar / 60 N/cm$^2$) in the same manner. When you have finished, unload the sensors and click the stop and store button. If you would like you may save this trial, though it is not required.
i. NOTE: The new configuration file will default to the configuration file settings in the default configuration. It may be necessary to modify the measurement frequency, sync settings, color code or cutoff etc. This should be done prior to the first data collection. Please see **Section 4.3** for additional information.

### 4.4.2.1.1 Removing or turning off sensors

This menu can be useful when a sensor has become damaged, but calibration needs to be performed. If an error occurs during the calibration procedure, calibration lines cannot be created. This option may also be used to reduce the number of sensors in the matrix for an increase in sampling rate. However, the pliance online menu item **Data Acquisition | Define sensors to be scanned** is more commonly used for this (see **Section 4.2.3 – f**).

1. Follow **Steps 1-6** described in **Section 4.4.2.1** to set up the system for calibration
2. Click **DOWNLOAD** to connect the system to the pliance
3. If a **t** appears in one of the sensors, or this sensor has been identified previously or during the Settings procedure as problematic it may be necessary to remove it:



a. Click on sensors to be removed. The sensor will turn gray with an '**x**' in it.
b. When all sensors to be removed are selected, click the **remove selected sensors** button.
c. A message will appear in the message window that the sensors have been removed and to start the settings again.
d. Click **DOWNLOAD** again.
**NOTE:** it is best to remove sensors *before* completing the settings procedure, so the procedure will not have to be repeated.

4. Complete **Steps 9-13** as described in **Sections 4.4.2.1 and 4.4.2.2**

### 4.4.2.1.2 Changing the maximum load of the sensor mat

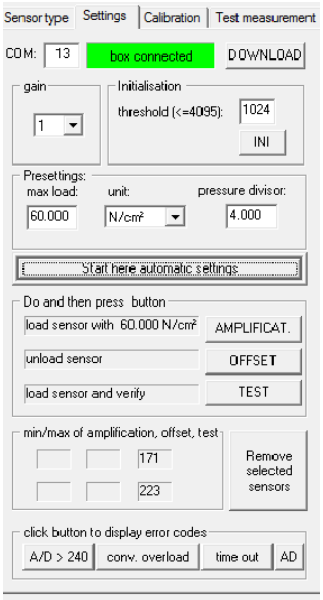
Usually your pliance sensors are specified for a typical pressure range. You may want to produce a second calibration with a lower maximum pressure, for instance 300kPa instead of 600kPa. This way, you'll receive a higher resolution of displayed pressure values. Because the pliance-x system has a fixed gain amplification, changing the maximum load will change the measurement resolution. However, because the sensors are not perfectly linear, decreasing or increasing the maximum load will not double or halve your measurement resolution. Please consult your novel representative to determine the proper load, or alternative sensor types for your application.

If it has been determined that an alternative load is appropriate the following steps should be performed:

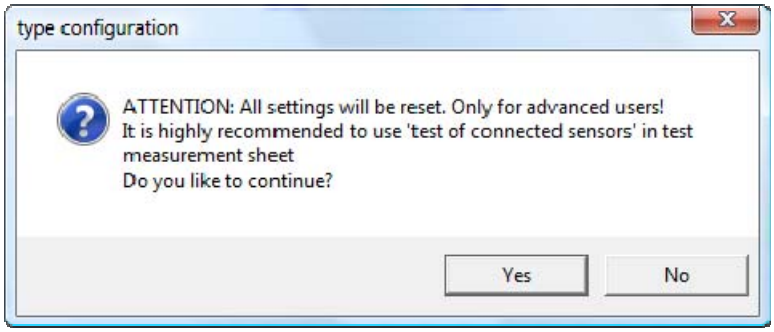
1. Follow **Steps 1-6** described in **Section 4.4.2.1** to set up the system for calibration
2. Click **DOWNLOAD** to connect to the pliance system
3. Within the Expert Mode under **max load** enter the desired calibration level for the pliance sensor mat. Again, the default listed will be 600kPa. Change this to the desired maximum load. You may also change the units of measurement. The pressure divisor will be automatically calculated.
4. Complete **Steps 9-13**. The level which will be obtained for the Settings procedure and the levels of calibration for the step-wise Calibration procedure will be altered automatically. Follow these steps accordingly.

## 4.4.2.1.3 Additional tools of the Expert Level Calibration

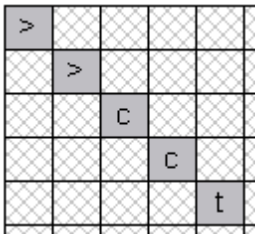
1. Settings Tab:



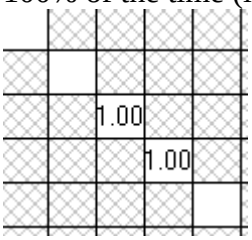

   a. **Gain:** The gain of the system can be changed if amplification can't be reached with a gain setting of "1" (the default value). This may occur when smaller sensor mat are used, with smaller sensor area and lower capacitance. It may also be needed if the sensor mat is calibrated to a maximum pressure lower than the standard 600 kPa. To discuss alternative mat types, please contact your novel representative.
   b. **INI button:** Click this button to reset all values to the initial settings. This feature is typically not used, as all predefined settings are stored with the sensor mat. The following message will appear:




   c. When the **Test** button is clicked, or when the **Download** button is first clicked one of the following error codes may display: **>**, **c**, **t**



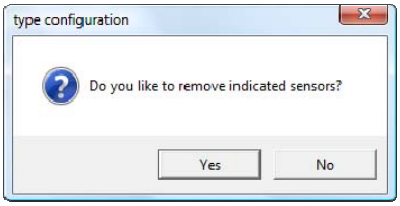

Click the following buttons to display the error codes. The data window will display the amount of time in which the error occurred. A value of 1.00 equals 100% of the time (i.e. a value of 0.41 = 41% of the time).




i. **A/D >240, symbol is ">":** Any sensors that exceeded an AD value above 240 will be automatically set to 240. This message is only a warning and does not affect the precision of the calibration. This error code is common during the **Test** mode when the sensor mat is loaded to the maximum level.

ii. **conv. Overload, symbol is "c":** Any sensor which reads higher than the amplification previously performed will display. This error code should be avoided and the amplification process should be repeated if it occurs.

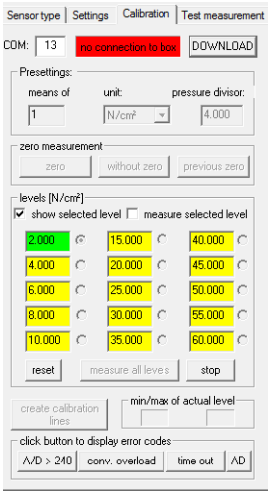
iii. **Time out, symbol is "t":** Any sensor which is not being read properly will display. When the button is clicked a message will appear asking to remove the selected sensors:




The sensor may be removed. See **Section 4.4.2.2.2** for additional information.

**iv.** **AD:** This button will switch back the display of the data from

2. Calibration Tab:

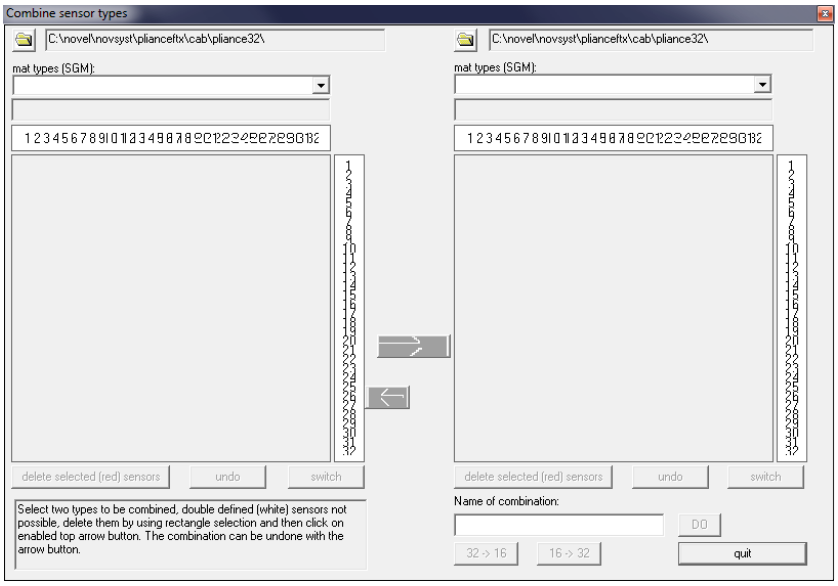


a. **zero:** use this setting to repeat a zero. This may be done if the calibration procedure was stopped reset to begin again.
b. **without zero:** do not use during calibration process.
c. **previous zero**: do not use during calibration process.
d. **measure selected level:** if a level was not correct when measuring all levels, a single level can be re-measured using this feature. This process may not produce the same results as completing the level during a step wise calibration. It may be necessary to repeat the entire step-wise procedure. **See Section 4.4.2.1 #10** for additional information.
    i. Select this option.
    ii. Select the level to be repeated.
    iii. Load mat to selected level.
    iv. Hit the enter key to make measurement.
    v. Click 'create calibration lines' to view changes.
e. **reset:** Use this option to restart the process of measuring all levels to form calibration lines. Values are set to zero so the process can be repeated.
f. **stop:** Use this option to end the measure process during any portion of the process.
g. **min/max of actual level:** displays the minimum and maximum A/D values at each pressure level.

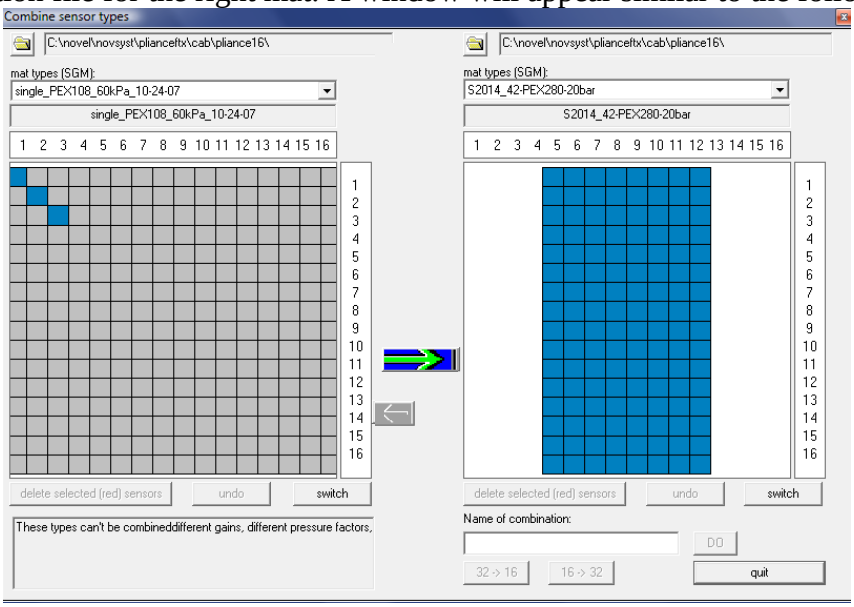
### 4.4.2.2 Combine sensor mat

This menu item is used to combine the calibration files from two different individual calibrations. The sensor mats you would like to combine must have been calibrated individually previously. This may be useful when calibrating mats in different configurations.

1. Within the pliance settings calibration program click **File | Combine sensor type** or click the  button. The following window will appear.



2. In the **mat types (DEF)** pull down menu on the left side of the window, choose the calibration file for the left mat.
3. In the **mat types (DEF)** pull down menu on the right side of the window, choose the calibration file for the right mat. A window will appear similar to the following:




4. Click the blue/green arrow located between the sensor windows to display both sensor mat on the right side.
5. The **name of combination** field located below the right set of sensor mat will automatically fill in with a name for the combined sensor file. The name will start with the mat type. Adjust the name (i.e. enter the date and remove the extra mat type indicator) if desired.
6. Click the **Do** button. The combined file will be created. A message will appear when the conversion is complete.
7. A second message will open asking to create a configuration file. Click **Yes** to create the configuration file. The configuration file will then open within the pliance online program and may be tested according to **Step 13** in **Section 4.4.2.1**.
8. Click **quit** to exit the program if a configuration file was not created.

### 4.4.2.3 Menu Options

1. File
   a. NEW Calibration  Select to create a new calibration. See **Section 4.4.2.1**
   b. REFRESH Calibration  Select to modify an existing calibration
   c. Combine Sensor Types  Select to combine single mat calibrations from two different mat sizes or sensor mat calibrated separately because of their large size. See Section **4.4.2.3**.
   d. Show calibration lines  Select to view calibration lines which were previously created

| | | |
|---|---|---|
| e. | Print screen | Select to print a copy of the screen |
| f. | Capture rectangle and print | Use to select a portion of the screen to be printed.<br>1. Select option from menu.<br>2. Set origin with the RIGHT mouse button.<br>3. Move the cursor to the end point and click the LEFT mouse button.<br>4. A print setup screen will appear. Choose options and then print. |
| g. | Exit | Select to exit the program |

2. Configuration

| | | |
|---|---|---|
| a. | Directories | Use this option to set default paths for files |
| b. | Simple/Expert Level | Select either the simple or expert user level for calibration. The calibration procedure for the Simple level may be found in **Section 4.4.2.1** and for the expert level in **Section 4.4.2.2**. |
| c. | Picture/Online Mode | The **Online mode** transfers data to the computer faster than the **Picture Mode.** The Online Mode should be used by default. If the computer does not display the data within the calibration windows correctly, the Picture mode can be used. |

3. Info

| | | |
|---|---|---|
| a. | About | Displays the software version information |

#####

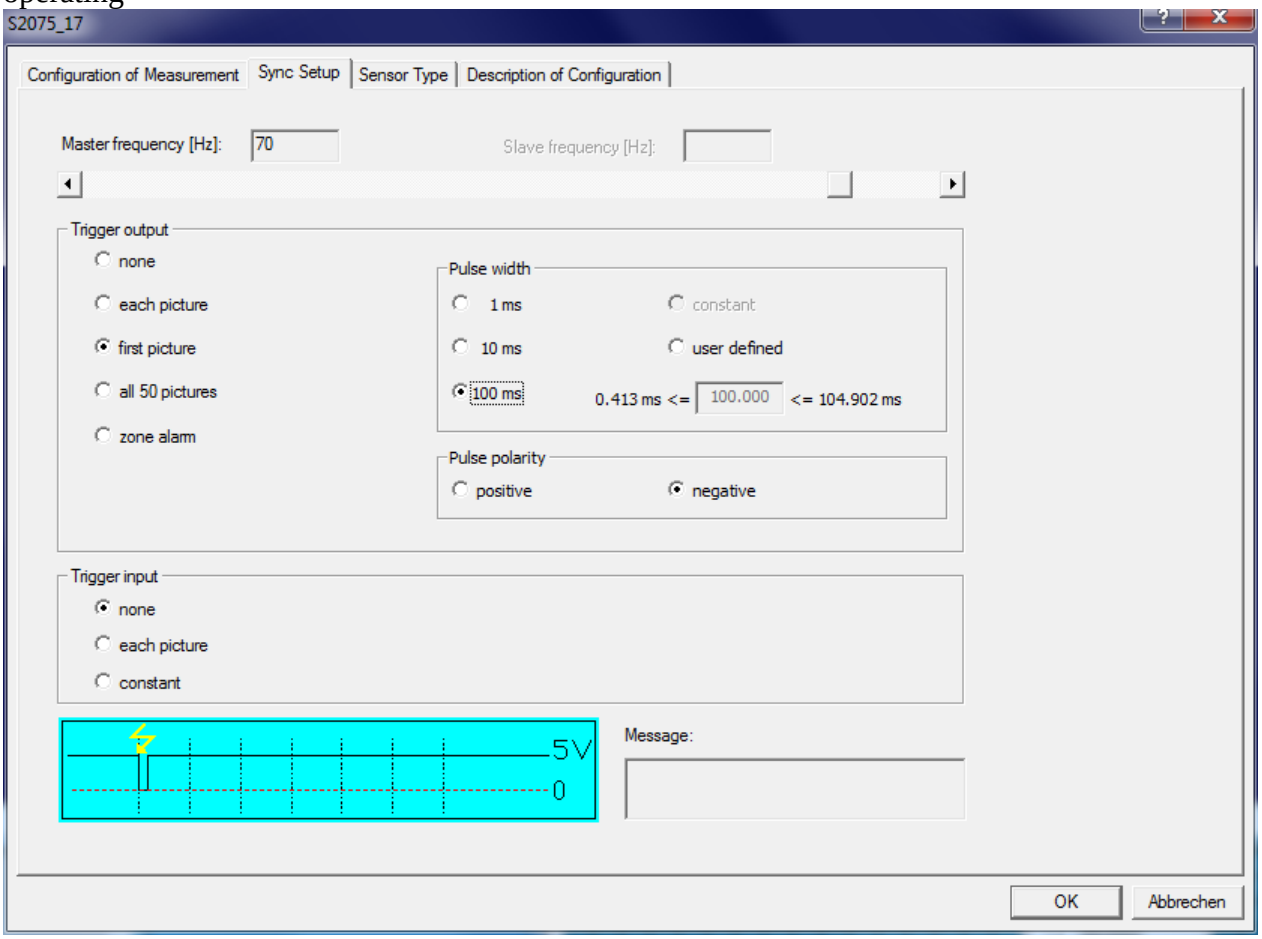
# 5 PLIANCE-X SYNC BOXES AND SOFTWARE

This section describes how to set up the synchronization of the pliance-X device with other devices such as video, EMG or motion capture systems. Generally there are two methods of synchronization:

1) Two (or more) devices run with their own clock. The pliance-X sends or receives a signal that is recorded/sent by the other devices (all devices run as "master").
2) One device sends clock pluses (the master); the other devices ("slaves") run with the clock speed defined by the master.

If one uses more than two devices methods 1) and 2) can be combined. The pliance-X sync box can either send or receive a signal, therefore making triggering an easy way to "sync" data streams from two different devices. Typically this is done by syncing the start of a trial. The two data streams are then exported for simultaneous analysis later. The pliance-X sync device can send a 5 V trigger to another device which can start that device's measurement, or be recorded as part the device's data stream. Similarly the pliance-X sync device can receive a signal triggering the data collection to start within the pliance-X online software.

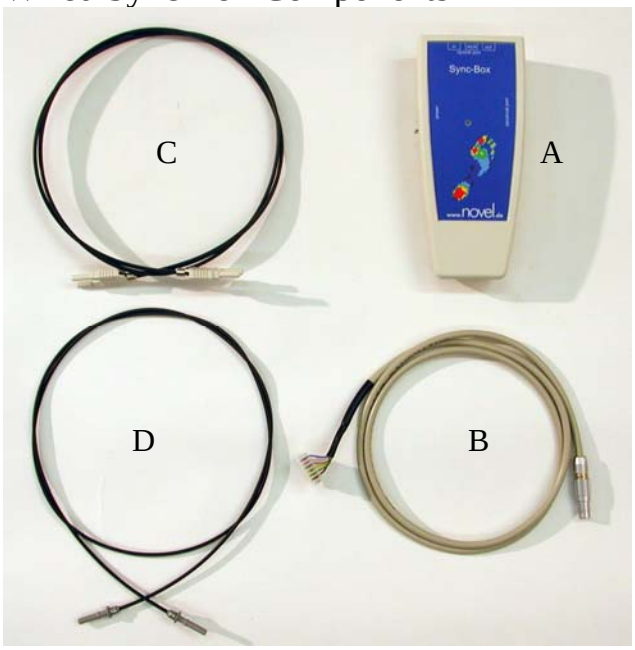
The modes of synchronization described in this manual are available in all operating



modes of the pliance-X (**online**, **online+flash**, **flash memory only**). When you use the last two modes (**online+flash**, **flash memory only**) the configuration must be stored in the flash memory of the pliance-X. For descriptions of each of these modes, please refer to **Section 2.4.3**.

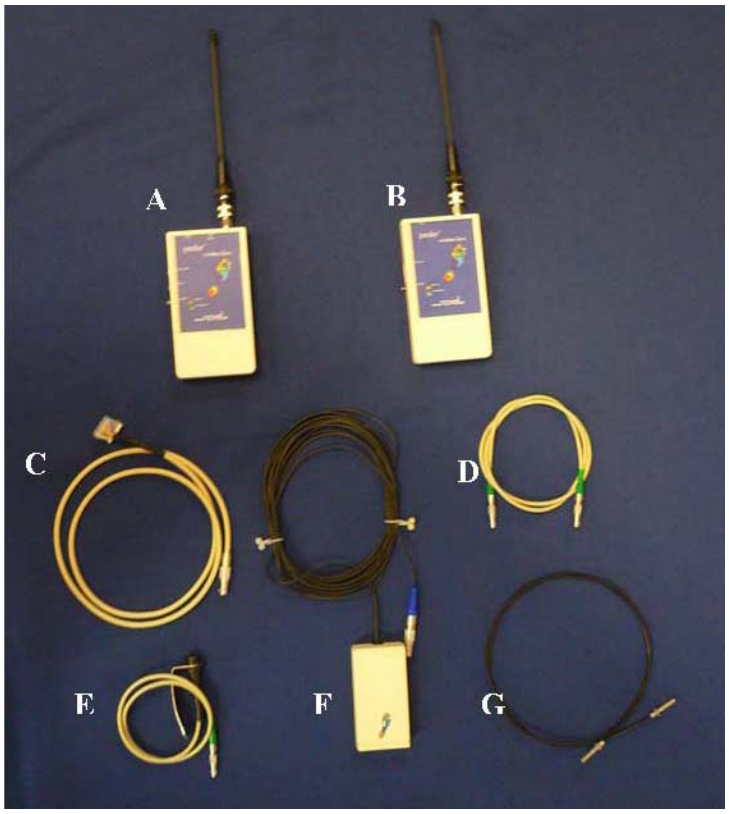
## 5.1 pliance-X Sync Box Components and Connections

### 5.1.1 Wired Sync Box Components



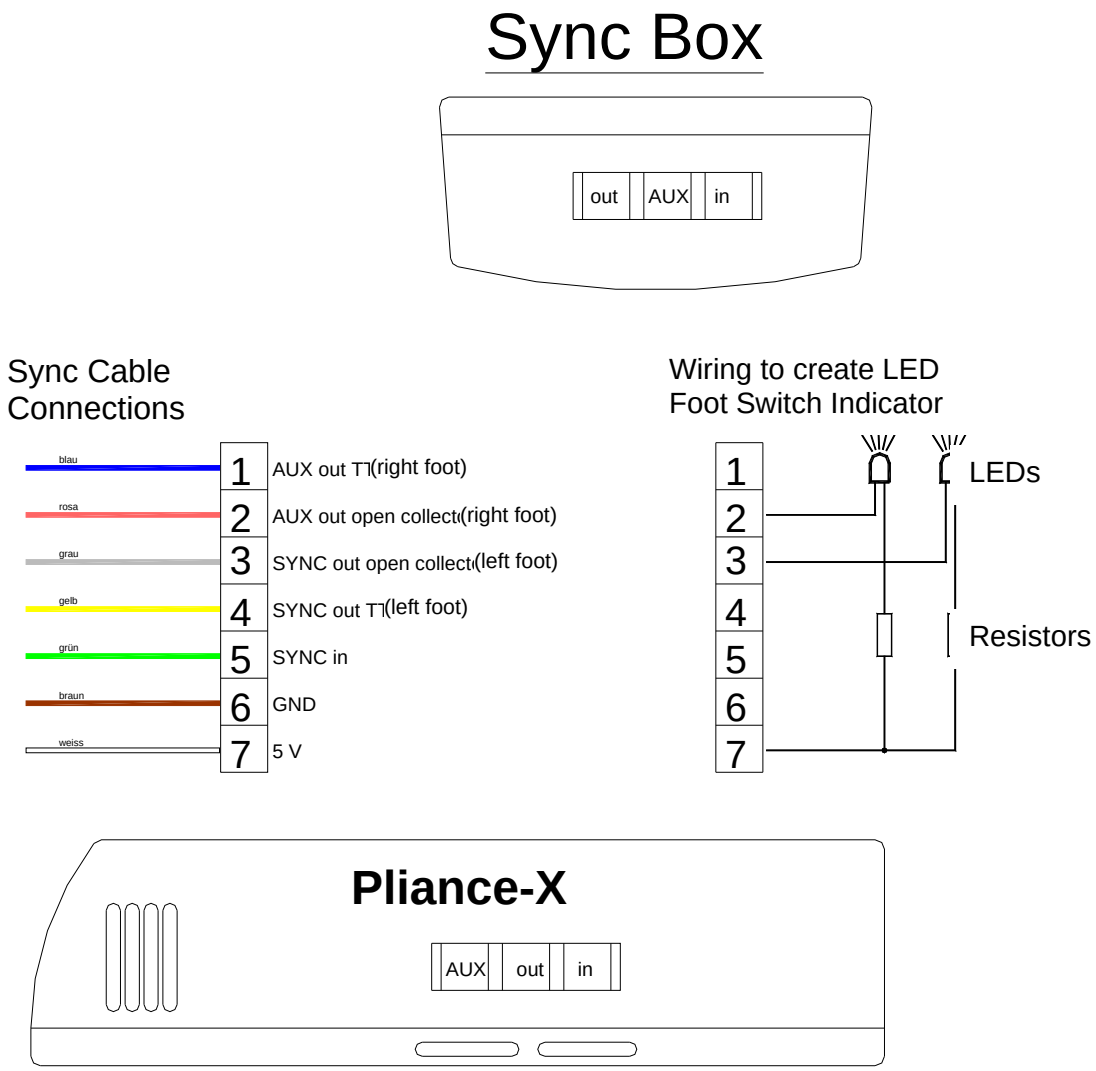

A) The wired synchronization unit
B) Electrical Signal Cable
C) Sync Out Fiber Optic Cable (2 pins)
D) Sync In Fiber Optic Cable (1 Pin)

## 5.1.2 Wireless Sync Unit Components



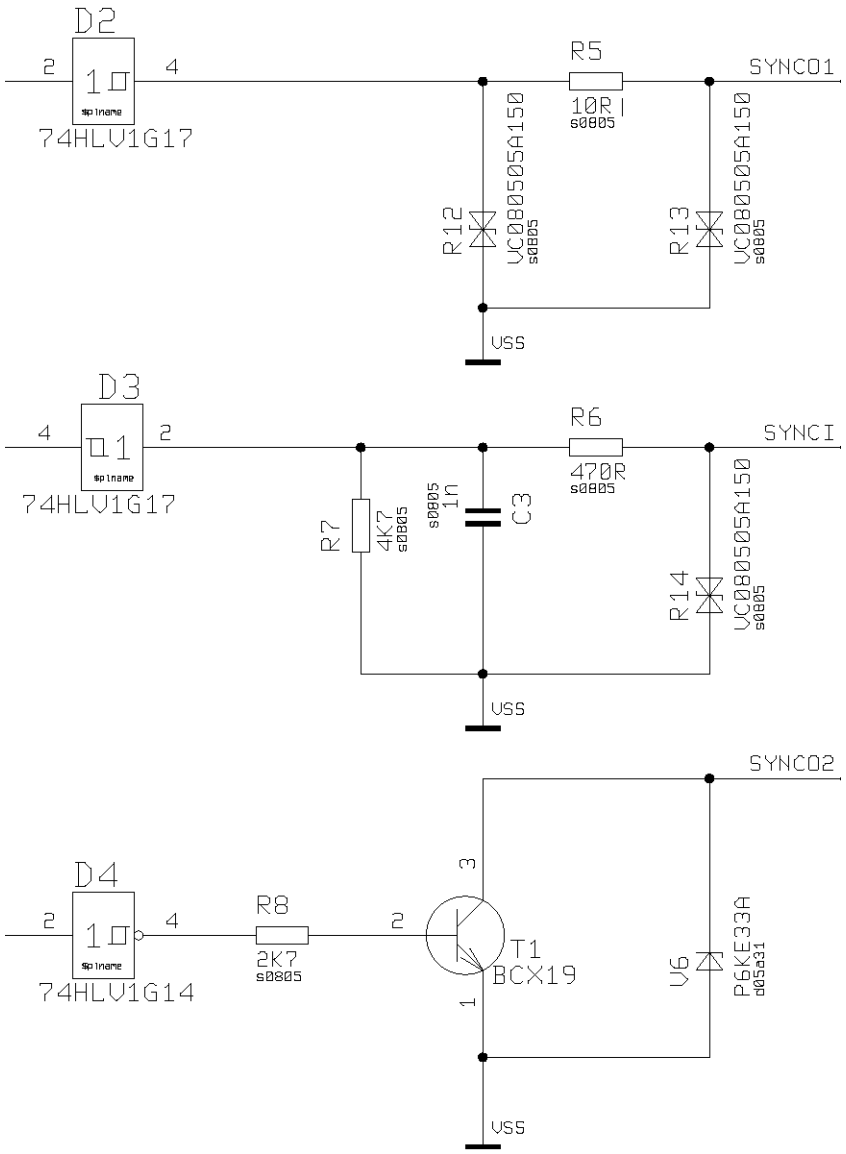

A) Wireless synchronization unit: PC Side
B) Wireless synchronization unit: pliance Side
C) Electrical Signal Cable
D) Sync Unit Trigger Cable
E) Trigger Cable
F) LED Flash Unit
G) Sync Fiber Optic Cable (1 Pin)

### 5.1.3 Connections of the Sync Boxes and pliance Box



## 5.1.4 Schematic diagram of the electrical ports of the Sync Box

SYNC01: TTL output
SYNCI: input
SYNC02: Open collector output



### 5.1.5 Connecting the wired sync box to the pliance device

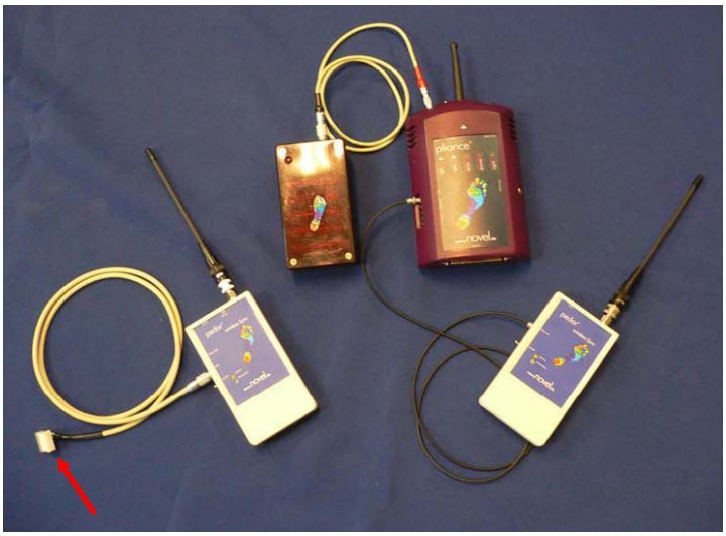



**Connections:**

(1) For a sync out connection:
  (a) For a sync out TTL signal: connect the 1 pin FOC cable to the sync out port on the pliance box and the sync box
  (b) For a sync out open connector in combination with an AUX connection (i.e. the foot switch option): connect the 2 pin FOC cable to the sync out and AUX ports on the pliance box and the sync box
(2) For a sync in connection connect the 1 pin FOC cable to the sync in port on the pliance box and sync box.
(3) Connect the electrical signal cable to the wired sync box.
(4) Utilize the pinouts described in **Section 5.1.3** to capture the electrical signal

## 5.1.6 Connecting the wireless sync box to the pliance device

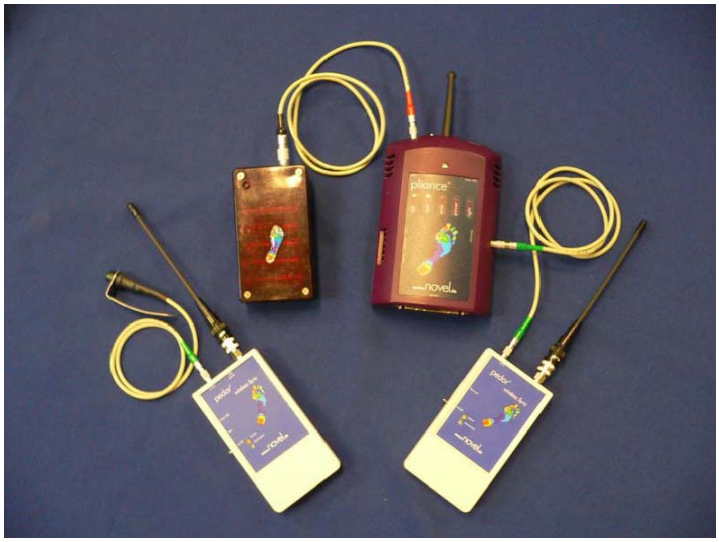
### 5.1.6.1 Electrical signal output/input



**Connections:**

(1) For a sync out connection:
   (a) For a sync out TTL signal: connect the 1 pin FOC cable to the sync out port on the pliance box and to the pliance side wireless sync box
   (b) For a sync out open connector in combination with a sync in connection: connect the 2 pin FOC cable to the sync out and sync in ports on the pliance box and the pliance side wireless sync box
(2) For a sync in connection connect the 1 pin FOC cable to the sync in port on the pliance box and pliance side wireless sync box.
(3) Connect the electrical signal cable to the PC side wireless sync box.
(4) Utilize the pinouts described in **Section 5.1.3** to capture the electrical signal

### 5.1.6.2 Remote trigger



**Connections:**

(1) Connect the sync unit trigger cable to the pliance side wireless sync box and to the start/stop port on the pliance box.
(2) Connect the start/stop cable to the switch port on the PC side wireless sync box.
(3) This setup may be used in combination with the electrical signal sync out option described in the previous section.

### 5.1.6.3 LED indicator

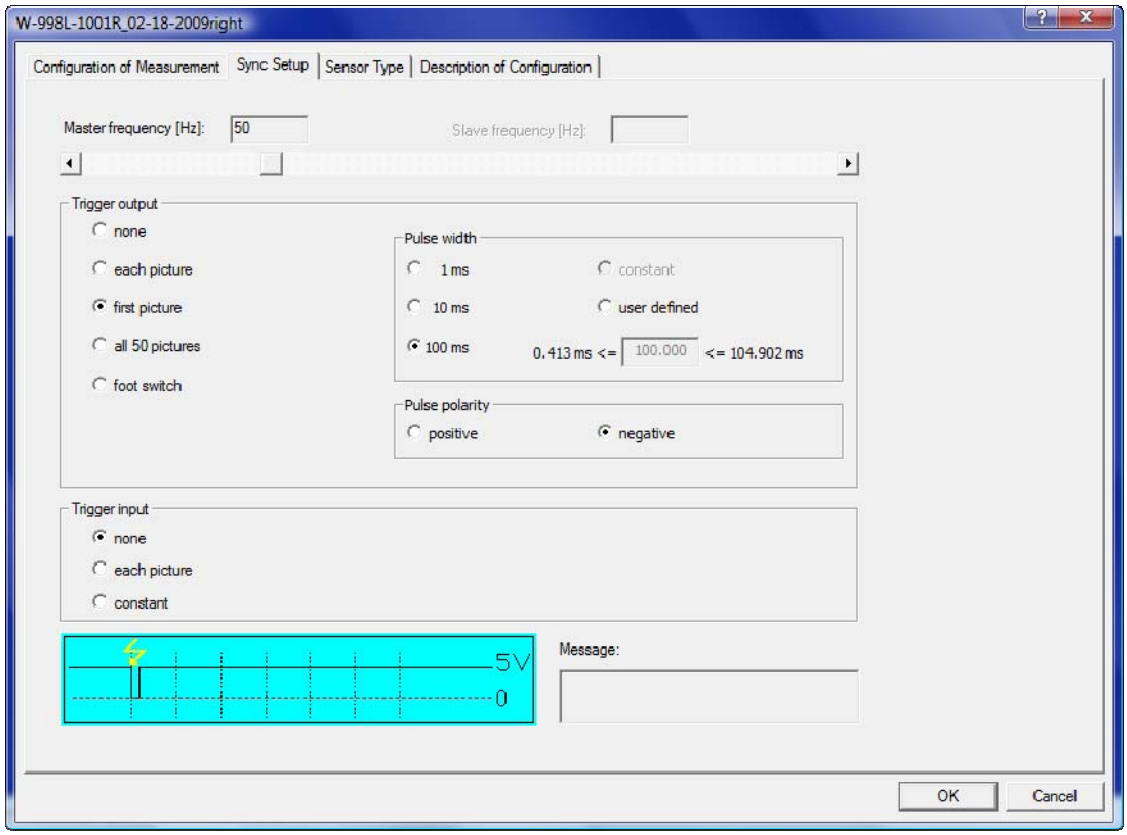


**Connections:**

(1) Connect the 1 pin FOC cable to the sync out port on the pliance box and to the pliance side wireless sync box.
(2) Connect the flash LED to the Flash LED port on the PC side wireless sync box.
(3) This option may be used in combination with the electrical signal sync out option described in the previous section.

## 5.2 The Sync Setup Tab in the pliance online Software

The settings for the synchronization signal are found within the mat configuration file. The configuration file can be accessed from the **Expert Settings | Edit mat configuration** or **Edit actual configuration** menu. Please see **Section 4.3** for additional information on the mat configuration file.




### 5.2.1 Master and Slave Frequency

The master frequency will typically have been set under the **Configuration of Measurement** tab (see **Section 4.3.1**). This setting can be adjusted here as well by moving the slide bar.

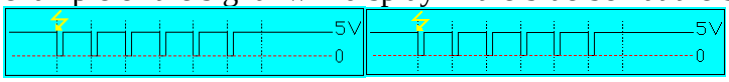
The slave frequency is used with the trigger input and an **each picture** setting. The pliance device will then wait to receive a signal from the "master" device.

### 5.2.2 Setting the Trigger Output

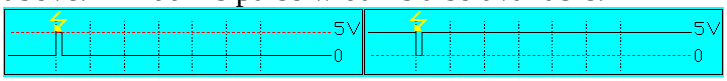
There are four options for a trigger output with the sync system:

**(1) Each picture:** With this option a trigger output will be sent with each frame of data collection from the pliance device. The pulse width must be defined: 1 ms, 10 ms, constant or user defined length. The pulse polarity must also be set, either positive or negative (rising or falling). This should be set according to the

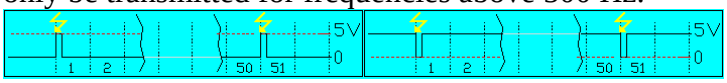
requirements and measurement rate of the device which is being triggered. An example of the signal will display in the blue box at the bottom of the screen:



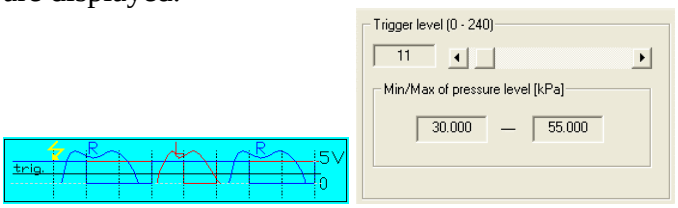

**(2) First picture:** With this option a trigger output signal is sent with the first frame of data collected by the pliance. The pulse width must be defined, as described above. A 100 ms pulse width is also available.




(3) **All 50 pictures:** With this option a trigger output signal will be sent with every 50th picture, i.e. 1st, 50th, 100th etc frames. The polarity and pulse width are defined as with the **each picture** option. NOTE: in this mode a constant pulse can only be transmitted for frequencies above 500 Hz.




(4) **Foot switch:** With this option trigger output signals are provided for the left and right foot contact, providing information similar to that of a foot switch. The trigger level to obtain a valid "step" must be set. The trigger level is not related to the calibrated pressure values, but to the raw data of the sensors (A/D values). Therefore the minimum and maximum pressure values that can trigger the contact are displayed.




## 5.2.3 Setting the Trigger input

There are two options for a trigger input with the sync system. The pliance is either triggered on the rising (positive) or falling (negative) edge of the signal. The duration is not important, as long as it is not below 1 ms:

(1) **Each picture:** This option is used when the pliance device runs in the "slave" mode. With each pulse (either a positive or negative polarity), the pliance will take a measurement. The frequency of measurement must be set as the "slave" frequency. Ensure that the frequency is not above the maximum frequency for the sensor layout.

(2) **Constant:** With the constant option the pliance will record with a constant input, either positive or negative polarity. The pliance will collect at the frequency set by the master frequency. To begin recording the **Measure** button must be clicked within the pliance online software. The device will wait for the trigger pulse. The **stop and store** button must be clicked to stop the measurement.

(a) **Edge-triggered:** Select this option to trigger both the start and stop of the measurement with a pulse, either positive or negative polarity. The advantage of this option is that only two pulses are needed to control the start and stop of the measurement.

**NOTE:** The trigger output options available with the trigger input change depending on the settings which are chosen.

############

# 6 TROUBLESHOOTING BLUETOOTH CONNECTION

## 6.1 General Bluetooth Information

The pliance-x system transmits wirelessly to the computer with the pliance online software via Bluetooth (BT) transmission. A BT dongle is provided to all novel customers which is rated to transmit for 100 meters. Many laptops also have a BT connection, though this should not be used, and possibly disabled, so as not to interfere with the pliance BT transmission. Please see **Section 2.4.6** for information on installing the BT dongle and connecting the pliance-x box to the computer via BT.

Bluetooth transmission uses "line-of-sight", which means that they pliance box must be able to "see" the BT dongle connected to the computer. However, it does not need direct line of sight, but rather can use the walls and ceilings to bounce the signal off of. It is generally a good idea to have the BT dongle up and out in the open as best as possible. You may want to utilize the extension cable provided with the dongle or a longer extension cable if your data collection computer is surrounded by walls, etc. It is also important that the BT dongle's antennae point in the same direction that the pliance box's antennae points. For example, if you are using the standard pliance belt, the pliance box will sit vertically on the subject. Therefore ensure that your BT antennae points vertically as well.

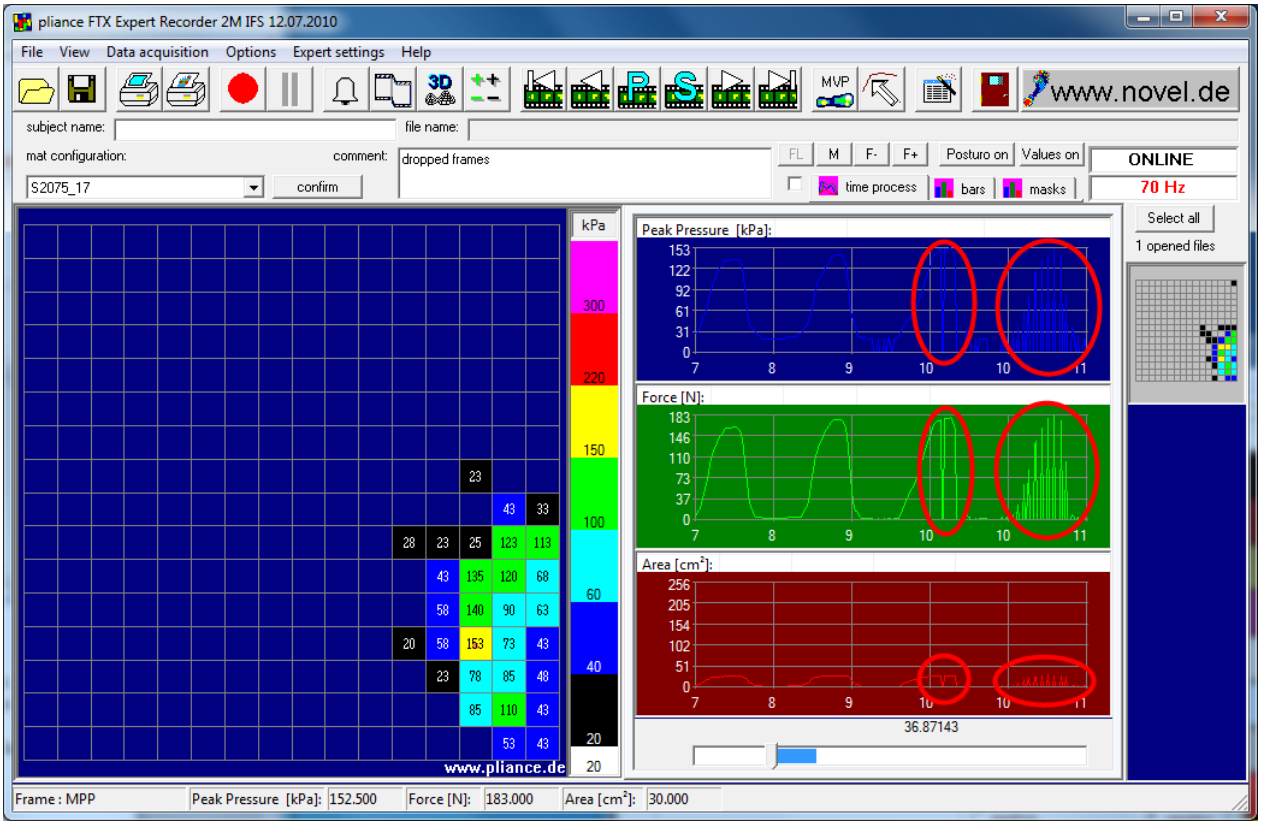
The quality of the Bluetooth transmission is highly dependent on the environment and the number of other signals which are also present. The following are items which can interfere with the BT transmission. It is recommended that, when possible, these items are removed from the field for data collection:

- Other Bluetooth devices such as
  - Mice, keyboards, headsets
  - Cell phones
  - Other computers transmitting BT
- Wireless Internet routers
- Media devices which transmit a Internet POC

To see a listing of other BT devices which are in the area, please see **Section 2.4.6**.

## 6.2 Possible Solutions for Dropped Frames

If dropped frames are occurring regularly within the BT transmission, you will see the pressure/force/area values drop to “0” when pressure should be displayed, as shown in the following screen capture:

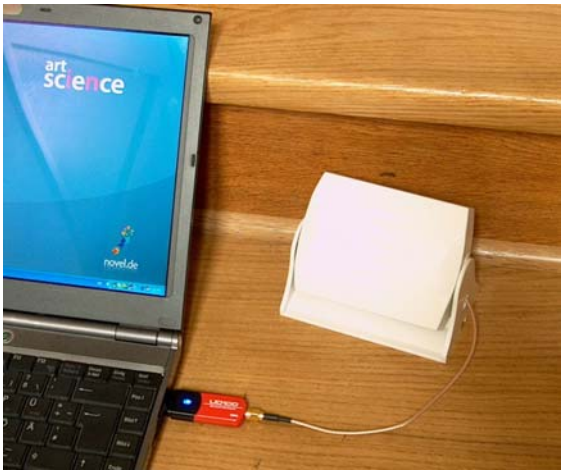



To try to resolve this issue consider the following:

1) Determine what other wireless devices (please see the previous section on devices which may cause interference) are within the data collection space and try to remove these from the space.
2) If a wireless connection is present on the computer you are collecting data, disable this for data collection.
3) Try relocating the BT dongle to a space which is more out in the open utilizing an extension cable if necessary
4) If you are transmitting outdoors, or if the subject’s body is often blocking the antenna/computer line of sight, consider using a pliance removable antenna (please see picture below). This antenna allows the user to place the antennae on the shoulder, top of the head or another location which will always have a line of sight to the BT dongle. If you do not have a removable antenna, please contact your novel representative for information on obtaining such a device.



5) novel also offers a stronger long range patch antenna which can significantly boost the BT signal for both outdoor and indoor activities in which the space is very open, or has many other wireless signals. Please contact your novel representative for information on obtaining such a device.




6) Consider using the **online+flash** mode for data collection. With this mode the data will also be captured to the flash memory and can be downloaded after collection if dropped frames have occurred. For more information on this mode of data collection please see **Section 2.4.3** for additional information.
7) Turn on the BT error correction option. This option must be selected prior to data collection under the Options tab (please see **Section 4.2.4-e**)
When using the **online + flash** mode, this option will automatically fill the dropped frames which occur to the software from the flash memory of the device.

## 6.3 If Bluetooth COM Ports are Not Created Automatically

**Problem Description:**

One of the following issues may arise when connecting the pliance-x box to a Windows XP SP 2 or 3 or Windows Vista computer via a Bluetooth Connection:

- After completing the **Add New Bluetooth Device** wizard there are no COM ports created. Typically an outgoing and incoming COM port will be listed after completing the Add New BT Device wizard. COM ports can also be found under the COM ports tab.
- When adding a COM port, you receive the message **access is denied**

**Solution:**

One of two issues could be the problem:

1) The first issue can be read about in the following Microsoft Knowledge Base article:
   http://support.microsoft.com/kb/894035/

   a) To work around this issue, follow these steps:
      1. Open the Microsoft Management Console (MMC) snap-in for Services. To do this in Windows, click Start, type **services.msc** in the **Start Search** box, and then click **Services** in the **Programs** list. Windows Vista: If you are prompted for an administrator password or for a confirmation, type the password, or click **Continue**.
         To open the Microsoft Management Console (MMC) snap-in for Services in Windows XP, click **Start**, click **Run**, type **services.msc**, and then click **OK**.
      2. Double-click the **Bluetooth Support** service.
      3. If the Bluetooth Support service is stopped, click **Start**.
      4. On the **Startup type** list, click **Automatic**.
      5. Click the **Log On** tab.
      6. Click **Local System account**.
      7. Click **OK**.
      8. If you prompted to restart the computer, click **Yes**

   b) After completing these steps, remove and re-add the pliance-x device within Bluetooth Devices. An outgoing COM port should be created. Test the connection with the pliance-x Online software.

2) If the above process does not work, the second issue can be read about in the following Microsoft Knowledge Base article:
   http://support.microsoft.com/kb/892891/en-us

   a) To work around this issue, follow these steps:

      1. Click **Start**, click **Run**, type **cmd**, and then click **OK**.
      2. At the command prompt, type
         **esentutl /g %windir%\security\database\secedit.sdb**
         You should receive output that is similar to the following output:

```
Microsoft(R) Windows(TM) Database Utilities
Version 5.1
Copyright (C) Microsoft Corporation. All Rights Reserved.
Initiating INTEGRITY mode...
        Database: C:\WINDOWS\security\database\secedit.sdb
```

```
  Temp. Database: TEMPINTEG1704.EDB
Checking database integrity.
                    Scanning Status (% complete)

          0    10   20   30   40   50   60   70   80   90  100
          |----|----|----|----|----|----|----|----|----|----|
          ..................................................
Integrity check successful.
Operation completed successfully in 1.592 seconds.
```

If the Windows security database is corrupted, the output will contain the following line:

```
This operation may find that this database is corrupt.
```

To work around this problem, follow these steps:

1. If both the Edb.log file and the Edb.chk file are in the %windir%\security directory, follow these steps to repair the security database:
   a. Click **Start**, click **Run**, type **cmd**, and then click **OK**.
   b. At the command prompt, type **esentutl /r edb /l %windir%\security /s %windir%\security**
2. If either the Edb.log file or the Edb.chk file is in the C:\WINDOWS\security directory, or if neither file exists in the directory, follow these steps to repair the security database: Click **Start**, click **Run**, type **cmd**, and then click **OK**.
   a. At the command prompt, type **esentutl /p %windir%\security\Database\secedit.sdb**
   b. Click **OK** when you receive the following message:
      You should only run Repair on damaged or corrupted databases. Repair will not apply information in the transaction log files on the database, and may cause information to be lost. Do you wish to proceed?
3. After you repair the security database, follow these steps to modify the registry so that you can access the registry key that is required to configure the Bluetooth device:
   a. Create the C:\bt.inf file that has the following text.

```
[version]
signature="$CHICAGO$"
revision=1
DriverVer=07/01/2001,5.1.2600.1106
[Registry Keys]
"MACHINE\SYSTEM\CurrentControlSet\Services\BTHPORT",0,"D:AR(A;CI;GRGW;;;PU)(A;CI;GRGW;;;LS)"
```

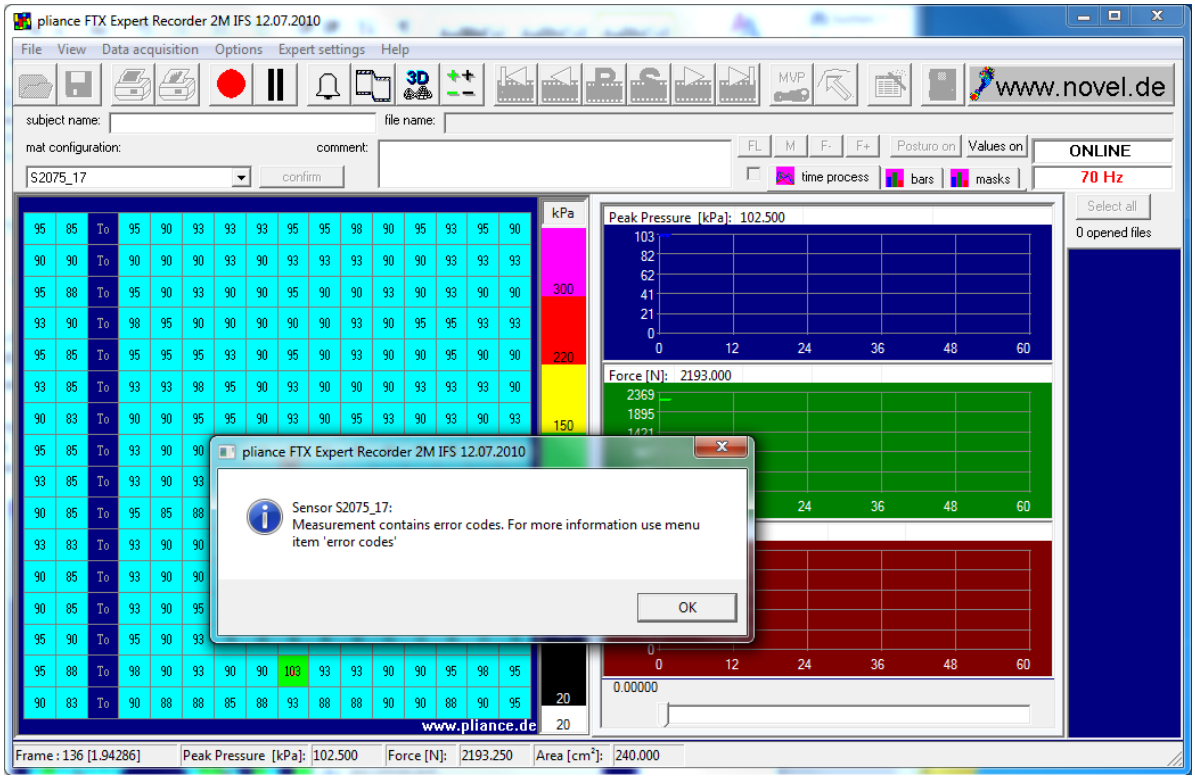
   b. Click **Start**, click **Run**, type **cmd**, and then click **OK**.
   c. At the command prompt, type **secedit /configure /cfg C:\bt.inf /db C:\bt.sdb**
   d. Delete the C:\bt.inf file and the C:\bt.sdb file.

b) After completing these steps, remove and re-add the pliance-x device within Bluetooth Devices. An outgoing COM port should be created. Test the connection with the pliance-x Online software.

# 7 TESTING SENSOR MAT AND CABLE FOR ERRORS

If a frequent timeout error code is being read by the pliance online software, it may be useful to check where the error is coming from. This process can be helpful in reporting the information to your novel representative for further investigation or repair of the mat.

A typical error may occur in one sensor, one row, one column or multiple locations on the mat. Once an error occurs you will notice a TO in the sensor element and should receive the error code warning message, if you have this option turned on (see **Section 4.2.4 – e**).




The sensor mat may have an error in just one section or multiple sections. The following process will help to narrow down where the error is occurring, and therefore the possible cause of the error.

First, it may be helpful to ask these questions:

1) Does the error occur on only one mat, or multiple sensor mats?
2) If so, the issue could be an issue with the sensor cable, not the pliance mat itself. Please see below for additional information.

## 7.1 Testing the mat for errors

1) Connect the pliance device as described in **Section 2.4.4**. Connect the sensor mat in which the errors are occurring.
2) Lay the system flat on a table.
3) Open the pliance online software. In menu Options – Error Codes please uncheck the option With error correction. Choose the mat relevant configuration file, click the *Confirm* button, and when the message appears "Unload mat" click *Cancel*. Begin a recording. – A measurement will start, showing the basic sensor values without the deduction of zero values.

   See **Section 4.1.1** for additional information on collecting data within the pliance online software.
4) Starting from the mat connector, "wiggle" the mat back and forth, not to the point of bending.
5) Observe the pressure data which is displaying in the 2D pressure picture within the pliance online software. Determine if the movement you are making causes the error to occur/stop.
6) Continue this process moving down the lead of the mat. Typically an error in the lead will cause more than one sensor (i.e. an entire row or column) to show a TO.
7) Continue the "wiggle" process from the heel to the toes. Again, make note if a certain location causes the error to occur, or stop occurring.
8) Once the entire mat is checked, stop the recording.

If the error occurs close to the mat connector, this could because too much stress is being placed on the mat lead. Ensure that the cable is attached well and that the connector does not "flap" around as the subject is moving.

If the error occurs at the juncture of the mat lead and measuring mat, this could be due to the fact that there is not enough "slack" being placed in the mat lead.

## 7.2 Testing the mat cable for errors

1) Inspect the mat cable. Look at the connector which connects to the pliance box. Are there any pins bent? This can occur if the cable is not pushed straight in and pulled straight out (i.e. if there is a lot or torque placed on the cable during the connection process). Also inspect the mat connector to determine if any of these pins are bent. If pins are bent, please contact your novel representative for repair instructions. If not, please continue to step #2.
2) Connect the pliance device as described in **Section 2.4.4**. Connect any pair of sensor mat.
3) Lay the system flat on a table.
4) Open the pliance online software. In menu Options – Error Codes please uncheck the option With error correction. Choose the mat relevant configuration file, click the *Confirm* button, and when the message appers "Unload left mat" click *Cancel*. Begin a recording. – A measurement will start, showing the basic sensor values without the deduction of zero values.

   See **Section 4.1.1** for additional information on collecting data within the pliance online software.

5) Starting from the connector which attaches to the pliance box, “wiggle” the cable for which the mat is receiving errors (either right or left) back and forth to the point of bending, but not crimping the cable.
6) Observe the pressure data which is displaying in the 2D pressure picture within the pliance online software. Determine if the movement you are making causes the error to occur/stop.
7) Continue this process moving down the entire length of the cable. Typically an error in the cable will cause more than one sensor (i.e. an entire row or column) to show a TO.
8) If the error is occurring on the other pliance mat as well, repeat this process for that side’s cable.
9) Once the entire cable is checked, stop the recording.
10) Repeat such a test with another mat. If exactly the same columns or lines are instable or missing, it indicates a damage of the cable.

Once you have determined that the cable is faulty, please contact your novel representative for repair instructions.

# 8 DEFINITION OF VARIABLES

**Maximum Pressure Picture (MPP)**
Displays the highest pressure that was reached for each sensor at any time during the data collection period. This is therefore an artificial picture that provides a summary of the highest pressures that were reached over the entire surface of the mat – it does not represent any one step or event.

**Mean Value Picture (MVP)**

***over all frames:***
Displays the mean over all frames of the data collection process is calculated (even if the mat was completely unloaded during a frame) for each sensor within the mat.

***without unloaded frames*:**
Displays the average pressure recorded by each sensor during the contact phases of the data collection period, calculated by summing the pressures for each sensor and dividing by the number of frames that the sensor is loaded.

***without unloaded sensors*:**
For each sensor the mean is calculated over the frames during the data collection period when the sensor was loaded, i.e. all zero values are ignored.

**Force (N)**

***for the MPP:***
The maximum force is shown. This value will depend on how the MPP was calculated. In the pliance-x online software, it is the maximum force for the entire data collection.

***for the MVP:***
The mean force is shown. This value depends on the MVP calculation. If the MVP was calculated over all frames, then the force will be the mean over all frames. If the MVP was calculated without unloaded sensors, then the force will be the mean of the steps only.

***for any other picture:***
The force is calculated as the sum of forces experienced by each sensor loaded at that time.

**Area ($cm^2$)**
Area is calculated as the sum of the area of all loaded sensors shown in the picture or frame.

**Peak Pressure (kPa)**
The maximum pressure (sometimes referred to as maximum pressure) is the highest pressure experienced by any one sensor in the picture/frame.

**Mean Pressure (kPa)**
The average pressure is calculated for each picture/frame by summing the pressures in all sensors and dividing by the number of sensors.

**Force-time Integral (N*s)**

This value is calculated once for each trial. It is the summation of the force at each frame multiplied by the time per frame, or more simply, the area under the force-time curve.

**Pressure-time Integral (kPa*s)**

This value is calculated once for each trial. It is the summation of the peak pressure at each frame multiplied by the time per frame, or more simply, the area under the peak pressure-time curve.

#####

# 9 DEFINITION OF FILE TYPES

## 9.1 pliance Data Files

By default, the data is stored in the folder C:\novel\novfile\data\pliance\. This directory can be changed from the **Expert Settings** menu of from the **Configuration** program.

- ***.mat** – pliance data file. This file can be opened in the pliance online software as well as in many of the novel scientific software programs (please refer to these manuals directly for additional information).

## 9.2 pliance Calibration and Configuration Files

By default, the calibration files are located at C:\novel\NOVSYST\pliancex\cab\pliance and the configuration files are located at C:\novel\NOVSYST\pliancex\mfg. These directories can be changed from the **Expert Settings** menu of from the **Configuration** program. These files can only be accessed from the pliance online software. All six calibration files must be located within the defined directory in order for a calibration to be accessed and used.

- **Calibration Files:**
  - ***.def:** Contains information regarding the values at each step used during the calibration process.
  - ***.cab:** Contains the calibration curves for each sensor.
  - ***.tbl:** Contains information regarding the arrangement of sensors and which sensors are activated.
  - ***.set:** Contains information regarding the pressure range of the mat and the span of the range across the A/D values.
  - ***.geo:** Contains information regarding the shape and size of the sensors.
  - ***.chl:** Not implemented with the pliance-x system.
- **Configuration File:**
  - ***.mfg:** Contains the calibration file, as well as the information on the configuration of measurement, synchronization settings and sensor layout. Please see **Section 4.3** for additional information.

## 9.3 pliance Exported files

These files can be exported and created from the .sol files. They are tab delimited files which can be open in programs such as Excel, Notepad, etc. These files can be saved manually from the pliance online software program, or exported automatically when the file is saved immediately after data collection. Please see **Section 4.2.1 -f and 4.2.4 -c** for additional information

- ***.asc:** This is the raw ascii data from the pliance data collection file. It contains the individual pressure values for each of the sensor elements throughout time. A header is also included. A portion of an example output is shown here:

| file name: | date/time 17.02.11 17.32 | | | | |
|---|---|---|---|---|---|
| sensor type: S2075_17 | | | | | |
| total time [secs]: 0.986 | time per frar | scanning rate [Hz]: 70 | | | |
| pressure values in kPa | | | | | |
| | | | | | |
| electrical: | 0 | 1 | 2 | 3 | 4 |
| time[secs] | 1 | 2 | 3 | 4 | 5 |
| 0.01429 | 90.000 | 90.000 | 87.500 | 90.000 | 87.500 |
| 0.02857 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.04286 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.05714 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.07143 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.08571 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.10000 | 92.500 | 90.000 | 87.500 | 90.000 | 87.500 |
| 0.11429 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.12857 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.14286 | 92.500 | 90.000 | 90.000 | 90.000 | 90.000 |
| 0.15714 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.17143 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |
| 0.18571 | 92.500 | 90.000 | 87.500 | 90.000 | 90.000 |

- ***.fgt:** This exported file contains the force and gaitline (center of pressure) for the left and right mat throughout time. A header is also included. A portion of an example output is shown here:

| file name: | date/time 17.02.11 17.32 | | |
|---|---|---|---|
| sensor type: S2075_17 | | | |
| force time integral[N*s]: 2312.732 | | | |
| pressure time integral [kPa*s]: 100.857 | | | |
| total time [secs]: 0.986 | time per frar | scanning rate [Hz]: 70 | |
| | | | |
| | | | |
| time[secs] | force[N] | x[mm] | y[mm] |
| | | | |
| 0.01429 | 2.343.750 | 79.949 | 80.170 |
| 0.02857 | 2.346.000 | 79.931 | 80.167 |
| 0.04286 | 2.346.500 | 79.944 | 80.153 |
| 0.05714 | 2.346.250 | 79.957 | 80.142 |
| 0.07143 | 2.345.250 | 79.960 | 80.143 |
| 0.08571 | 2.347.500 | 79.948 | 80.180 |
| 0.10000 | 2.346.000 | 79.984 | 80.185 |
| 0.11429 | 2.345.250 | 79.955 | 80.165 |
| 0.12857 | 2.346.000 | 79.972 | 80.175 |
| 0.14286 | 2.347.500 | 79.938 | 80.163 |
| 0.15714 | 2.346.750 | 79.970 | 80.159 |

- ***.mva:** This exported file contains data throughout time for the masked parameters. (For more information on creating masks please see **Section 4.2.4 – p**. The parameters included are force, peak pressure, mean pressure and standard deviation of pressure. A header is also included. A portion of an example output is shown here:

| | | | | | | |
|---|---|---|---|---|---|---|
| file name: | date/time 17.02.11 17.32 | | | | | |
| sensor type: S2075_17 | | | | | | |
| total time [secs]: 0.986 | time per frar | scanning rate [Hz]: 70 | | | | |
| pressure values in kPa | | | | | | |
| | | | | | | |
| mask definition: | | | | | | |
| left front: 105,106,107,108,109,110,111,112,121,122,123,124,125,126,127,128,137,138,139,140,141,142,143 | | | | | | |
| left hind: 9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32,41,42,43,44,45,46,47,48,57,58,59,60,61,62,63,64, | | | | | | |
| right front: 97,98,99,100,101,102,103,104,113,114,115,116,117,118,119,120,129,130,131,132,133,134,135,1 | | | | | | |
| right hind: 1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24,33,34,35,36,37,38,39,40,49,50,51,52,53,54,55,56,65,66,6 | | | | | | |
| | | | | | | |
| | left front | | | | left hind | |
| | | | | | | |
| time[secs] | force[N] | max pressur | mean pressu | +-% of mean | force[N] | max pressur |
| 0.01429 | 730.500 | 97.500 | 91.313 | 2.33 | 435.000 | 97.500 |
| 0.02857 | 730.500 | 97.500 | 91.313 | 2.37 | 435.500 | 97.500 |
| 0.04286 | 731.000 | 97.500 | 91.375 | 2.37 | 436.000 | 97.500 |
| 0.05714 | 730.750 | 97.500 | 91.344 | 2.37 | 436.000 | 97.500 |
| 0.07143 | 730.750 | 97.500 | 91.344 | 2.45 | 435.500 | 97.500 |
| 0.08571 | 732.250 | 97.500 | 91.531 | 2.35 | 435.250 | 97.500 |

## 9.4 Other pliance Files

- ***.sgm:** This file is the sensor geometry file. It is located at C:\novel\novfile\defin\mats. This file is used in definition of the mat size and layout.
- ***.msk:** This is the mask file which is created during the Read/Define Masks process or during the Define sensors to be scanned process. Please see **Sections 4.2.4 – p and 4.2.3 – f** for additional information.
- ***.far:** This file contains a group of measurements which were collected to the flash memory of the pliance device. This file may be opened using the menu item File | Open FAR file (see **Section 4.2.1-c**). The file is created from the menu item Data Acquisition | Read Flash | Save to FAR file (see **Section 4.2.3-j**).

## 9.5 Measurement conversions

**Pressure Units:**

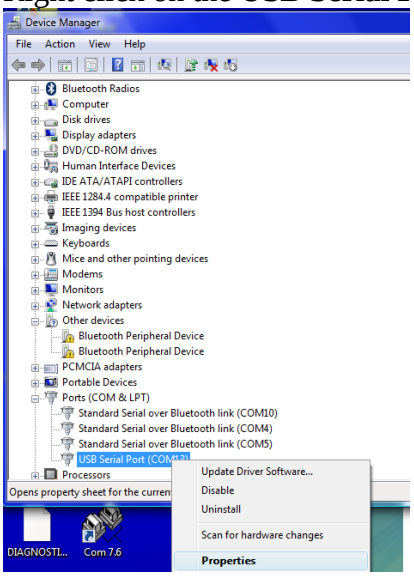
1 N/cm$^2$ = 0.1 Bar
10 kPa
74.006 mmHg
1.4504 Psi

#####

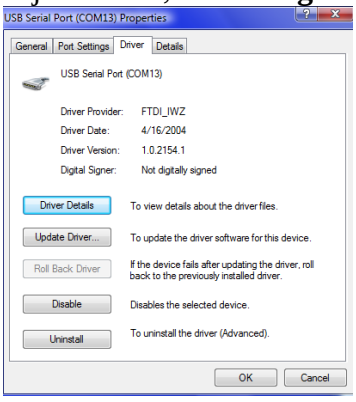
# 10 ISSUES WITH THE FIBER OPTIC CABLE DRIVER INSTALLATION

If the Fiber Optic Cable drivers installed automatically without your control, or you are not making a connection with the pliance online software, please follow these steps to check that the proper drivers have installed.

1. Check which driver is installed
    a. plug in the USB converter
    b. Go to Device Manager
    c. Choose **Ports COM&LPT**
    d. Right click on the **USB Serial Port** and choose **Properties**




    e. Under the **Driver** tab the Driver provider should be **FTDI_IWZ**. If it is not, and is just **FTDI**, the **wrong** drivers are installed.




    f. Close out of these properties

2. Browse to C:\novel\NOVSYST\driver\usb

3. Run the program **FTdiunin.exe**.
    a. It will first ask you to unplug the USB converter
    b. Click Continue

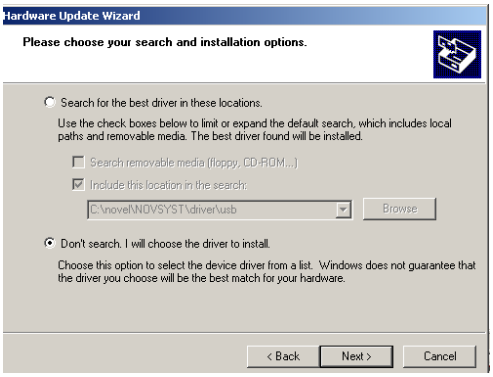
i. Possible error: the program may ask to be in a different location. If so copy the necessary file to the location requested and re-run the program

c. Click **Finish** once successfully uninstalled

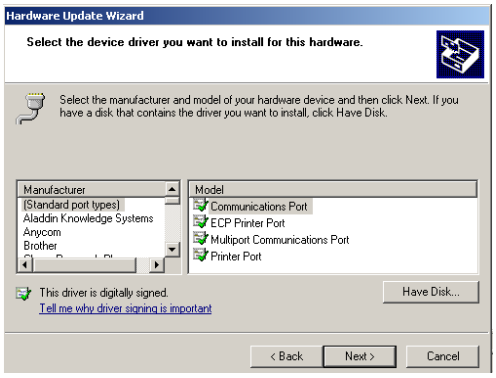
4. Plug back in the USB converter
   a. Follow the steps outlined in **Section 2.4.5.1**.
5. Once this driver is installed go back to the device manager and check for the FTDI_IWZ is now listed
6. Note the COM port listed
7. Open the pliance online software and test the connection. Please see **Sections 4.1.1 and 4.2.3 – a** for additional information.

If the above process still does not install the proper driver, it may be required to manually install the driver via this process:

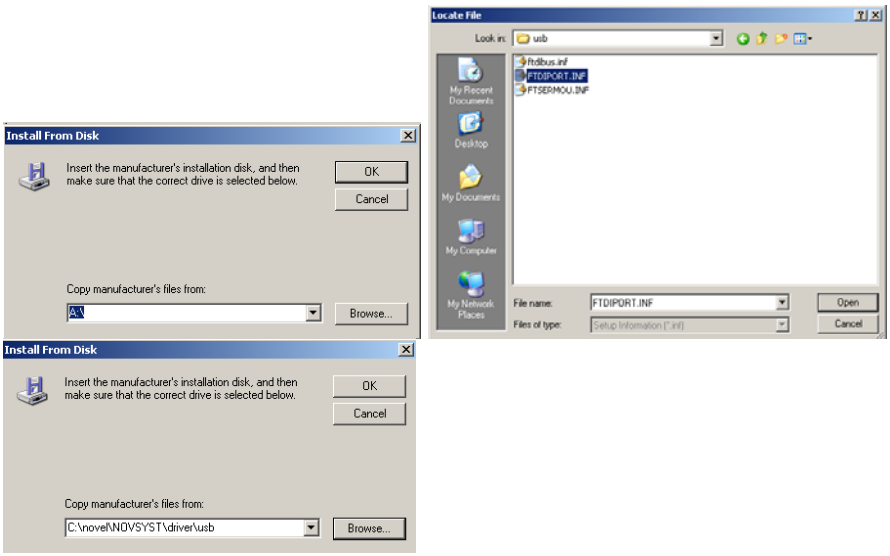
1. Follow step #1 above to check which driver is installed.
2. Choose **Update Driver**
3. The **Found New Hardware Wizard** will appear. This time choose **Don't search, I will choose the driver to install**



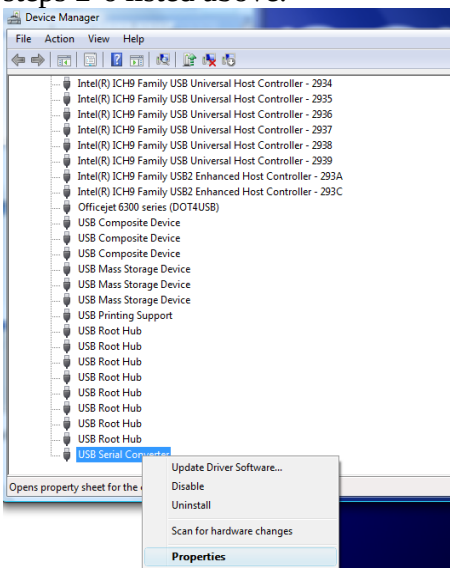

4. Choose **Have Disk**




5. Browse to the C:\novel\NOVSYST\driver\usb directory and choose the **FTDIPORT.inf** file. Click **Open**, then **OK**, and then click **Next**.




6. Click **Finish** to complete the installation.
7. It may be necessary to complete this process for the second driver as well. If so, choose the **USB Serial Converter** driver under the **Universal Serial Bus Controllers** and follow steps 2-6 listed above.




8. Open the pliance online software and test the connection. Please see **Sections 4.1.1 and 4.2.3 – a** for additional information.

#######